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MIL-M-28787/348(SH)
5 December 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1553 DUMB RTU MODULE, KEY CODE UKT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1553 dumb Remote Terminal Unit (RTU) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.81 failures per million operating hours for class I or 12.75 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be UKT.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 115 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A12	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B14	Continuity	III	X	0	1	0	1	0	1	Ohm
C1-C3	Power supply stress	IV	S	---------- Module shall pass all remaining tests ----------						
D1,D2	Transformer short circuit test	V	S	---------- Module shall pass all remaining tests ----------						
E1-E21	Ibd	VI	S	0	100	0	100	0	100	uA
E22	Ibd	VI	S	0	150	0	150	0	150	uA
E23,E24	Ibd	VI	S	0	200	0	200	0	200	uA
E25,E26	Ibd	VI	S	0	400	0	400	0	400	uA
E27-E31	Ibd	VI	S	0	150	0	150	0	150	uA
E32-E34	Ibd	VI	S	0	40	0	40	0	40	uA
E35-E42	Ibd	VI	S	0	20	0	20	0	20	uA
E43	Ibd	VI	S	0	40	0	40	0	40	uA
F1-F16	Ios	VII	S	-225	-40	-225	-40	-225	-40	mA
F17-F21	Ios	VII	S	-100	-20	-100	-20	-100	-20	mA
G1-G21	Iih	VIII	X	0	20	0	20	0	20	uA
G22	Iih	VIII	X	-250	-40	-250	-40	-250	-40	uA
G23,G24	Iih	VIII	X	0	40	0	40	0	40	uA
G25,G26	Iih	VIII	X	0	80	0	80	0	80	uA
G27-G32	Iih	VIII	X	-400	-10	-400	-10	-400	-10	uA
G33,G34	Iih	VIII	X	-250	-40	-250	-40	-250	-40	uA
G35-G42	Iih	VIII	X	-20	20	-20	20	-20	20	uA
G43	Iih	VIII	X	-40	40	-40	40	-40	40	uA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H21	Iil	IX	X	-0.4	0.0	-0.4	0.0	-0.4	0.0	mA
H22	Iil	IX	X	-0.8	-0.04	-0.8	-0.04	-0.8	-0.04	mA
H23,H24	Iil	IX	X	-0.8	0.8	-0.8	0.8	-0.8	0.8	mA
H25,H26	Iil	IX	X	-1.6	0.0	-1.6	0.0	-1.6	0.0	mA
H27-H31	Iil	IX	X	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	mA
H32	Iil	IX	X	-0.8	-0.04	-0.8	-0.04	-0.8	-0.04	mA
H33,H34	Iil	IX	X	-0.4	-0.04	-0.4	-0.04	-0.4	-0.04	mA
H35-H42	Iil	IX	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
H43	Iil	IX	X	-0.04	0.04	-0.04	0.04	-0.04	0.04	mA
J1-J26	Vic	X	S	-1.5	-0.3	-1.5	-0.3	-1.5	-0.3	V
K1-K40	Voh	XI	X	2.4	4.55	2.4	4.55	2.4	4.55	V
L1-L40	Vol	XII	X	0.0	0.4	0.0	0.4	0.0	0.4	V
M1-M16	Iozh	XIII	X	-20	20	-20	20	-20	20	uA
N1-N16	Iozl	XIV	X	-20	20	-20	20	-20	20	uA
P1	Ioo	XV	X	30	300	30	300	30	300	mA
P2,P3	Icc	XV	X	25	250	25	250	25	250	mA
Q1,Q2	Functional	XVI	X	See table XVI						
R1,R2	Dynamic	XVII	S/Q	See table XVII						
S1	PSFC	XVIII	S	0.33	0.61	0.33	0.61	0.33	0.61	uF
S2,S3	PSFC	XVIII	S	4.8	8.8	4.8	8.8	4.8	8.8	uF
T1-T43	Cin	XIX	Q	2	50	2	50	2	50	pF
U1-U16	Cout	XX	Q	2	50	2	50	2	50	pF

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
V1-V3	PSTA	XXI	Q	0	400	0	480	0	400	mV
W1-W4	Transformer impedance	XXII	S	2.5	5.0	2.5	5.0	2.5	5.0	Vrms
X1-X12	Receiver common mode	XXIII	S	See table XXIII						
Y1-Y20	Transmitter functional	XXIV	X	See table XXIV						
Z1-Z4	Receiver no response test	XXV	Q	See table XXV						
AA1-AA6	Output symmetry	XXVI	Q	-90	90	-90	90	-90	90	mVpk
AA7-AA12	Output symmetry	XXVI	Q	-250	250	-250	250	-250	250	mVpk
BB1-BB40	Output X-TALK, high level	XXVII	Q	0	400	0	480	0	400	mV
CC1-CC40	Output X-TALK, low level	XXVIII	Q	0	400	0	480	0	400	mV
DD1-DD16	Receiver functional	XXIX	X	See table XXIX						

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test tolerance table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/348-1 for class I and M28787/348-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, (V+) of 15.0 ± 1.5 V dc, and (V-) of -15.0 ± 1.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

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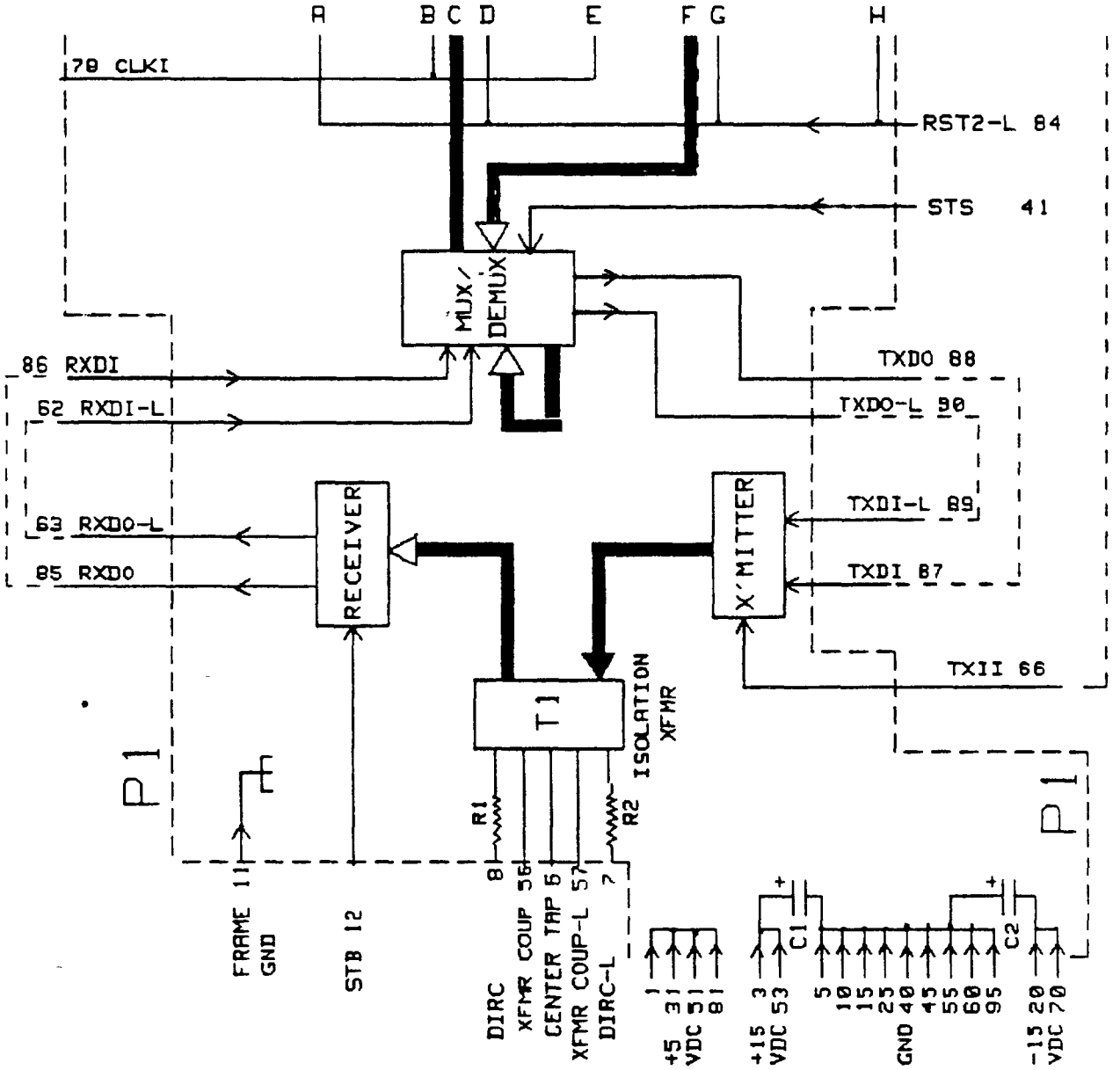


FIGURE 1. UKT functional block diagram.

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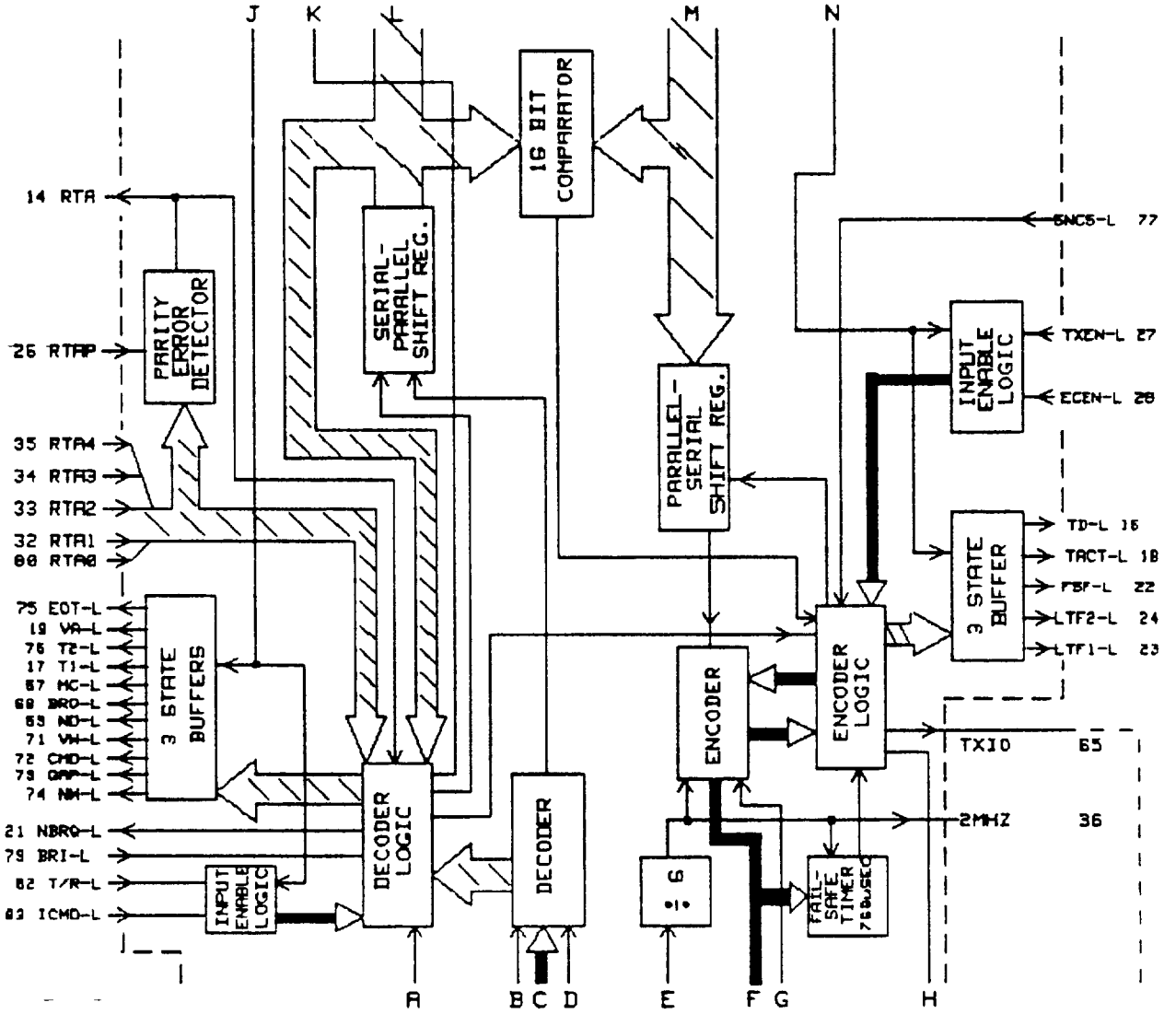


FIGURE 1. UKT functional block diagram - Continued.

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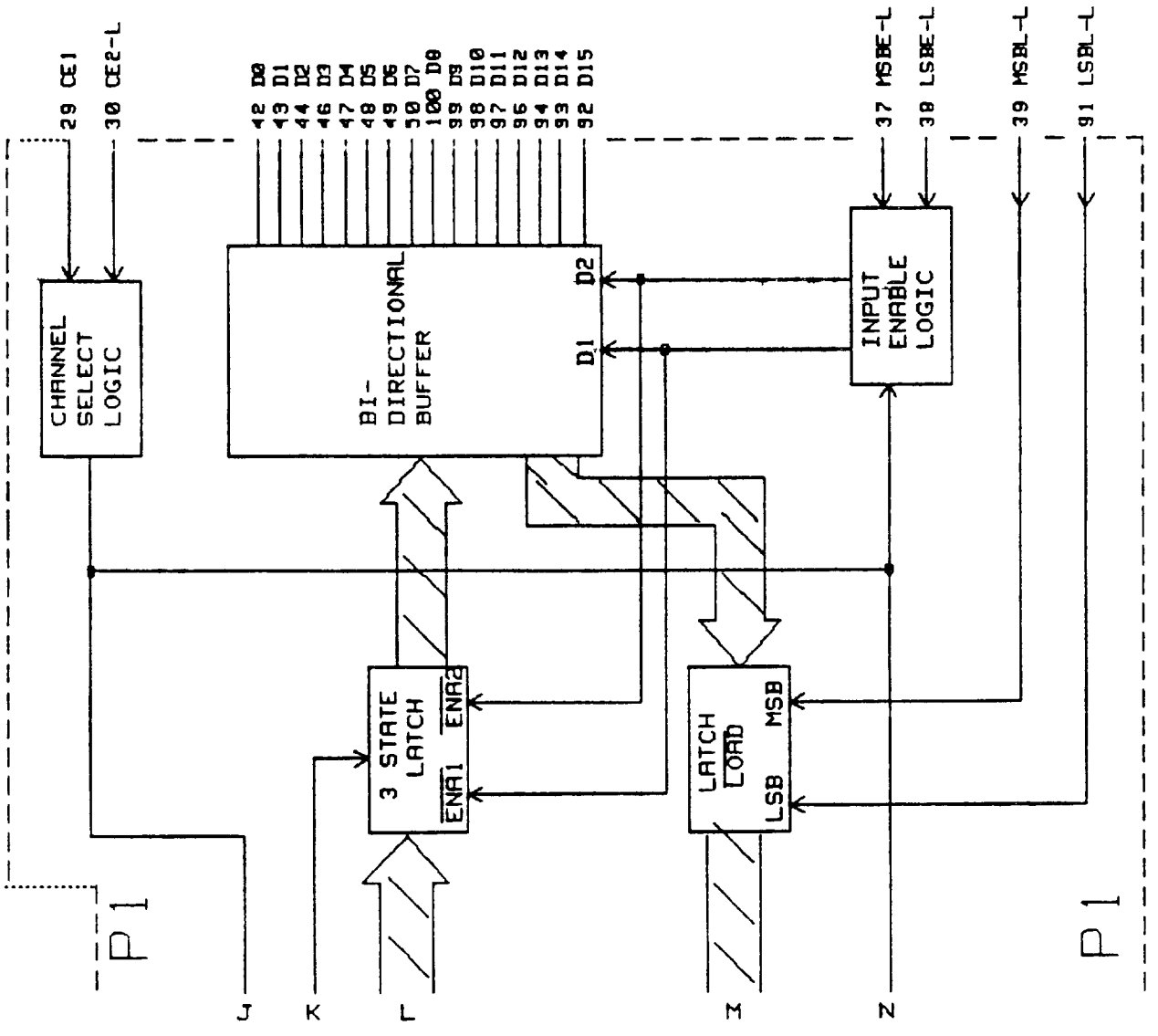


FIGURE 1. UKT functional block diagram - Continued.

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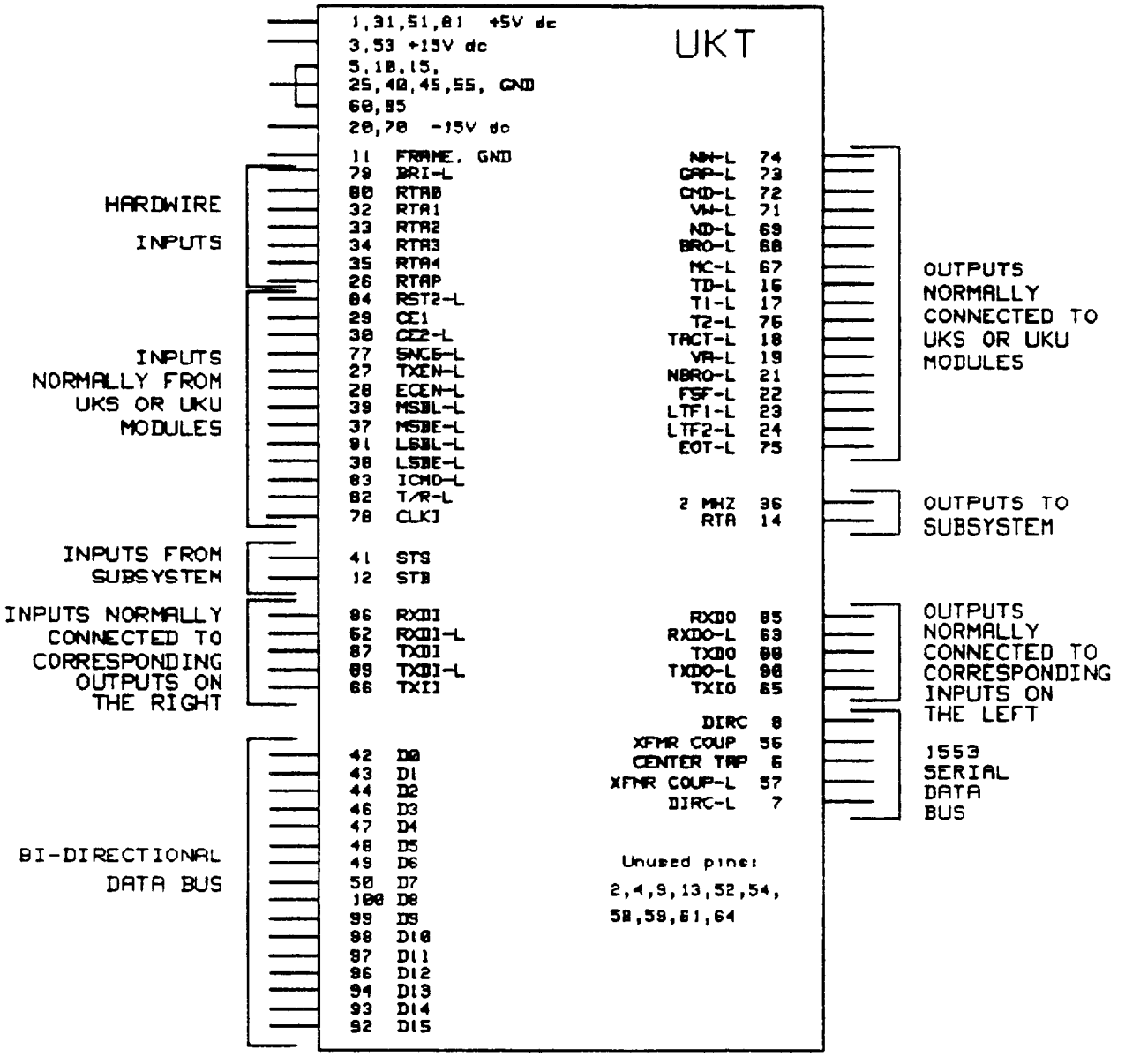


FIGURE 2. UKT pin assignment.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-STD-28787.

4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. Test conditions A, C, D, E, F and J are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, F or J are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, F, and J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through DD in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G, through S and V through DD shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25 ± 5°C ambient.

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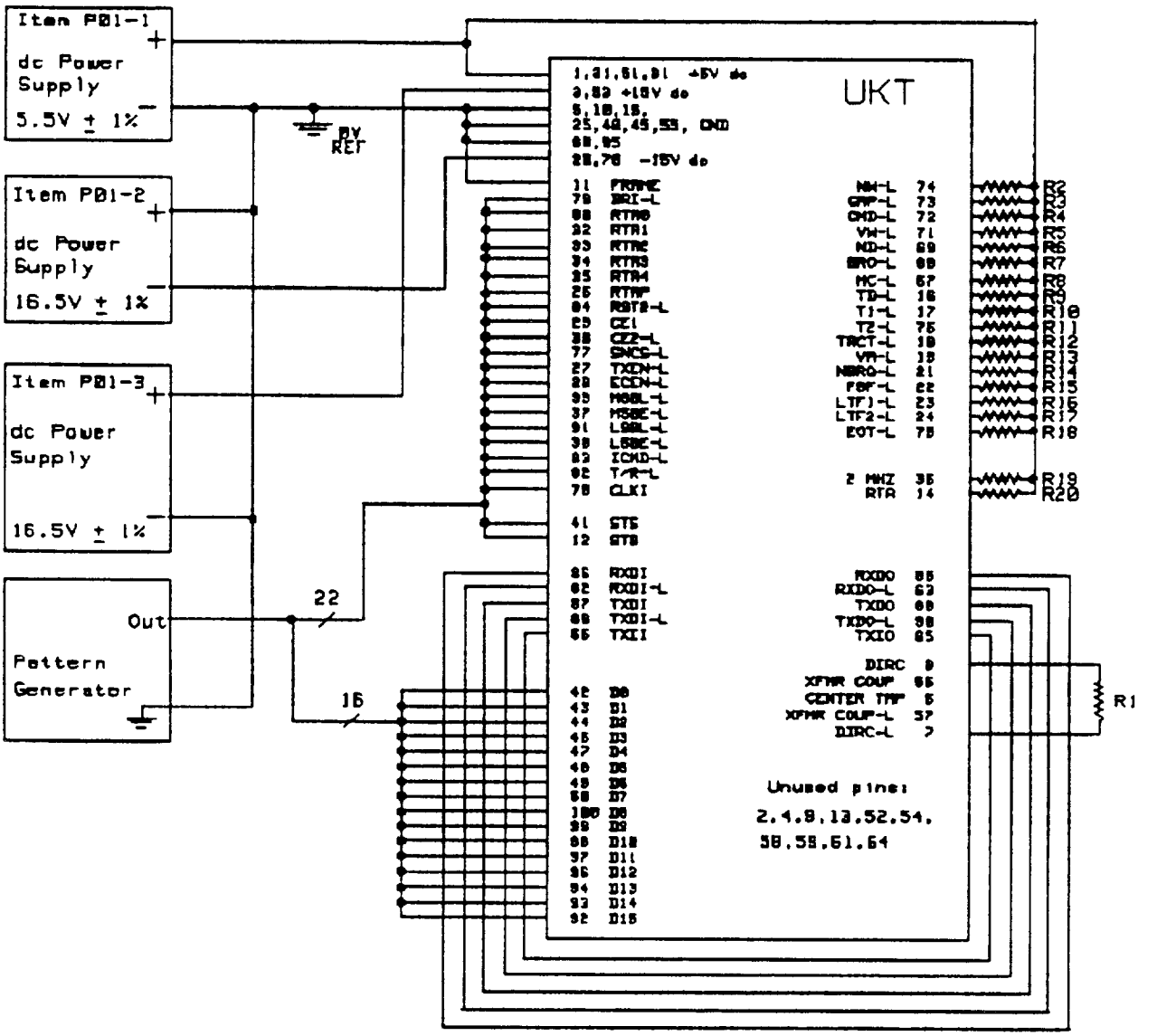


FIGURE 3. UKT operating temperature, component temperature and life test circuit.

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NOTES:

1. R1 = 35 Ohm ± 2 percent, 2 W; R2-R18 = 3.24 KOhm ± 5 percent, 1/4 W; R19 = 1.3 KOhm ± 5 percent 1/4 W; R20 = 1.43 KOhm ± 5 percent, 1/4 W.

2. D0-D15 are bidirectional signals. Each will have a 432 Ohm ± 5 percent, 1/4 W resistor tied to Vcc (5.5 V) when programmed to be an output.

3. Set the pattern generator to a pattern rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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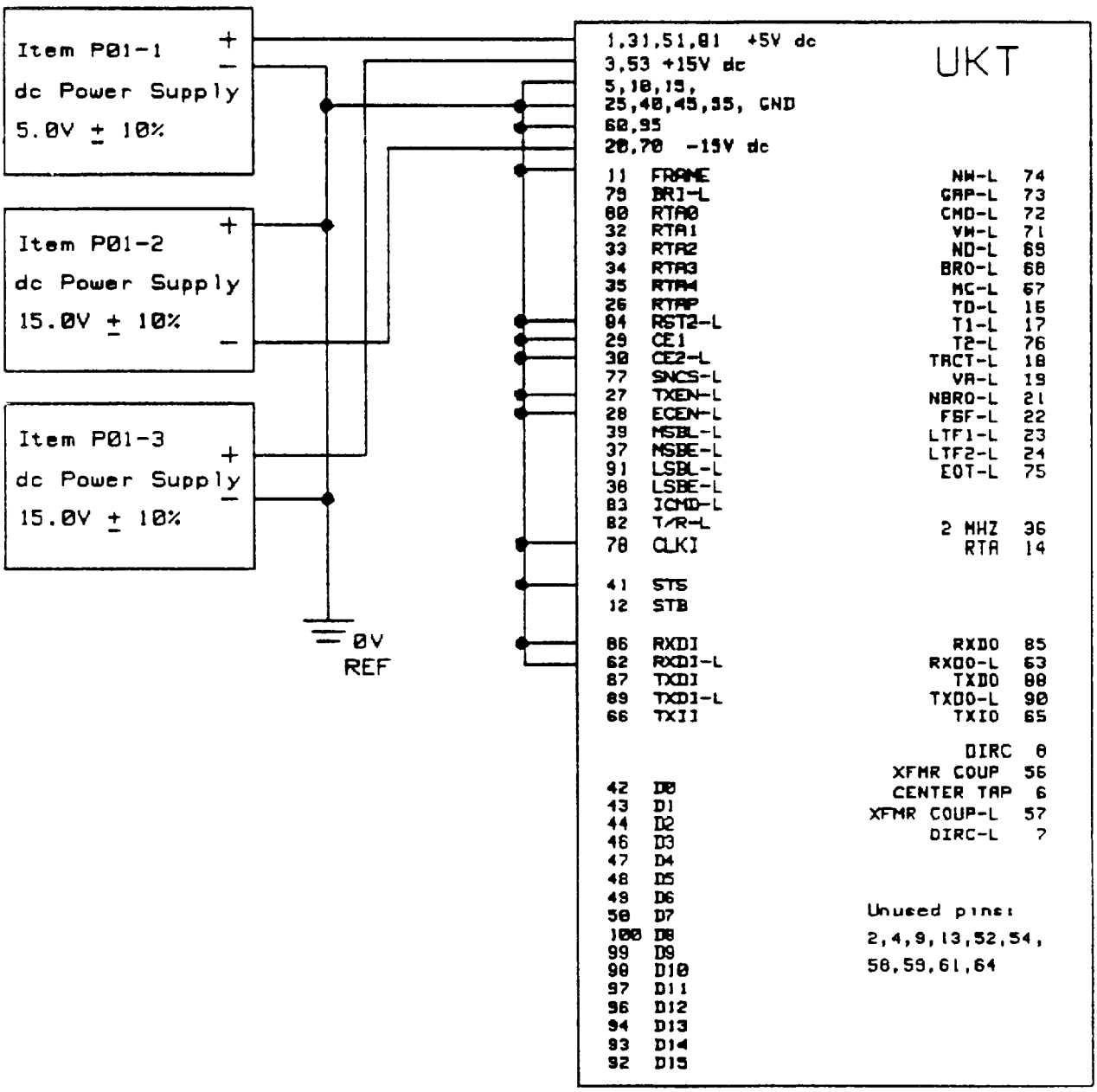


FIGURE 4. UKT vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G, through S and V through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G, through S and V through DD shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vout = output forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vin = input forcing voltage.
(w) X-TALK = crosstalk.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the class I and II end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 9.9 W (max).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
RXDI	NA	0.8	2.4	±0.02	± 20	20	50
RXDI-L	NA	0.8	2.4	±0.02	± 20	20	50
STS	NA	0.8	2.4	±0.02	± 20	20	50
TXEN-L	NA	0.8	2.4	±0.02	± 20	20	50
ECEN-L	NA	0.8	2.4	±0.02	± 20	20	50
CE2-L	NA	0.8	2.4	±0.02	± 20	20	50
CE1	NA	0.8	2.4	±0.02	± 20	20	50
BRI-L	NA	0.8	2.4	-0.8	-400	40	50
T/R-L	NA	0.8	2.4	-0.4	-250	40	50
ICMD-L	NA	0.8	2.4	-0.4	-250	40	50
MSBE-L	-1.5	0.7	2.0	-0.4	20	100	50
LSBE-L	-1.5	0.7	2.0	-0.4	20	100	50
MSBL-L	-1.5	0.7	2.0	-0.4	20	100	50
LSBL-L	-1.5	0.7	2.0	-0.4	20	100	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
SNCS-L	-1.5	0.7	2.0	-0.4	20	100	50
TXII	-1.5	0.7	2.0	-0.8	40	200	50
TXDI	-1.5	0.7	2.0	-1.6	80	400	50
TXDI-L	-1.5	0.7	2.0	-1.6	80	400	50
STB	-1.5	0.7	2.0	-0.8	40	200	50
RTA0	NA	0.8	2.0	-1.2	-400	150	50
RTA1	NA	0.8	2.0	-1.2	-400	150	50
RTA2	NA	0.8	2.0	-1.2	-400	150	50
RTA3	NA	0.8	2.0	-1.2	-400	150	50
RTA4	NA	0.8	2.0	-1.2	-400	150	50
RTAP	-1.5	0.8	2.0	-0.8	-250	150	50
RST2-L	NA	0.7	2.4	$\pm$0.04	$\pm$ 40	40	50
CLKI	NA	0.5	Vcc-0.5	$\pm$0.02	$\pm$ 20	20	50
D0-D15	-1.5	0.8	2.0	-0.4	20	100	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DIRC	NA	1.2-20Vp-p		NA	NA	NA	NA
DIRC-L	NA	1.2-20Vp-p		NA	NA	NA	NA
XFMR COUP	NA	0.86-14Vp-p		NA	NA	NA	NA
XFMR COUP-L	NA	0.86-14Vp-p		NA	NA	NA	NA

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)	Ios max (mA)
TXDO	2.4	0.4	-0.4	2.4	NA	NA	NA	NA
TXDO-L	2.4	0.4	-0.4	2.4	NA	NA	NA	NA
NBRQ-L	2.4	0.4	-0.4	1.6	NA	NA	NA	NA
NW-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
GAP-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
CMD-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
VW-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
ND-L	2.4	0.4	-0.4	1.6	20	-20	50	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)	Ios max (mA)
ND-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
BRO-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
MC-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
TD-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
T1-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
T2-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
TACT-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
VA-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
FSF-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
LTF1-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
LTF2-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
EOT-L	2.4	0.4	-0.4	1.6	20	-20	50	NA
2MHZ	2.4	0.4	-0.4	4.0	NA	NA	NA	-100
TXIO	2.4	0.4	-0.4	4.0	NA	NA	NA	-100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Cout max (pF)	Ios max (mA)
RXDO	2.4	0.4	-0.4	4.0	NA	NA	NA	-100
RXDO-L	2.4	0.4	-0.4	4.0	NA	NA	NA	-100
RTA	2.4	0.4	-0.38	3.6	NA	NA	NA	-100
D0-D15	2.4	0.7	-3.0	12.0	NA	NA	NA	-225
DIRC	6-9 V p-p		NA	NA	NA	NA	NA	NA
DIRC-L	6-9 V p-p		NA	NA	NA	NA	NA	NA
XFMR COUP	18-27 V p-p		NA	NA	NA	NA	NA	NA
XFMR COUP-L	18-27 V p-p		NA	NA	NA	NA	NA	NA

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
1	tpd	CLKI to NW-L (+)	-	-	-	325	325	325	nS	5a
2	tpd	CLKI to NW-L (-)	-	-	-	325	325	325	nS	5a

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
3	tpd	CLKI to GAP-L (+)	-	-	-	325	325	325	nS	5a
4	tpd	CLKI to GAP-L (-)	-	-	-	325	325	325	nS	5a
5	tpd	CLKI to CMD-L (+)	-	-	-	325	325	325	nS	5a
6	tpd	CLKI to CMD-L (-)	-	-	-	325	325	325	nS	5a
7	tpd	CLKI to VW-L (+)	-	-	-	325	325	325	nS	5a
8	tpd	CLKI to VW-L (-)	-	-	-	325	325	325	nS	5a
9	tpd	CLKI to ND-L (+)	-	-	-	325	325	325	nS	5a
10	tpd	CLKI to ND-L (-)	-	-	-	325	325	325	nS	5a
11	tpd	CLKI to BRO-L (+)	-	-	-	325	325	325	nS	5a
12	tpd	CLKI to BRO-L (-)	-	-	-	325	325	325	nS	5a
13	tpd	CLKI to MC-L (+)	-	-	-	325	325	325	nS	5a

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
14	tpd	CLKI to MC-L (-)	-	-	-	325	325	325	nS	5a
15	tpd	CLKI to TD-L (+)	-	-	-	325	325	325	nS	5a
16	tpd	CLKI to TD-L (-)	-	-	-	325	325	325	nS	5a
17	tpd	CLKI to T1-L (+)	-	-	-	325	325	325	nS	5a
18	tpd	CLKK to T1-L (-)	-	-	-	325	325	325	nS	5a
19	tpd	CLKI to T2-L (+)	-	-	-	325	325	325	nS	5a
20	tpd	CLKI to T2-L (-)	-	-	-	325	325	325	nS	5a
21	tpd	CLKI to TACT-L (+)	-	-	-	325	325	325	nS	5a
22	tpd	CLKI to TACT-L (-)	-	-	-	325	325	325	nS	5a
23	tpd	CLKI to VA-L (+)	-	-	-	325	325	325	nS	5a
24	tpd	CLKI to VA-L (-)	-	-	-	325	325	325	nS	5a
25	tpd	CLKI to LTF1-L (+)	-	-	-	325	325	325	nS	5a

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
26	tpd	CLKI to LTF1-L (-)	-	-	-	325	325	325	nS	5a
27	tpd	CLKI to LTF2-L (+)	-	-	-	325	325	325	nS	5a
28	tpd	CLKI to LTF2-L (-)	-	-	-	325	325	325	nS	5a
29	tpd	CLKI to EOT-L (+)	-	-	-	325	325	325	nS	5a
30	tpd	CLKI to EOT-L (-)	-	-	-	325	325	325	nS	5a
31	tpd	CLKI to NBRQ-L (+)	-	-	-	325	325	325	nS	5a
32	tpd	CLKI to NBRQ-L (-)	-	-	-	325	325	325	nS	5a
33	tpd	CLKI to TXDO (+)	-	-	-	325	325	325	nS	5a
34	tpd	CLKI to TXDO (-)	-	-	-	325	325	325	nS	5a
35	tpd	CLKI to TXDO-L (+)	-	-	-	325	325	325	nS	5a

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
36	tpd	CLKI to TXDO-L (-)	-	-	-	325	325	325	nS	5a
37	tpd	CLKI to 2MHZ (+)	-	-	-	125	125	125	nS	5a
38	tpd	CLKI to 2MHZ (-)	-	-	-	125	125	125	nS	5a
39	tpd	TXEN-L to TXIO	-	-	-	100	100	100	nS	5b
40	tp (hl)z	CE1 (-) to outs H or L to Z 3/	-	-	-	110	110	110	nS	5c
41	tp (hl)z	CE2-L (+) to outs H or L to Z 3/	-	-	-	110	110	110	nS	5c
42	tpz (hl)	CE1 (+) to outs (except data bus outs) Z to H or L	-	-	-	90	90	90	nS	5c
43	tpz (hl)	DATA BUS out Z to H or L	-	-	-	250	250	250	nS	5c

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
44	tpz (hl)	CE2-L (-) to outs (except) data bus outs) Z to H or L	-	-	-	90	90	90	nS	5c
45	tpz (hl)	DATA BUS out Z to H or L	-	-	-	250	250	250	nS	5c
46	pw	NBRQ-L 4/	400	400	400	-	-	-	nS	5a
47	pw	NW-L 4/	400	400	400	-	-	-	nS	5a
48	pw	T1-L 4/	400	400	400	-	-	-	nS	5a
49	pw	LTF1-L 4/	400	400	400	-	-	-	nS	5a
50	pw	LTF2-L 4/	400	400	400	-	-	-	nS	5a
51	pwh	CLKI	37.47	37.47	37.47	-	-	-	nS	5a
52	pwl	CLKI	37.47	37.47	37.47	-	-	-	nS	5a
53	prr	CLKI	83.33	83.33	83.33	-	-	-	nS	5a
54	pw	RST2-L	300	300	300	-	-	-	nS	5a
55	pw	MSBL-L	100	100	100	-	-	-	nS	8b
56	pw	LSBL-L	100	100	100	-	-	-	nS	8b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
57	tpd	MSBE-L (-) to D8-D15 valid	-	-	-	150	150	150	nS	8a
58	tpd	LSBE-L (-) to D0-D7 valid	-	-	-	150	150	150	nS	8a
59	tpz	MSBE-L (+) to D8-D15 tri-state 3/	-	-	-	40	40	40	nS	8a
60	tpz	LSBE-L (+) to D0-D7 tri-state 3/	-	-	-	40	40	40	nS	8a
61	tsu	D8-D15 valid to MSBL-L (+)	100	100	100	-	-	-	nS	8b
62	th	D8-D15 valid to MSBL-L (+)	100	100	100	-	-	-	nS	8b
63	tsu	D0-D7 valid to LSBL-L (+)	100	100	100	-	-	-	nS	8b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
64	th	D0-D7 valid to LSBL-L (+)	100	100	100	-	-	-	nS	8b
65	tsu	RXDI to CLKI 5/	100	100	100	-	-	-	nS	5e
66	th	CLKI to RXDI 5/	0	0	0	-	-	-	nS	5e
67	tsu	RXDI-L to CLKI 5/	100	100	100	-	-	-	nS	5e
68	th	CLKI to RXDI-L 5/	0	0	0	-	-	-	nS	5e
69	pw	TXDO 6/	485	485	485	515	515	515	nS	5a
70	pw	TXDO-L 6/	485	485	485	515	515	515	nS	5a
71	pw	RXDI	100	100	100	-	-	-	nS	5e
72	pw	RXDI-L	100	100	100	-	-	-	nS	5e
73	tpd	RXDI to RXDI-L 7/	380	380	380	-	-	-	nS	5e
74	tsu	ECEN-L to CLKI 5/	100	100	100	-	-	-	nS	5d

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum 1/ limit			Maximum 1/ limit			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
75	th	ECEN-L to CLKI 5/	400	400	400	-	-	-	nS	5d
76	pw	ECEN-L	1100	1100	1100	-	-	-	nS	5d
77	tpd	RXDI to RXDI-L 8/	-	-	-	50	50	50	nS	5e
78	tsu	SNCS-L to CLKI	100	100	100	-	-	-	nS	5d
79	th	SNCS-L to CLKI	400	400	400	-	-	-	nS	5d
80	pw	SNCS-L	1100	1100	1100	-	-	-	nS	5d
81	prr	RXDI/ 9/ RXDI-L MIN/MAX REPRATE	925	925	925	1032	1032	1032	nS	5e

- See footnotes on next page.

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1/ Class I limits at 0°C and 60°C are the same as class II limits for -55°C and 85°C, respectively.

2/ (+) following a signal name represents a positive going edge and (-) following a signal name represents a negative going edge.

3/ Qualification (Q) test.

4/ Limit is for 12 MHz input clock. For slower operation, limit should be increased using the formula: limit = (6x CLKI Prr) - 100nS.

5/ Not for 12 MHz free running clock operation (system operation).

6/ Limits are for 12 MHz operation. For slower operation, limits should be changed using the formula: min = (6x CLKI Prr) - 10nS, max = (6x CLKI Prr) + 10nS.

7/ Limit is for 12 MHz operation. For slower operation, limit should be changed using the formula: limit = 4x Prr + 47nS.

8/ Limit is for 12 MHz operation. For slower operation, limit should be changed using the formula: limit = CLKI Prr - 33nS.

9/ Limits are for 12 MHz operation. For slower operation, limits should be changed using the formula: min = (10.5x CLKI Prr) + 5nS, max = (13x CLKI Prr) - 50nS.

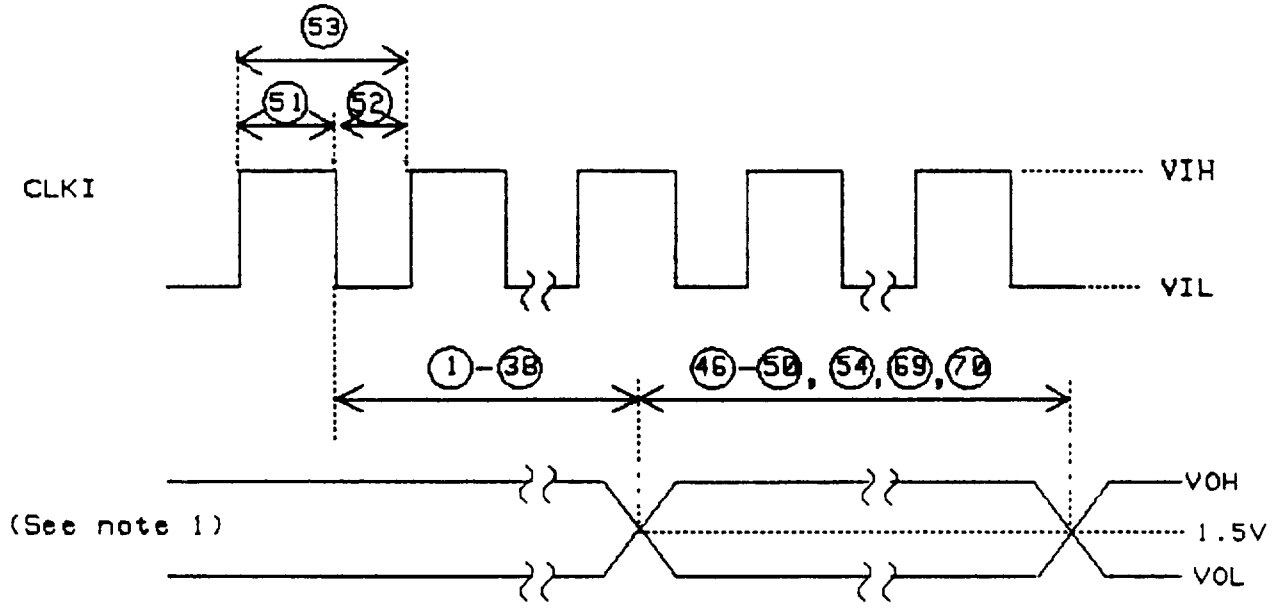
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6.4 Module function. The UKT module contains the MIL-STD-1553B bus interface driver/receiver circuitry referred to as a MIL-STD-1553B dumb Remote Terminal Unit (RTU).

6.4.1 Module timing. The UKT module timing is illustrated on figures 5 through 8.

6.4.2 1553 configuration. The UKT be used with the UKU module to perform a MIL-STD-1553B dual redundant RTU. (2 UKT and 1 UKU modules are required. See figure 9.) A MIL-STD-1553B bus controller can be implemented utilizing the UKT and UKS modules. (2 UKT and 1 UKS modules are required. See figure 10.) A dual redundant MIL-STD-1553B bus controller function can be accomplished using the UKT, UKU and UKS modules. (2 UKT, 1 UKU and 1 UKS modules are required. See figure 11.)

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NOTE:

1. NW-L, GAP-L, CMD-L, VW-L, ND-L, BRO-L, MC-L, TD-L, T1-L, T2-L, TACT-L, VA-L, LTF1-L, LTF2-L, EOT-L, NBRQ-L, TXDO, TXDO-L, 2MHZ, RST2-L.

FIGURE 5a. UKT timing waveforms.

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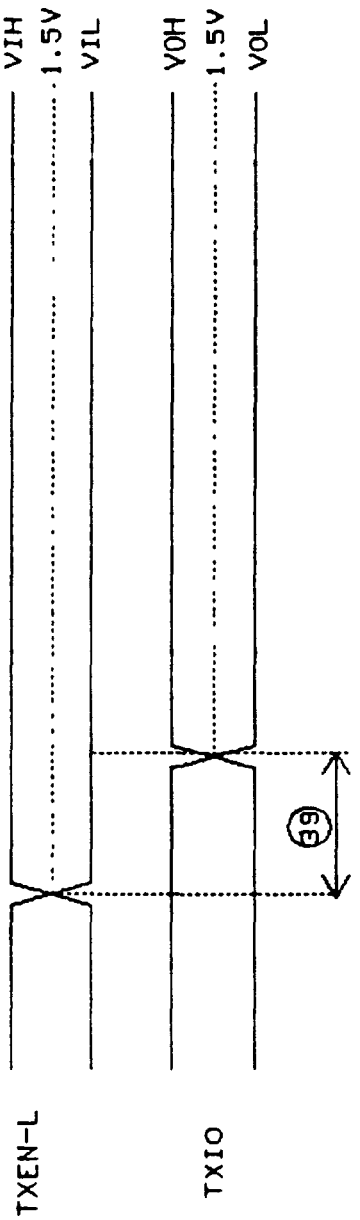
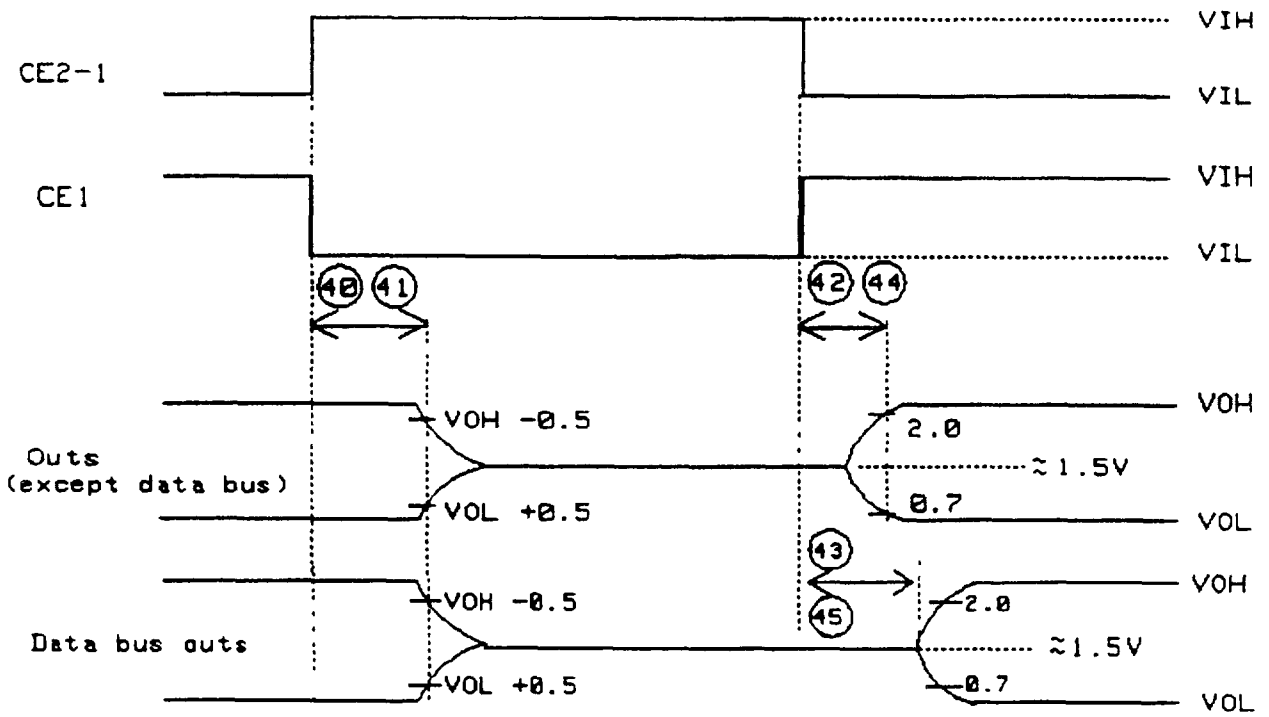


FIGURE 5b. UKT timing waveforms.

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NOTE:

1. "Outs" in reference numbers 40 thru 44 refer to the outputs normally connected to the UKS or UKU modules as shown on figure 2.

2. These outputs are as follows: NW-L, GAP-L, CMD-L, VW-L, ND-L, BRO-L, MC-L, TD-L, T1-L, T2-L, TACT-L, VA-L, FSF-L, LTF1-L, LTF2-L, EOT-L.

FIGURE 5c. UKT timing waveforms.

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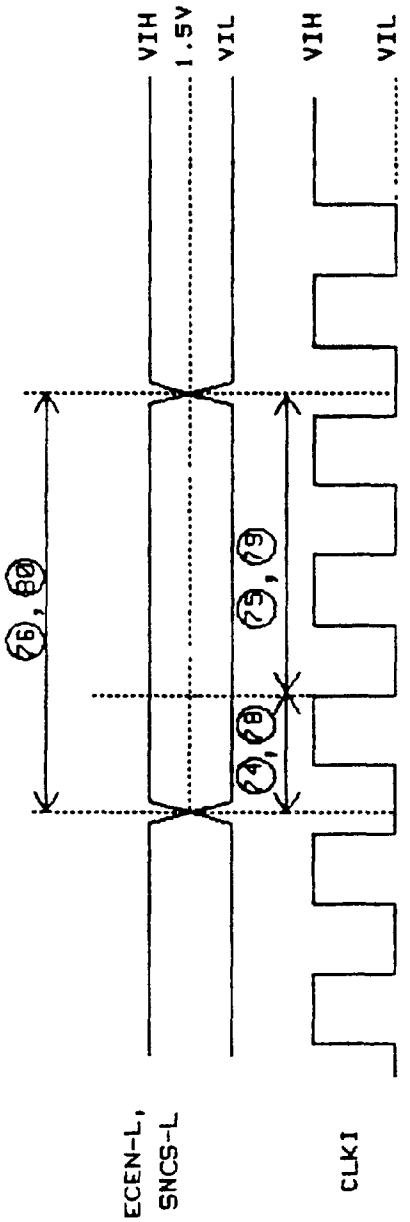


FIGURE 5d. UKT timing waveforms.

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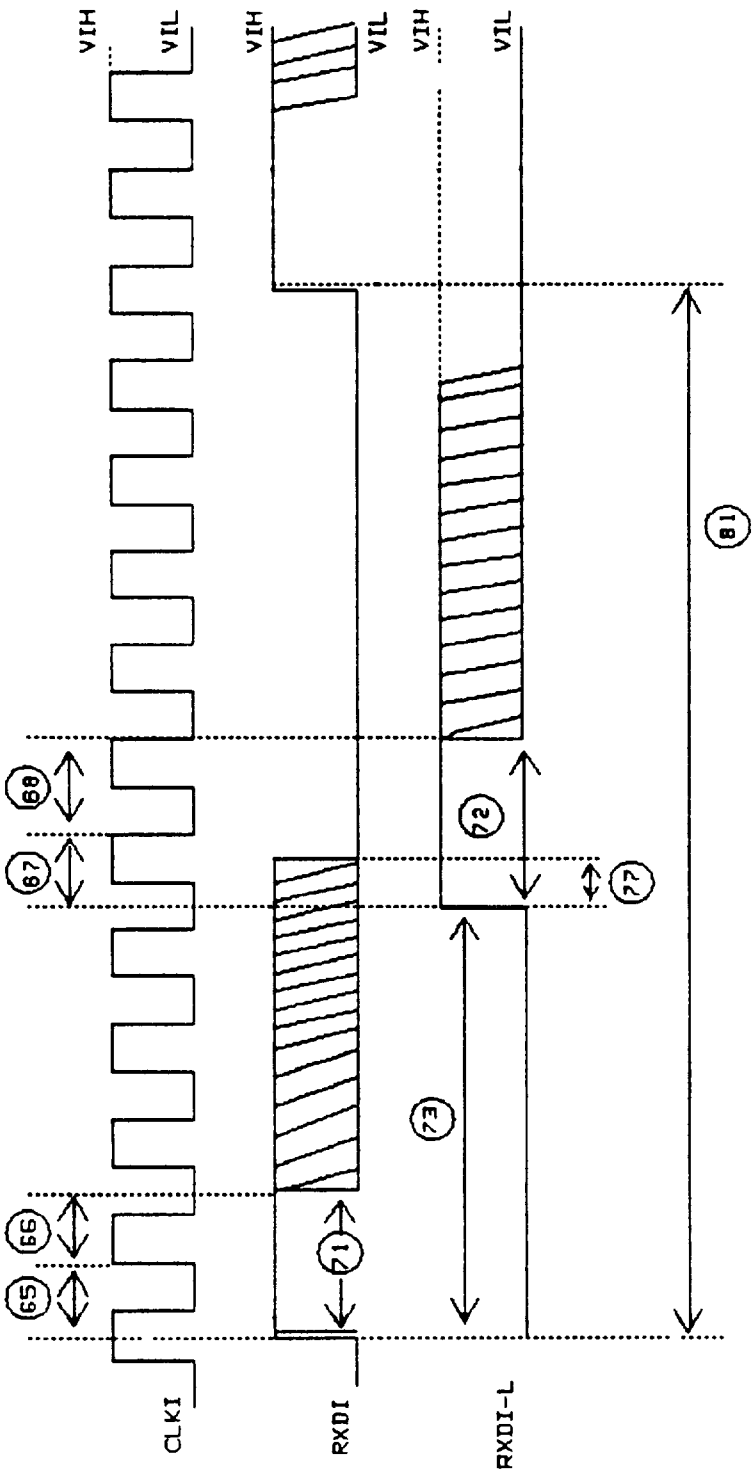


FIGURE 5e. UKT timing waveforms.

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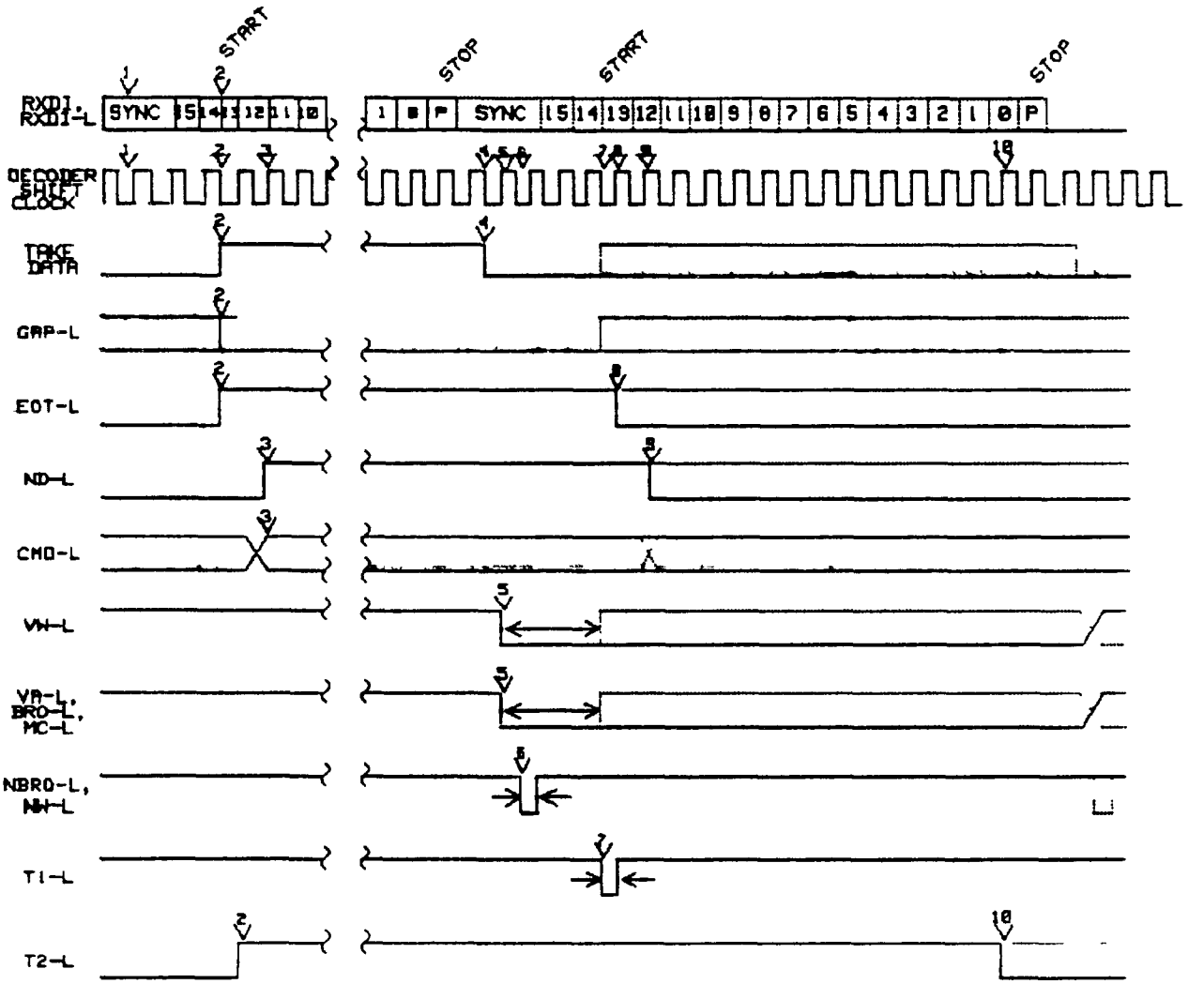


FIGURE 6. UKT receive sequence timing waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:57Z
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NOTES:

1. DECODER SHIFT CLOCK and TAKE DATA are internal signals shown for reference only.

2. Dashed lines shown represent the sequence of a second word being received continuous with the first word.

3. It is assumed that either CE1 or CE2-L is active. If neither one are active, all outputs shown here except NBRQ-L, are in the high impedance mode.

4. Sequence of events (see numbers marked over a 'V' on timing waveforms).

(a) "1" The start of a valid sync field has been detected on the RXDI and RXDI-L inputs and the decoder shift clock does the first stage synchronization (the clock cycle is lengthened or shortened by up to 3 decoder clock cycles to place the switching edge in the correct position).

(b) "2" A valid sync field and two data bits have been detected. TAKE DATA now rises (indicating serial data is ready to be shifted out of the decoder) and the decoder shift clock is corrected for phase (if it was 180 degrees out). TAKE DATA rising strobes in GAP-L, clears T2-L counters and forces the EOT-L signal high.

(c) "3" The second rising edge of the decoder shift clock after TAKE DATA went high clocks the ND-L flip-flop high and updates the CMD-L flip-flop with the type of sync received.

(d) "4" TAKE DATA, which has been high for 16 uS falls, indicating all the serial data has been shifted out.

(e) "5" The first rising edge of the decoder shift clock after TAKE DATA falls, triggers VW-L output to go low (provided no Manchester or parity errors occurred) which in turn enables VA-L, BRO-L or MC-L as they apply to the word just received. VW-L will remain low for 20 uS unless another valid sync plus two data bits are received as shown by the dashed lines, which would reset it high.

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NOTES: - Continued.

(f) "6" The next low level of the decoder shift clock is used to generate a 0.5 uS pulse that would be either NBRQ-L or NW-L. If the word was a valid address or broadcast preceded by a gap, or there was a gap followed by a valid command word, followed by valid address (RT-RT transfer), the NBRQ-L would be pulsed. Otherwise, NW-L is pulsed.

(g) "7" Approximately 4 uS after TAKE DATA falls (the fifth low level decoder shift clock cycle), a 0.5 uS pulse, T1-L, is generated.

(h) "8" The fifth rising edge of the decoder shift clock after TAKE DATA fell generates EOT-L, which indicates that the start of another valid word did not occur in time to be continuous with the previous word.

(i) "9" The sixth rising edge of the decoder shift clock after TAKE DATA went low clocks the ND-L flip-flop, which in turn enables the T2-L counters to start counting.

(j) "10" 17.5 uS after TAKE DATA went low (and stayed low) on the rising edge of the decoder shift clock, T2-L goes low.

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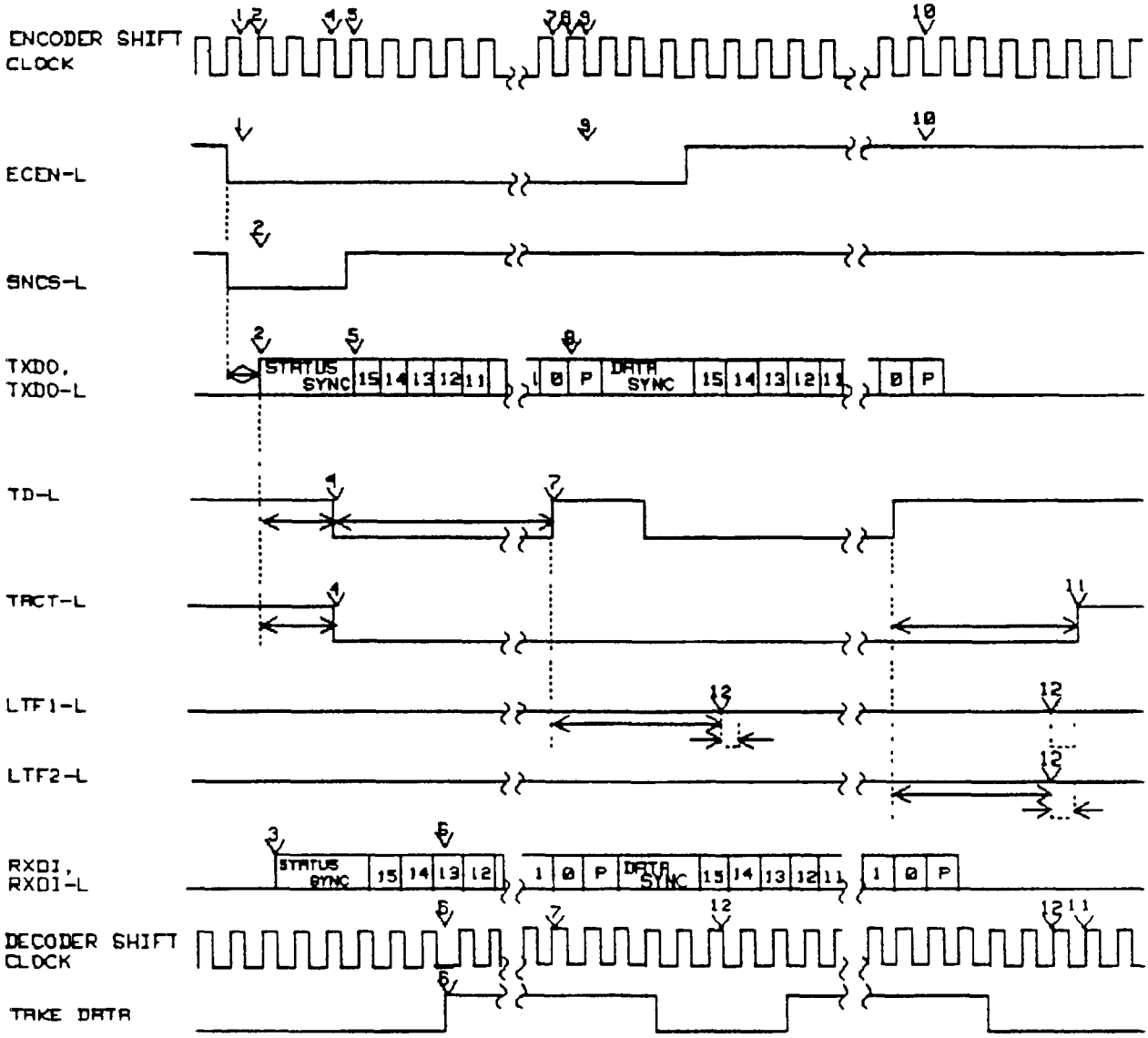


FIGURE 7. UKT transmit sequence timing waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:57Z
Check the source to verify that this is the current version before use.

NOTES:

1. Encoder shift clock, decoder shift clock and TAKE DATA are internal signals shown for reference only.

2. It is assumed that either CE1 or CE2-L is active to enable the ECEN-L input signal into the hybrid. If neither one was active, the hybrid would not see ECEN-L low and never start the transmit sequence. Also TD-L, TACT-L, LTF1-L and LTF2-L would be in the high impedance mode.

3. It is assumed that TXEN-L is low to enable the transceiver. If TXEN-L was high then data would not have been sent over the 1553 data bus. LTF1-L and LTF2-L would have pulsed in the corresponding locations.

4. Sequence of events (see numbers marked over a 'V' on the timing waveforms).

(a) "1" The first falling edge of the encoder shift clock after ECEN-L goes low initiates the transmit sequence for one word.

(b) "2" The next rising edge of the encoder shift clock samples the SNCS-L line and starts to output the corresponding sync type on TXDO and TXDO-L.

(c) "3" The information on TXDO and TXDO-L goes through the transceiver to the isolation transformer and is wrapped back around into RXDI and RXDI-L. This delay is less than 0.85 uS.

(d) "4" 0.5 uS before the first data bit is transmitted (on the falling edge of the encoder shift clock), TD-L goes low indicating the encoder is ready for the serial data to be transmitted. It will stay low for 16 uS. TD-L going low forces TACT-L to go low and transfers the information in the parallel registers to the serial shift registers. The parallel registers may now be updated again with the next word to be transmitted.

(e) "5" On the next 16 rising edges of the encoder shift clock, the serial data is sampled and transmitted over TXDO and TXDO-L.

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NOTES: - Continued.

(f) "6" Upon detecting a valid sync and two data bits on the RXDI and RXDI-L, the final synchronization is done on the decoder shift clock and TAKE DATA now rises. All output signals associated with the receive sequence still apply but are not shown here. Refer to the receive sequence for BUS67001 (D35622). The only exception is that NBRQ-L and NW-L are masked by the TACT-L signal.

(g) "7" On the first falling edge of the encoder shift clock after the 16 data bits have been shifted out, TD-L returns high.

(h) "8" On the next rising edge of the encoder shift clock, the parity bit is transmitted on TXDO and TXDO-L. The transmission cycle is almost completed.

(i) "9" On the next falling edge of the encoder shift clock, the ECEN-L signal is again sampled and since it is low, another transmission cycle is initiated. The sequence of events is the same as sequence 1 through 8.

(j) "10" After the transmission cycle is almost completed, the ECEN-L input is again sampled and since it is high, no more transmission cycles will be initiated until it is sampled low on the next or any following falling edges of the encoder shift clock.

(k) "11" After the last transition of TD-L from low to high, the sixth rising edge of the decoder shift clock (since TD-L is high) causes the TACT-L output to go high. With TACT-L high, the NBRQ-L or NW-L pulse is no longer masked.

(l) "12" The first low level of the decoder shift clock (after TD-L has gone from low to high and 5 rising edges of the decoder shift clock have occurred) is used to generate a 0.5 uS test pulse. If the word wrapped around from the isolation transformer had a sync type, Manchester or parity error, then this test pulse will be enabled on LTF1-L. On the last word transmitted, this pulse is also enabled on LTF1-L if the received word does not match the word sent. It is enabled on LTF2-L only on the last word transmitted, if the received word does not match or has a Manchester or parity error.

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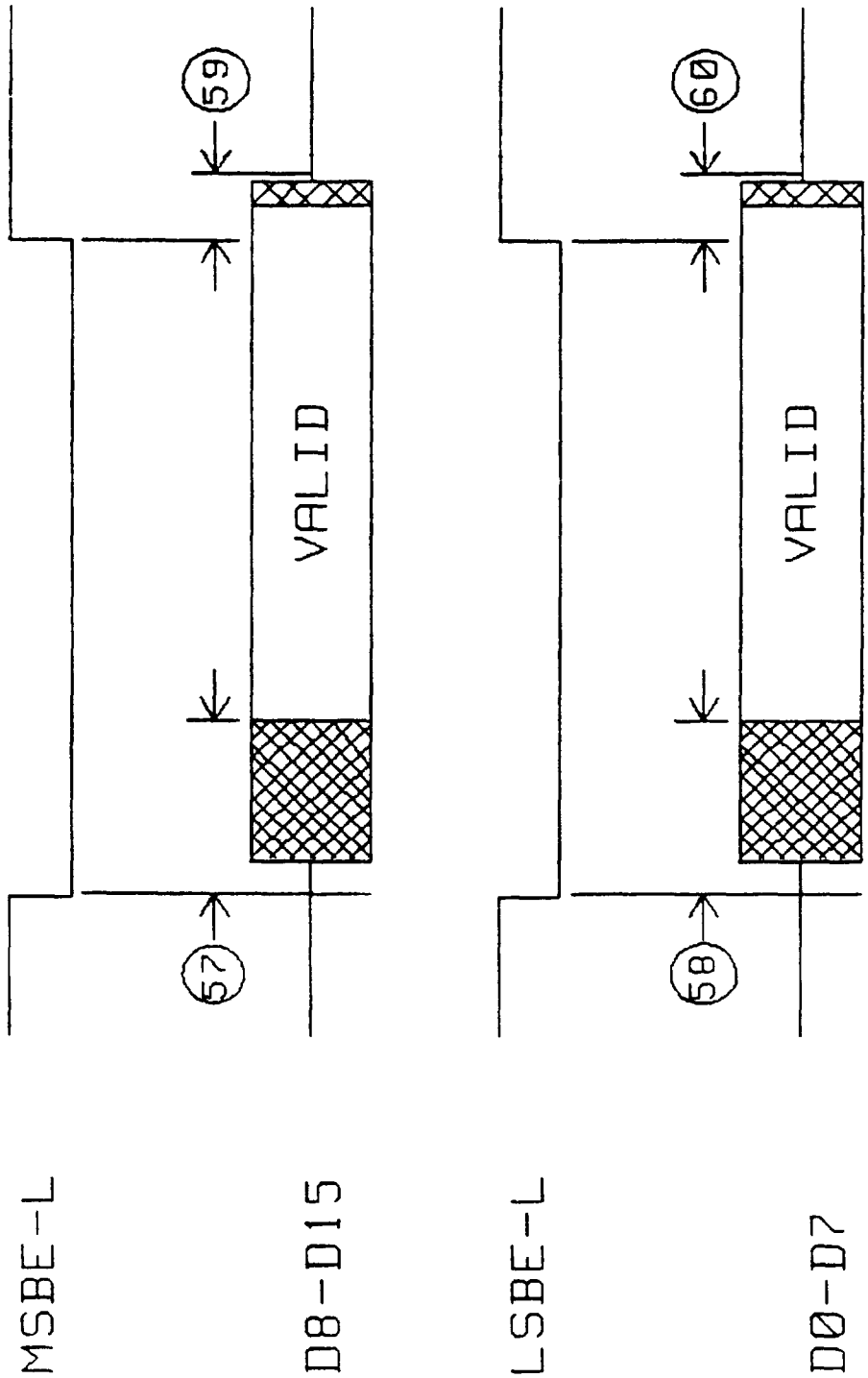


FIGURE 8a. UKT timing waveforms for driving data bus (assumed channel is selected).

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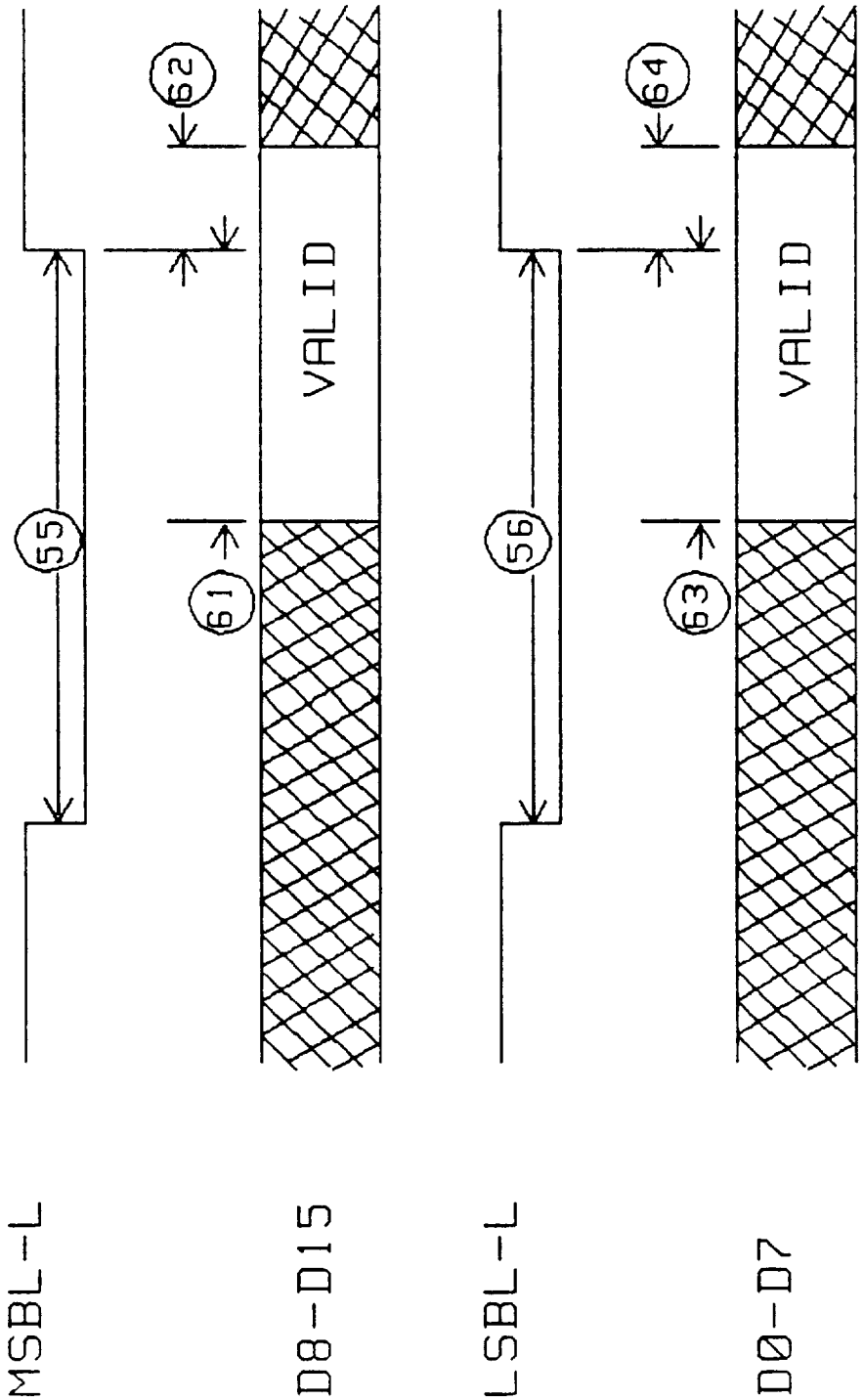


FIGURE 8b. UKT timing waveforms for latching data from data bus.

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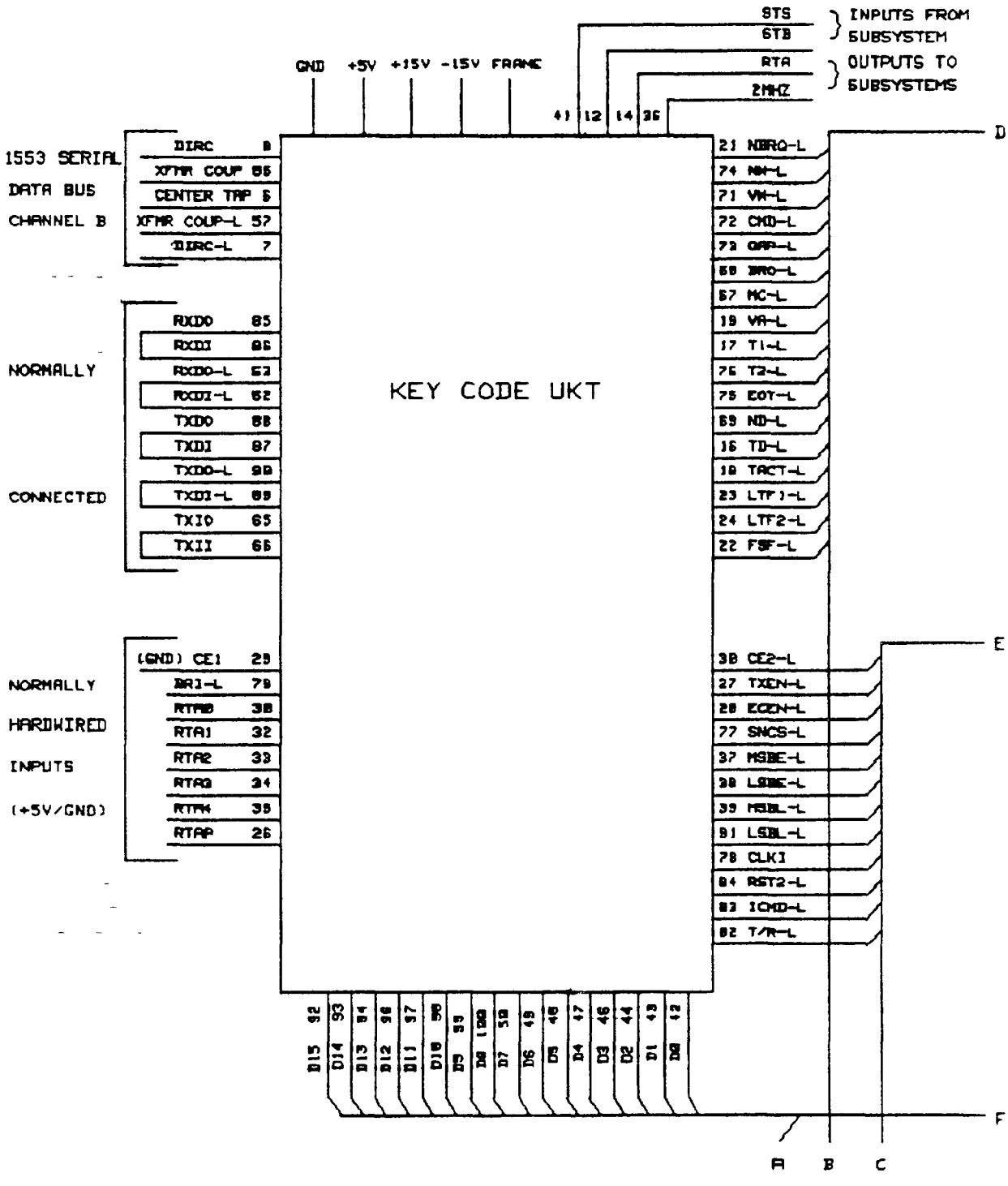


FIGURE 9. Dual redundant RTU block diagram.

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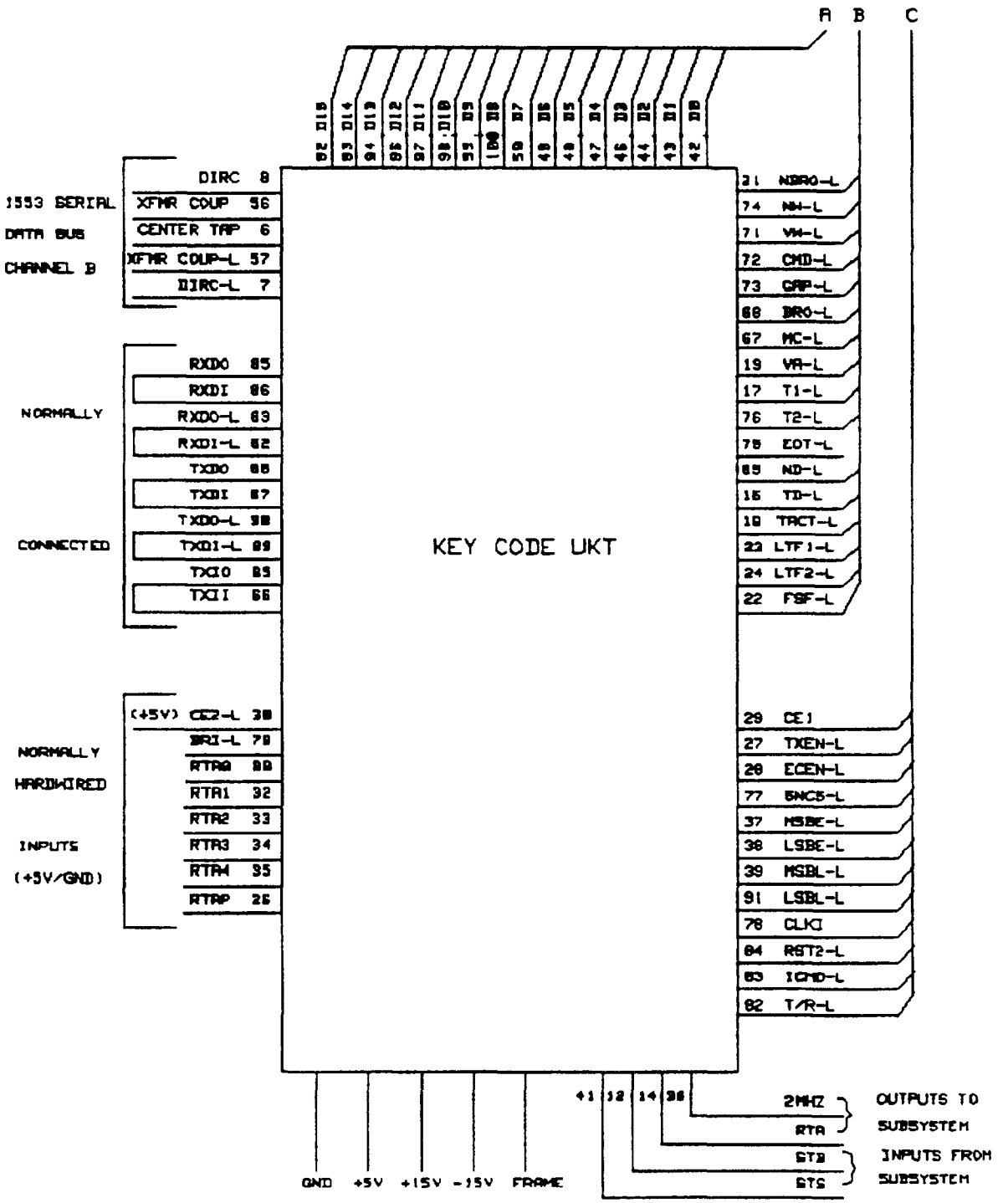


FIGURE 9. Dual redundant RTU block diagram - Continued.

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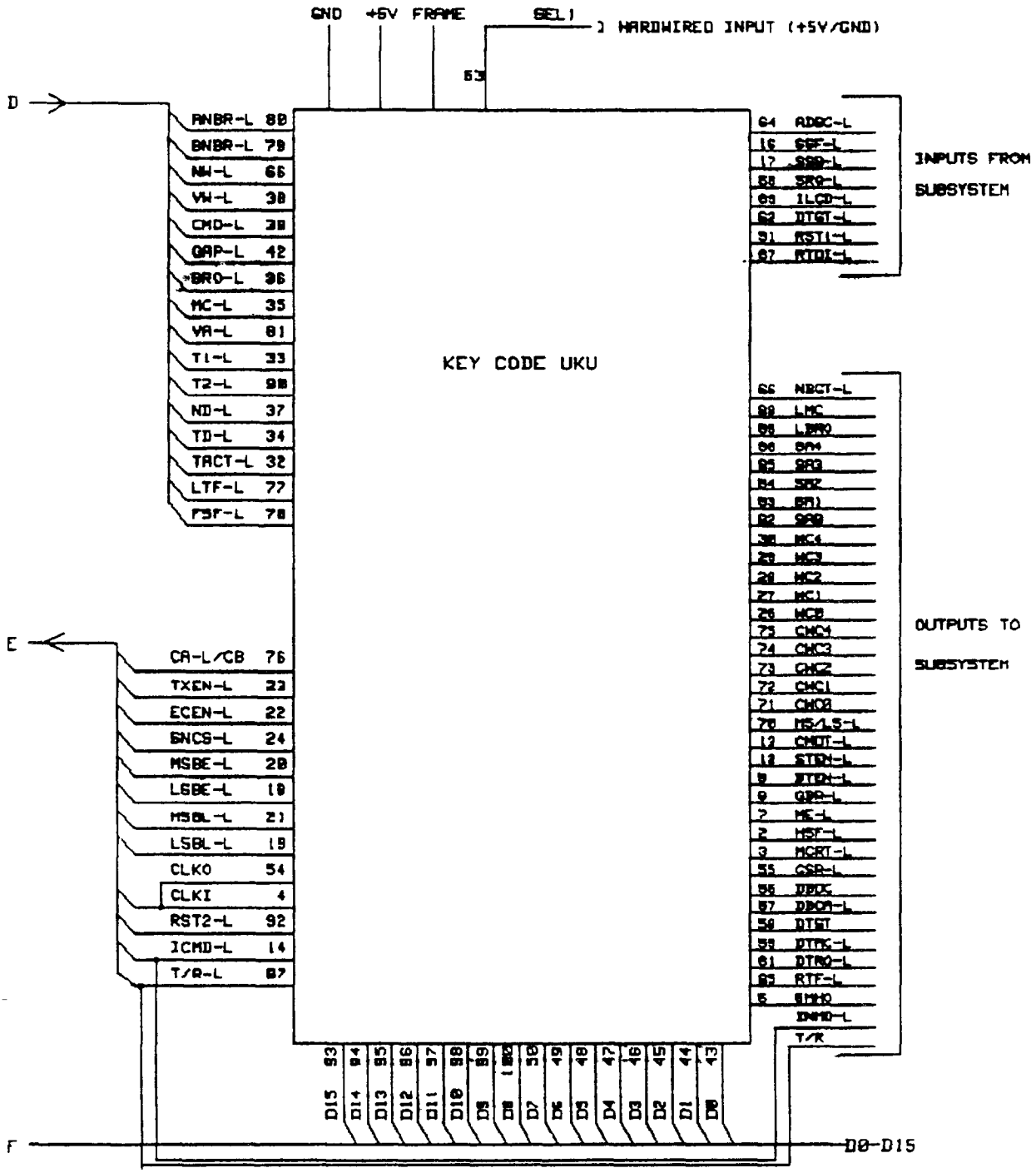


FIGURE 9. Dual redundant RTU block diagram - Continued.

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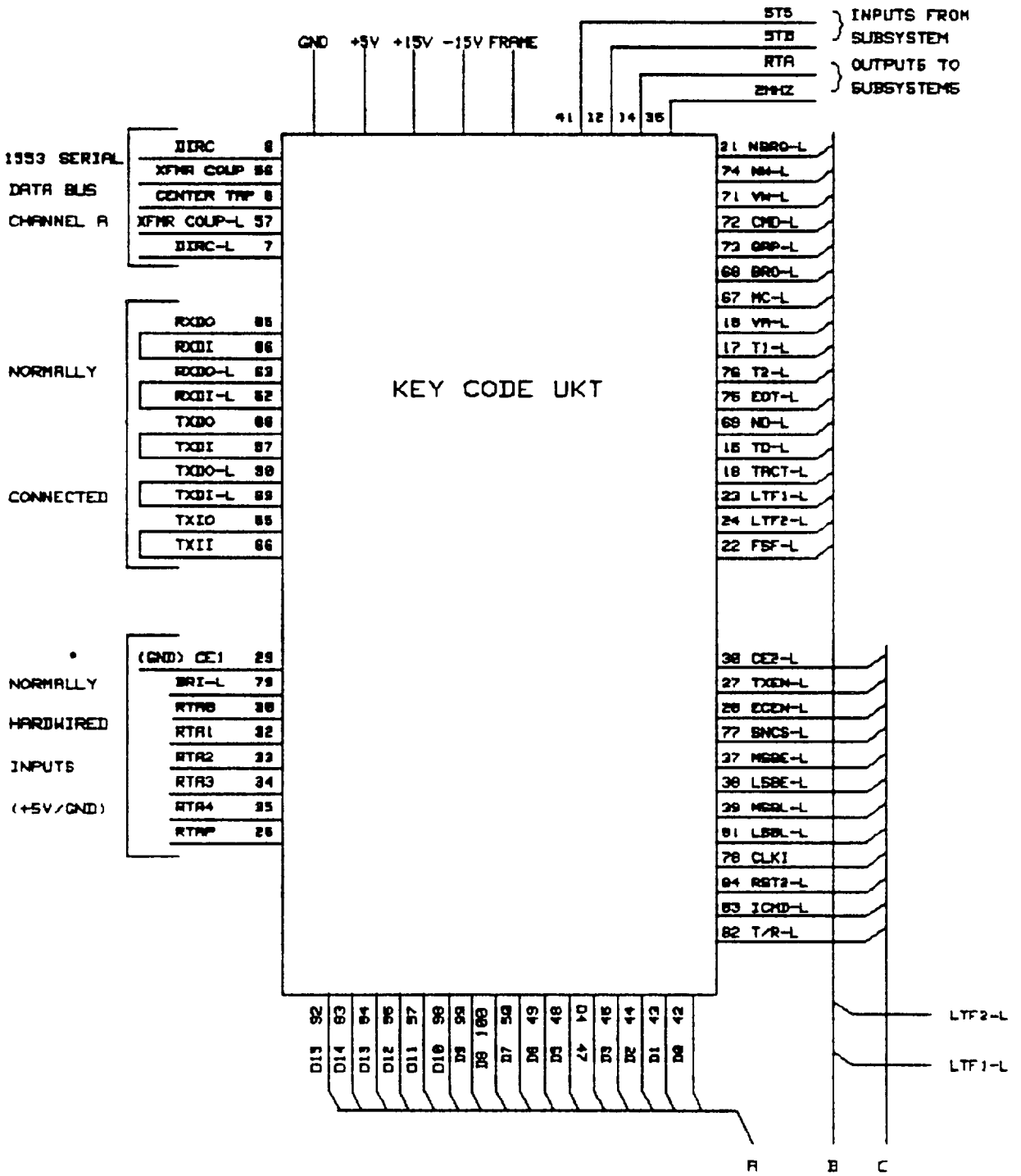


FIGURE 10. MIL-STD-1553B bus controller block diagram.

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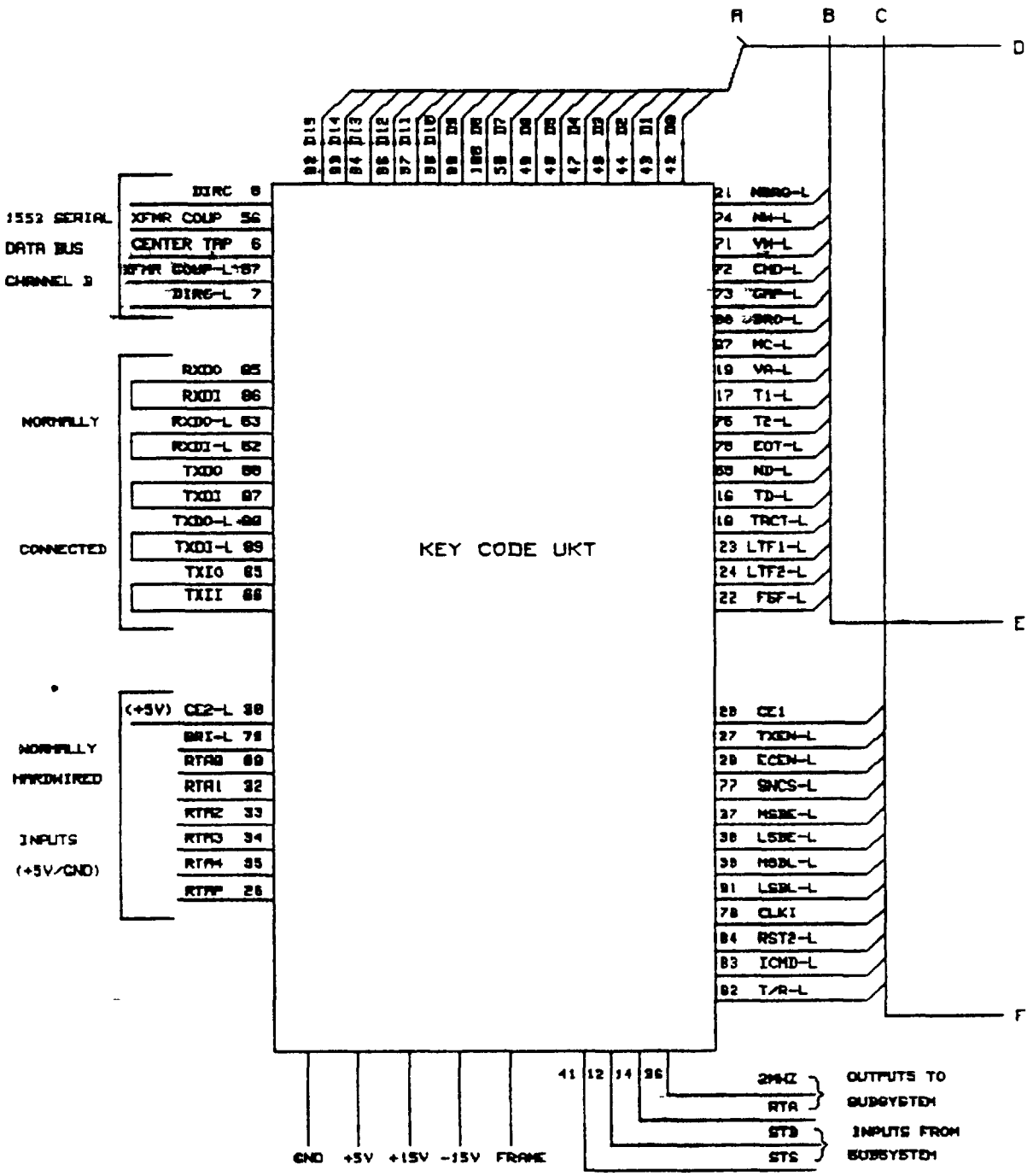


FIGURE 10. MIL-STD-1553B bus controller block diagram - Continued.

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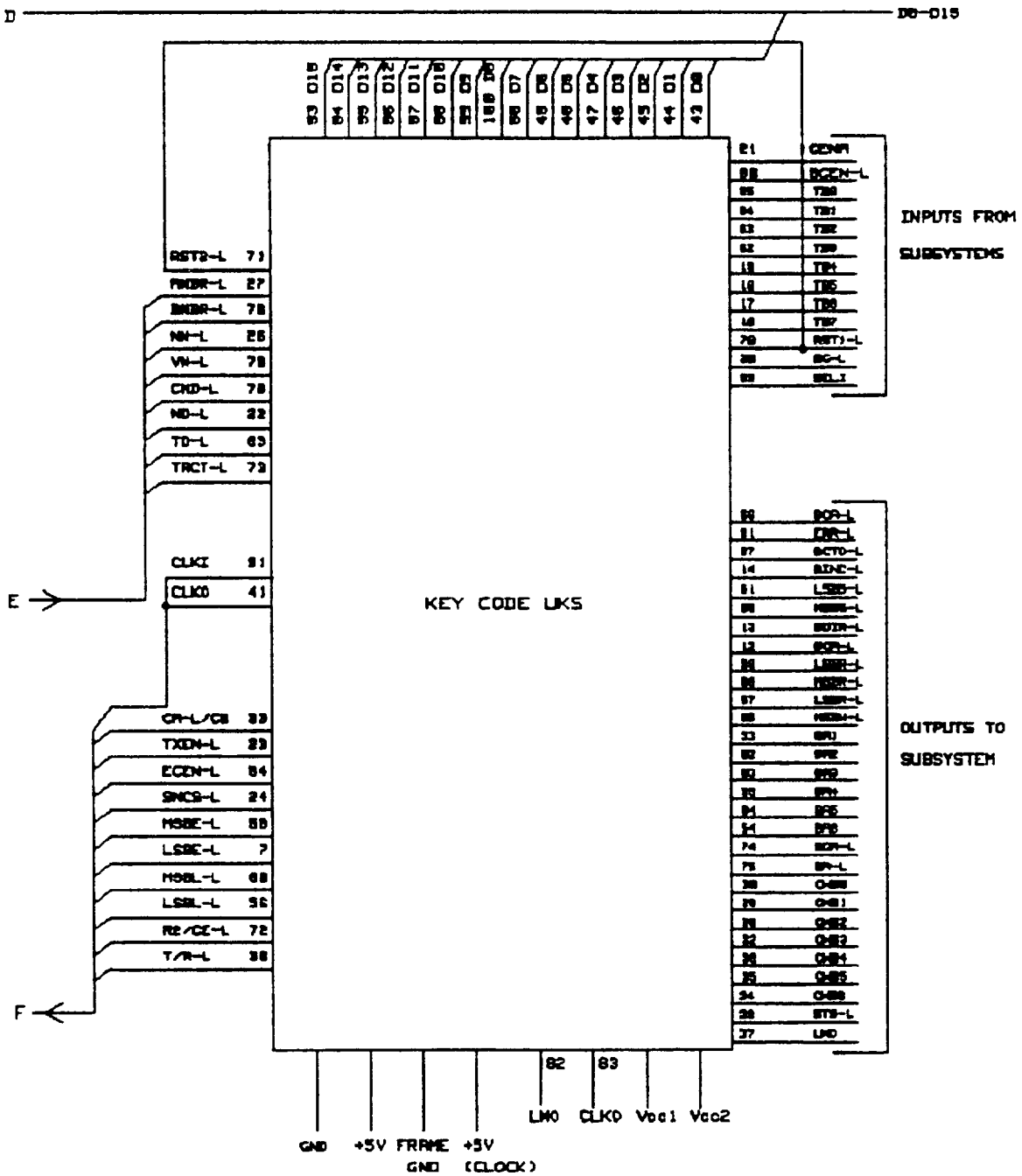


FIGURE 10. MIL-STD-1553B bus controller block diagram - Continued.

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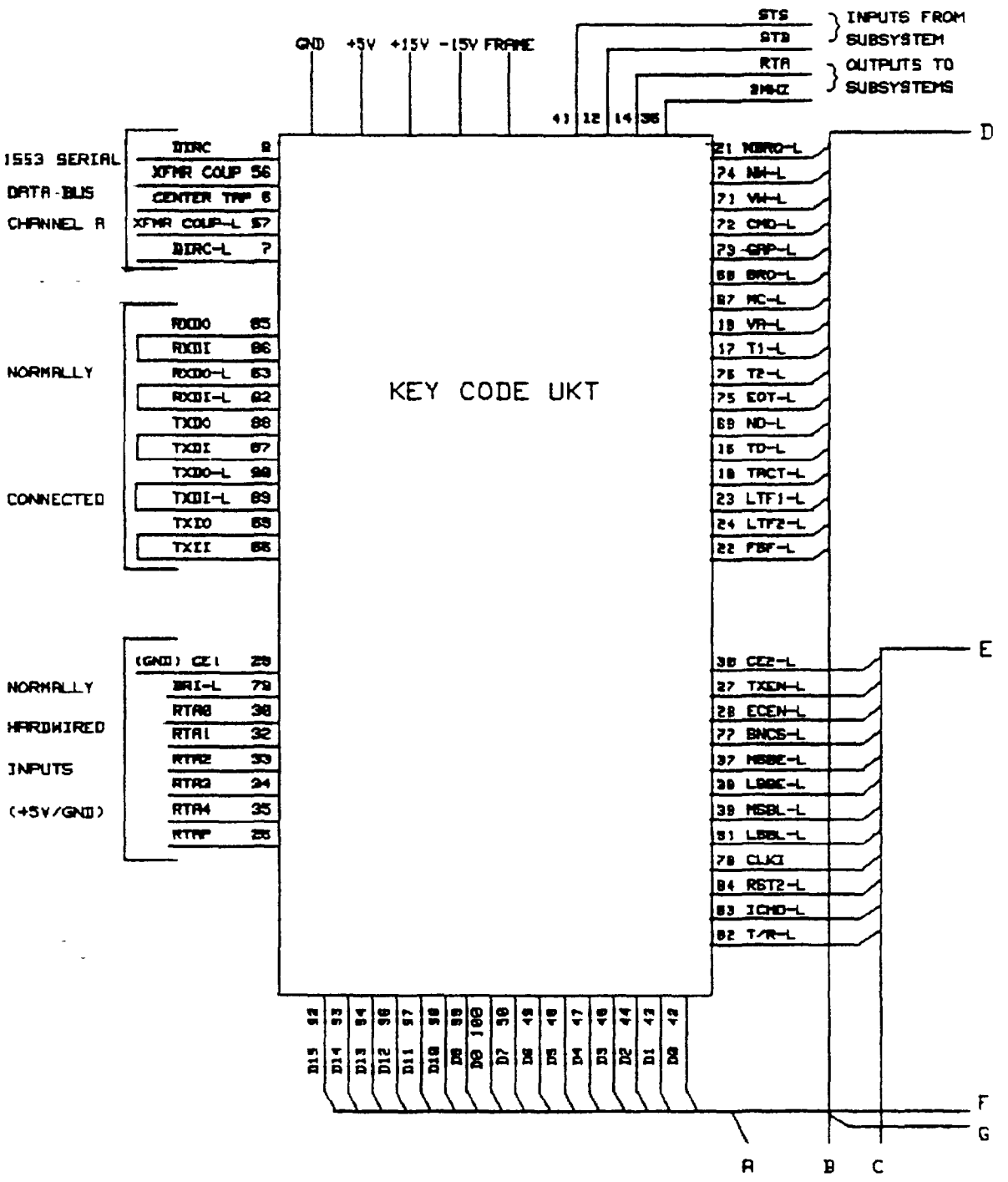


FIGURE 11. Dual redundant bus controller block diagram.

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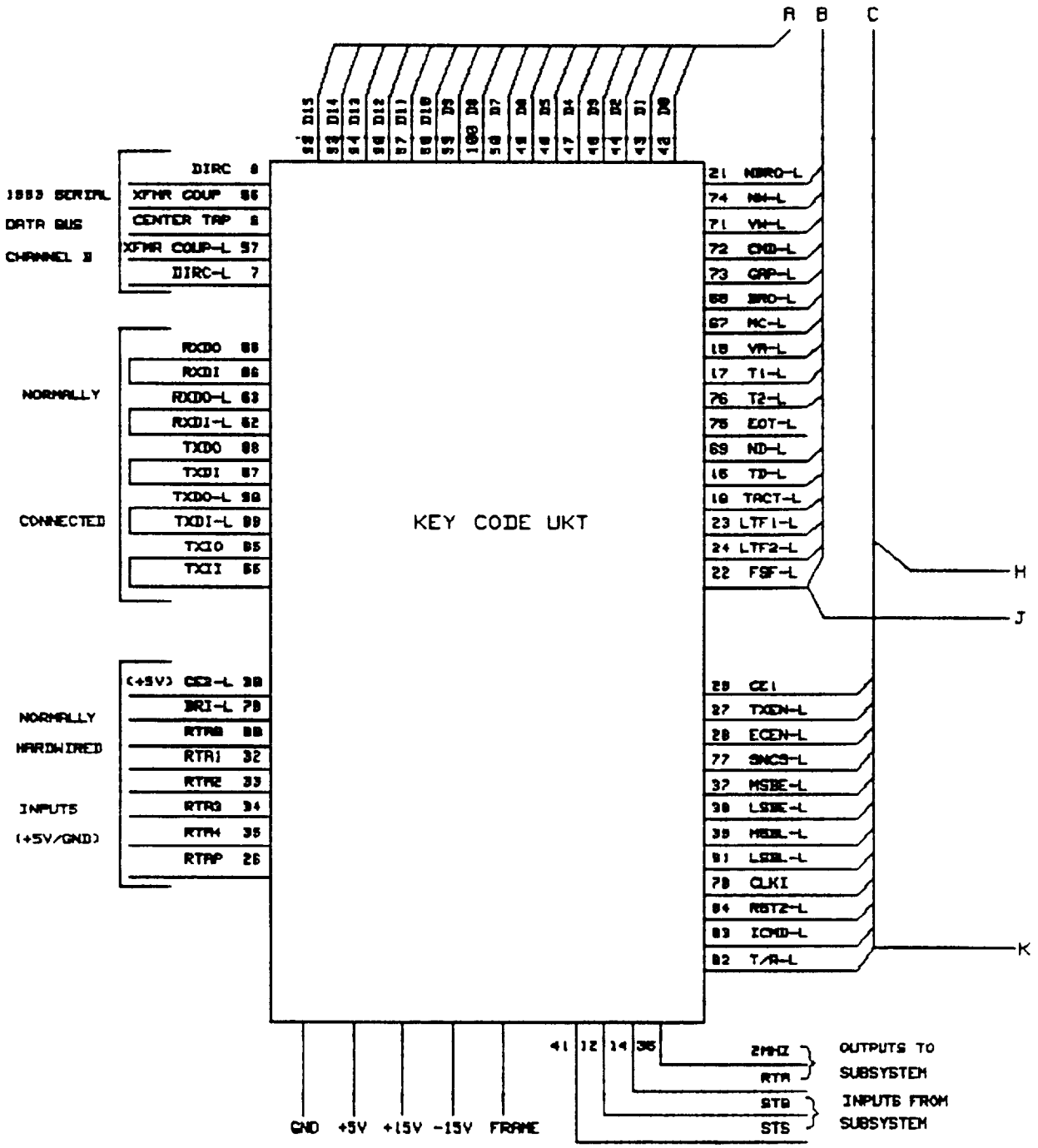


FIGURE 11. Dual redundant bus controller block diagram - Continued.

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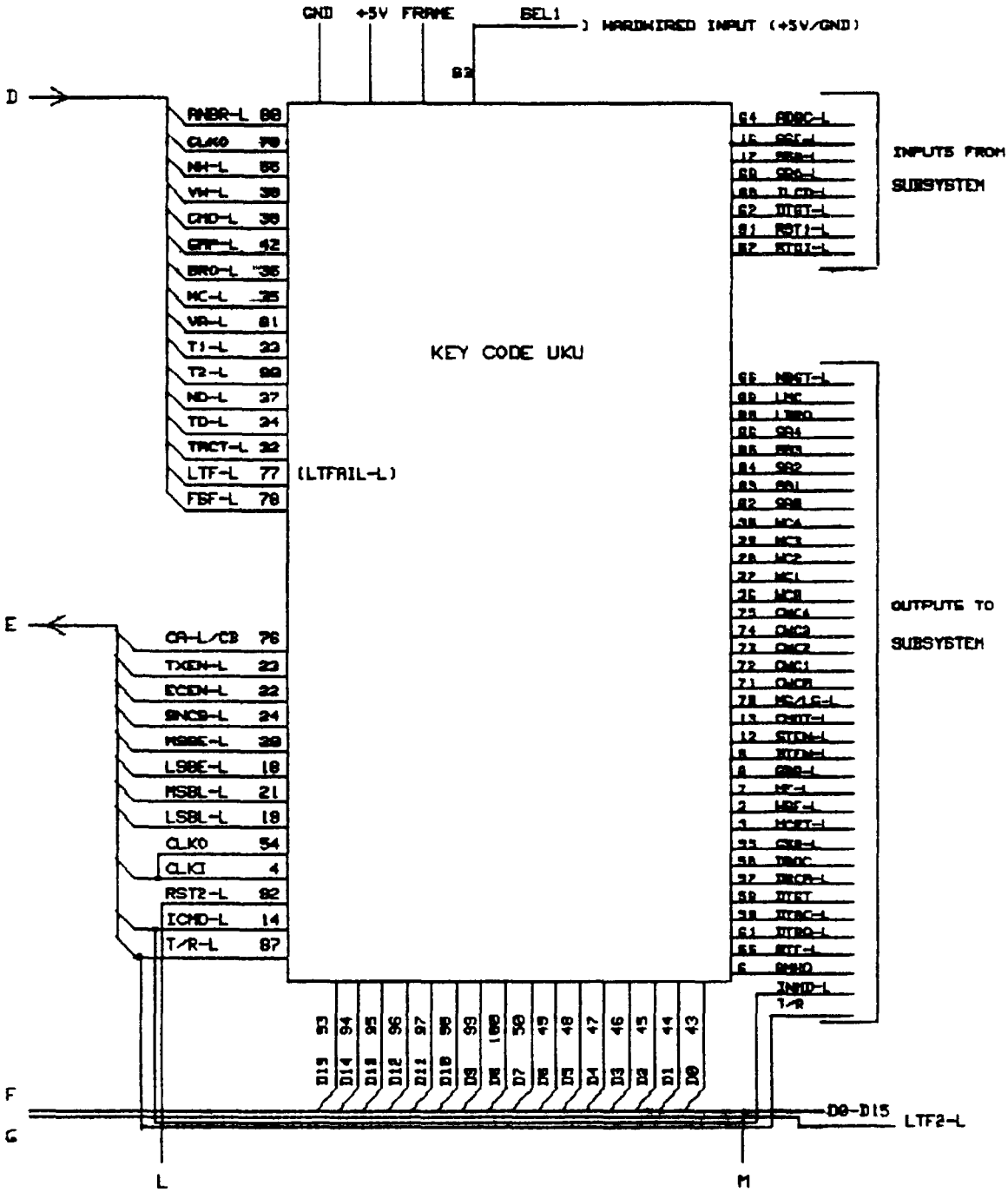


FIGURE 11. Dual redundant bus controller block diagram - Continued.

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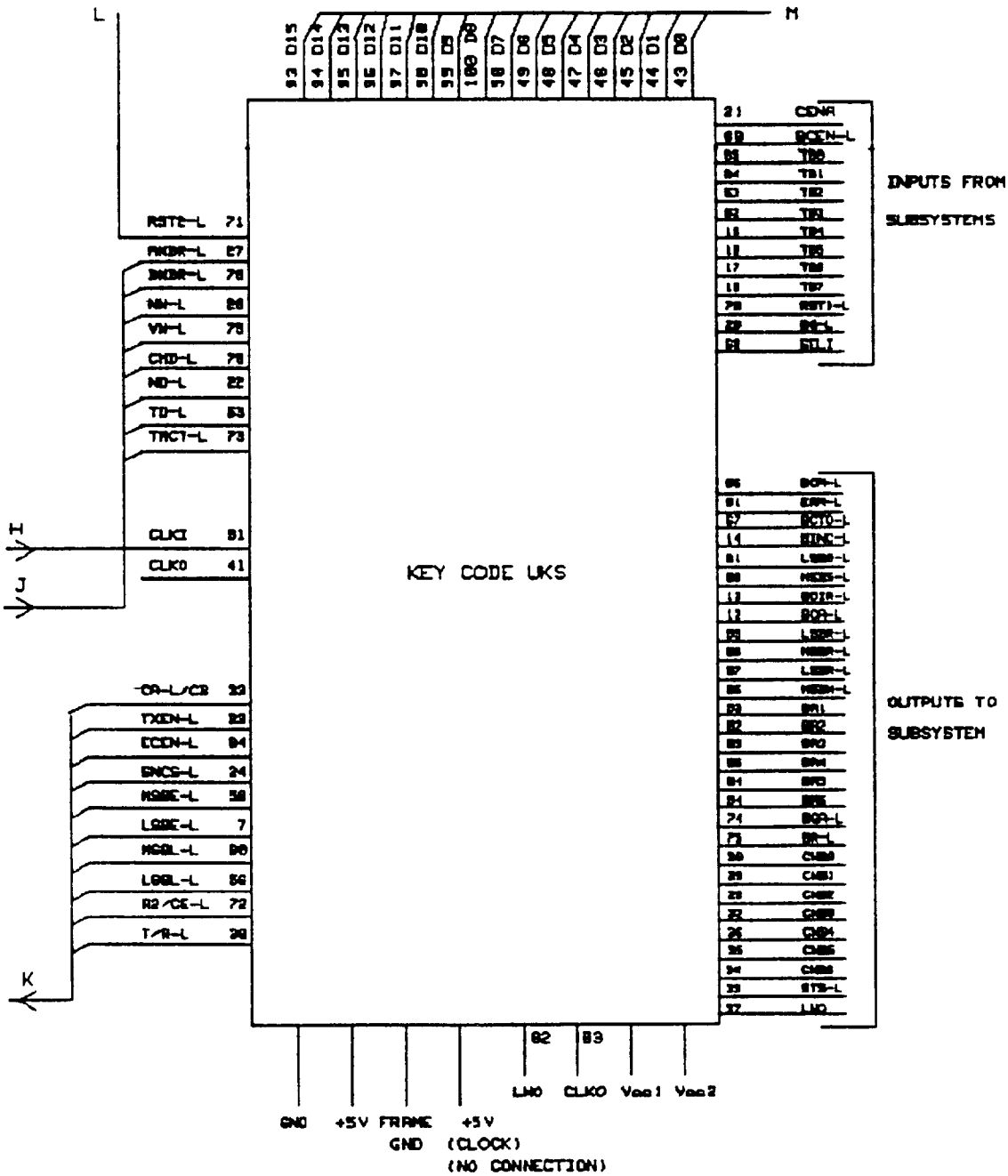


FIGURE 11. Dual redundant bus controller block diagram - Continued.

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6.4.3 Functional description of I/O pins.

Signal name	Pin number	I/O	Description
D0	42	I/O	Bi-directional parallel data bus bit 0 (Least Significant Bit (LSB)).
D1	43	I/O	Bi-directional parallel data bus bit 1.
D2	44	I/O	Bi-directional parallel data bus bit 2.
D3	46	I/O	Bi-directional parallel data bus bit 3.
D4	47	I/O	Bi-directional parallel data bus bit 4.
D5	48	I/O	Bi-directional parallel data bus bit 5.
D6	49	I/O	Bi-directional parallel data bus bit 6.
D7	50	I/O	Bi-directional parallel data bus bit 7.
D8	100	I/O	Bi-directional parallel data bus bit 8.
D9	99	I/O	Bi-directional parallel data bus bit 9.
D10	98	I/O	Bi-directional parallel data bus bit 10.
D11	97	I/O	Bi-directional parallel data bus bit 11.
D12	96	I/O	Bi-directional parallel data bus bit 12.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
D13	94	I/O	Bi-directional parallel data bus bit 13.
D14	93	I/O	Bi-directional parallel data bus bit 14.
D15	92	I/O	Bi-directional parallel data bus bit 15 (Most Significant Bit (MSB)).
MSBE-L	37	I	MSB enable - low level enables the most significant byte of the receive register onto the parallel data bus for transfer.
LSBE-L	38	I	LSB enable - low level enables the least significant byte of the receive register onto the parallel data bus for transfer.
MSBL-L	39	I	MSB load - low level enables the contents of the parallel data bus into the most significant byte of the pre-transmission register.
LSBL-L	91	I	LSB load - low level enables the contents of the parallel data bus into the least significant byte of the pre-transmission register.
SNCS-L	77	I	Sync select - input signal controls the word sync type to be transmitted. A logic low selects the command sync.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RTAP	26	I	Remote terminal address parity - odd parity line for the five terminal address lines (RTA0-RTA4). Must be utilized or remote terminal will not respond to its address.
TXII	66	I	Transmitter inhibit input - a logic high takes priority over the condition of the data inputs and disables the transmitter.
STB	12	I	Normal logic one input - a logic zero applied to this input will disable the receiver outputs.
TXDI	87	I	Transmitter data in - biphase TTL data to transmitter.
TXDI-L	89	I	Transmitter data in - biphase TTL data to transmitter.
RTA0	80	I	Remote terminal address line (LSB).
RTA1	32	I	Remote terminal address line.
RTA2	33	I	Remote terminal address line.
RTA3	34	I	Remote terminal address line.
RTA4	35	I	Remote terminal address line (MSB).
BRI-L	79	I	Broadcast inhibit - a logic low disables the broadcast output such that the ALL ONES will not be recognized.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ICMD-L	83	I	In command - input signal from UKU used in conjunction with T/R-L to clear the internal latch that enables NBRQ-L output during RT-RT type data transfers.
T/R-L	82	I	Input signal from UKU used in conjunction with ICMD-L clears the internal latch that enables NBRQ-L output during RT-RT type data transfers.
RXDI	86	I	Bi-phase TTL input to logic manchester decoding section - input must be at logic zero when no data is present.
RXDI-L	62	I	Bi-phase TTL input to logic manchester decoding section - input must be at logic zero when no data is present.
STS	41	I	Self test select - logic low for normal operation. Logic high places module into a wrap-around condition, by-passing the transceiver section of the module.
CLKI	78	I	12 MHz clock input.
ECEN-L	28	I	Encoder enable - active low input initiates a word transmission on the selected bus.
TXEN-L	27	I	Transmitter enable - active low input required to enable the transmitter (TXDO, TXDO-L, and transceiver) of that channel.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CE2-L	30	I	Channel enable 2 - a logic low on this pin will enable the tri-state outputs.
CE1	29	I	Channel enable 1 - a logic high on this pin will enable the tri-state outputs.
RST2-L	84	I	Reset 2 - a low level signal applied to this input resets front end module.
NW-L	74	O	New word - a low level output pulse (0.5 uS) that occurs 2.75 uS after the zero-crossing of a parity bit if the word just received did not meet the criteria of NBRQ-L. Masked if TACT-L is low.
GAP-L	73	O	Output signal that goes low upon detection of a valid sync and a 2 data bit if they were preceded by a minimum of 22 uS since the previous detection of a valid sync and 2 data bit (in other words, 2 uS minimum gap time or 4 uS minimum response time).
CMD-L	72	O	Command - low level output when the word just received had a command sync field.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
VW-L	71	O	Valid word - a low level output when a valid word is recognized (no Manchester or parity errors). Goes low approximately 2.25 uS from the zero-crossing of the parity bit. Goes high 20 uS later or upon detection of a valid sync and 2 data bits.
ND-L	69	O	No data - output that goes low whenever the start of a new word (valid sync and 2 data bits) has not been detected continuous with the previous word. Goes high 1 uS after detection of the start of a new word.
BRO-L	68	O	Broadcast - output signal that goes low while VW-L is low if the word just received had a command sync and an RT address field of all ones. Kept high if BRI-L is at a logic 0.
MC-L	67	O	Mode code - output signal that goes low while VW-L is low if the word just received had a command sync and the sub-address field was all zeroes or all ones.
TD-L	16	O	Transmitting data - output signal that goes low for 16 uS during each data word transmission while the serial data is being transferred inside the module to the encoder.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
T1-L	17	O	Low level output pulse (0.5 uS) that occurs approximately 5.75 uS after the zero-crossing of the PARITY BIT of the Data Word just received on RXDI and RXDI-L.
T2-L	76	O	Time 2 - output signal that goes low approximately 19.25 uS after the zero-crossing of the parity bit if the start of a new word has not yet been detected.
TACT-L	18	O	Transmitter active - delayed output signal that is low while the encoder is busy generating serial information for TXDO and TXDO-L.
VA-L	19	O	Valid address - output signal that goes low while VW-L is low if the word just received had a command sync and the RT address matched the RTA0-RTA4 inputs.
NBRQ-L	21	O	New bus request - a low level output pulse (0.5 uS) that occurs 2.75 uS after the zero-crossing of the parity bit if the word just received was preceded by a 2 uS gap and had a VA-L or BRO-L. Also present if there was a 2 uS gap, followed by a valid non-mode command word followed by a non-mode Valid Address (VA-L). Masked if TACT-L is low.
FSF-L	22	O	Fail safe flag - active low output signal indicating a transmitter timeout (768 uS) has occurred.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
LTF1-L	23	O	Loop test fail - a low level output pulse (0.5 us) that will occur after each data word transmission from UKT if a failure is detected with the word received. Checks each word for correct sync, Manchester errors and parity. Also checks the last word for data contents.
LTF2-L	24	O	A low level output pulse (0.5 us) that will occur after the last data word transmitted from UKT if a failure is detected with the word received. Checks for Manchester errors, parity and data content.
EOT-L	75	O	Output signal that goes low whenever the start of a new word (valid sync and 2 data bits) has not been detected continuous with the previous word. Goes high upon detection of a valid sync and 2 data bits.
2MHZ	36	O	2 MHz clock line output - Free running 2 MHz clock, synchronized with 12 MHz clock. (Not present reset.)
RTA	14	O	Logic one output indicates valid address with odd parity (RTA0-RTA4 and RTAP).
RXDO	85	O	BI-phase TTL output from receiver.
RXDO-L	63	O	BI-phase TTL output from receiver.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
TXDO	88	O	Transmitter data out - bi-phase TTL data from encoding section.
TXDO-L	90	O	Transmitter data out - bi-phase TTL data from encoding section.
TXIO	65	O	Transmitter inhibit output - output signal that goes low whenever the channel is selected (CE1 = logic 1 or CE2-L = logic 0) and the TXEN-L input is low.
DIRC	8	I/O	Bus connection from transformer, high side, manchester coded signal, direct coupled to/from isolation resistor on PC board.
XFMR COUP	56	I/O	Bus connection from transformer, high side, manchester coded signal, direction input/output from transformer (no isolation resistor).
CENTER TAP	6	I/O	Bus transformer center tap.
XFMR COUP-L	57	I/O	Bus connection from transformer, low side, manchester coded signal, direction input/output from transformer (no isolation resistor).
DIRC-L	7	I	Bus connection from transformer, low side, manchester coded signal, direct coupled to/from isolation resistor on PC board.
Vcc	1	I	+5 V power connection.
Vcc	31	I	+5 V power connection.
Vcc	51	I	+5 V power connection.
Vcc	81	I	+5 V power connection.

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6.4.3 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
V-	20	I	-15 V power connection.
V-	70	I	-15 V power connection.
V+	3	I	+15 V power connection.
V+	53	I	+15 V power connection.
GND	5	GND	Ground connection.
GND	10	GND	Ground connection.
GND	15	GND	Ground connection.
GND	25	GND	Ground connection.
GND	40	GND	Ground connection.
GND	45	GND	Ground connection.
GND	55	GND	Ground connection.
GND	60	GND	Ground connection.
GND	95	GND	Ground connection.
GND	11	GND	Frame ground.
	2,4,9, 13,52, 54,58, 59,61, 64		Unused pins.

6.4.3.1. Since the bus interface module has been designed to interconnect directly with the protocol module (UKU), refer to the UKU pinout and signal description for more information on input signals coming from UKU to UKT.

6.4.3.2. If CE1 is low and CE2-L is high, then the following output signals will be in the high impedance mode:

(a) NW-L, GAP-L, CMD-L, VW-L, ND-L, BRO-L, MC-L, TD-L, T1-L, T2-L, TACT-L, VA-L, FSF-L, LTF1-L, LTF2-L and EOT-L.

(b) Also, the following input signals will have no effect: TXEN-L, ECEN-L, MSBE-L, LSBE-L, ICMD-L and T/R-L.

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedure should be followed:

(a) Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up or static charge.

(b) Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

(c) Signals should not be applied to the inputs while the device power supply is off.

(d) All input signals and conditioning voltages must be removed before Vcc, V+ and V- are disconnected.

(e) The following input signals must not be left floating during testing: TXEN-L (27), ECEN-L (28), CE2-L (30) CE1 (29), RST2-L (84), CLKI (78), RXDI (86), RXDI-L (62), STS (41).

(f) When using the transceiver portion of the UKT module alone, the following inputs should be connected to the appropriate voltages:

Signal name	Pin no.	Tie to
BRI-L	79	GND
RTA0	80	GND
RTA1	32	GND
RTA2	33	GND
RTA3	34	GND

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Signal name	Pin no.	Tie to
RTA4	35	GND
RST2-L	84	GND
CE2-L	29	GND
CE1	30	GND
CLKI	78	GND
STS-L	41	GND
RXDI	86	GND
RXDI-L	62	GND
SNCS-L	77	+5V (through a 1-10
TXEN-L	27	+5V KOhm resistor)
ECEN-L	28	+5V |
MSBL-L	39	+5V |
MSBE-L	37	+5V |
LSBL-L	91	+5V (through a 1-10
LSBE-L	38	+5V KOhm resistor)

(g) Care should be exercised when using the UKT, UKU and UKS modules to prevent bus contention during normal operation or power up conditions. During software debugging where bus contentions are possible, current limiting of power supplies is a suggested method to prevent damage to modules.

Preparing activity:
Navy - SH

(Project 5963-N724)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Frame	Frame	C,2,4,9, 13,52, 54,58, 59,61, 64	10	INF	MOhm	1-7
A2	Unused to unused pins	2	4,9,13, 52,54, 58,59, 61,64				
A3	Unused to unused pins	4	9,13,52, 54,58, 59,61, 64				
A4	Unused to unused pins	9	13,52, 54,58, 59,61, 64				
A5	Unused to unused pins	13	52,54, 58,59, 61,64				
A6	Unused to unused pins	52	54,58, 59,61, 64				
A7	Unused to unused pins	54	58,59, 61,64				
A8	Unused to unused pins	58	59,61, 64				
A9	Unused to unused pins	59	61,64				
A10	Unused to unused pins	61	64	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A11	Used to unused pins	C,11	2,4,9, 13,52, 54,58, 59,61, 64	10	INF	MOhm	1-7
A12	Pin 11 to all other pins	11	C,2,4, 9,13, 52,54, 58,59, 61,64	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1
Megohm bridge voltage	30 V	± 10 percent	A2-A12
Isolation measurement	NA	± 500 KOhm	A1-A12
Isolation measurement (See note 6)	NA	± 50 KOhm	A1-A12

2. The ohmmeter used shall be item R06.

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NOTES: - Continued.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal, and 'C' shall correspond to all used pins shorted together except pin 11.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
B1	Continuity, Vcc	1	31	0	1	Ohm	1-5
B2	Continuity, Vcc	31	51				
B3	Continuity, Vcc	51	81				
B4	Continuity, V+	3	53				
B5	Continuity, V-	20	70				
B6	Continuity, GND	5	10				
B7	Continuity, GND	10	15				
B8	Continuity, GND	15	25				
B9	Continuity, GND	25	40				
B10	Continuity, GND	40	45				
B11	Continuity, GND	45	55	0	1	Ohm	1-5

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TABLE III. Continuity test - Continued.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
B12	Continuity, Vcc	55	60	0	1	Ohm	1-5
B13	Continuity, Vcc	60	95				
B14	Continuity, frame to pin 11	Frame	11	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B14

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. When performing test condition B14, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the positive terminal of the ohmmeter.

5. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits	Notes
C1	Vcc, stress	1,31,51 81	5,10,15, 25,40, 45,55, 60,95	Module shall pass all remaining tests.	1-6
C2	V+, stress	3,53	5,10,15, 25,40, 45,55, 60,95	Module shall pass all remaining tests.	1-6
C3	V-, stress	20,70	5,10,15, 25,40, 45,55, 60,95	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C1
(V+)	18.0 V	+0 mV,-180 mV	C2
(V-)	-18.0 V	-0 mV,+180 mV	C3
Power supply current limit (Vcc)	540 mA	NA	C1
(V+)	450 mA	NA	C2
(V-)	450 mA	NA	C3

2. There shall be no power supply voltage overshoot on this test.

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NOTES: - Continued.

3. In the setup conditions in table IV, 'A' shall correspond to the positive terminal of the power supply, and 'B' shall correspond to the negative terminal for test conditions C1 and C2. For test condition C3, 'A' shall correspond to the negative terminal of the power supply and 'B' shall correspond to the positive terminal.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. See 6.6 for module input handling.

6. Abbreviations used are defined in 6.1.

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TABLE V. Transformer short circuit stress test.

Test cond	Parameter	Connect A to pin	Initial limits	Figs
D1	DIRC, DIRC-L	7,8	Module shall pass all remaining tests.	12,13
D2	XFMR, XFMR-L	56,57	Module shall pass all remaining tests.	12,13

NOTES:

1. Transformer short circuit stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	+0 mV, -55 mV	D1,D2
(V+)	16.5 V	+0 mV,-165 mV	D1,D2
(V-)	-16.5 V	-0 mV,+165 mV	D1,D2
Power supply current limit			
(Vcc)	450 mA	NA	D1,D2
(V+)	375 mA	NA	D1,D2
(V-)	375 mA	NA	D1,D2

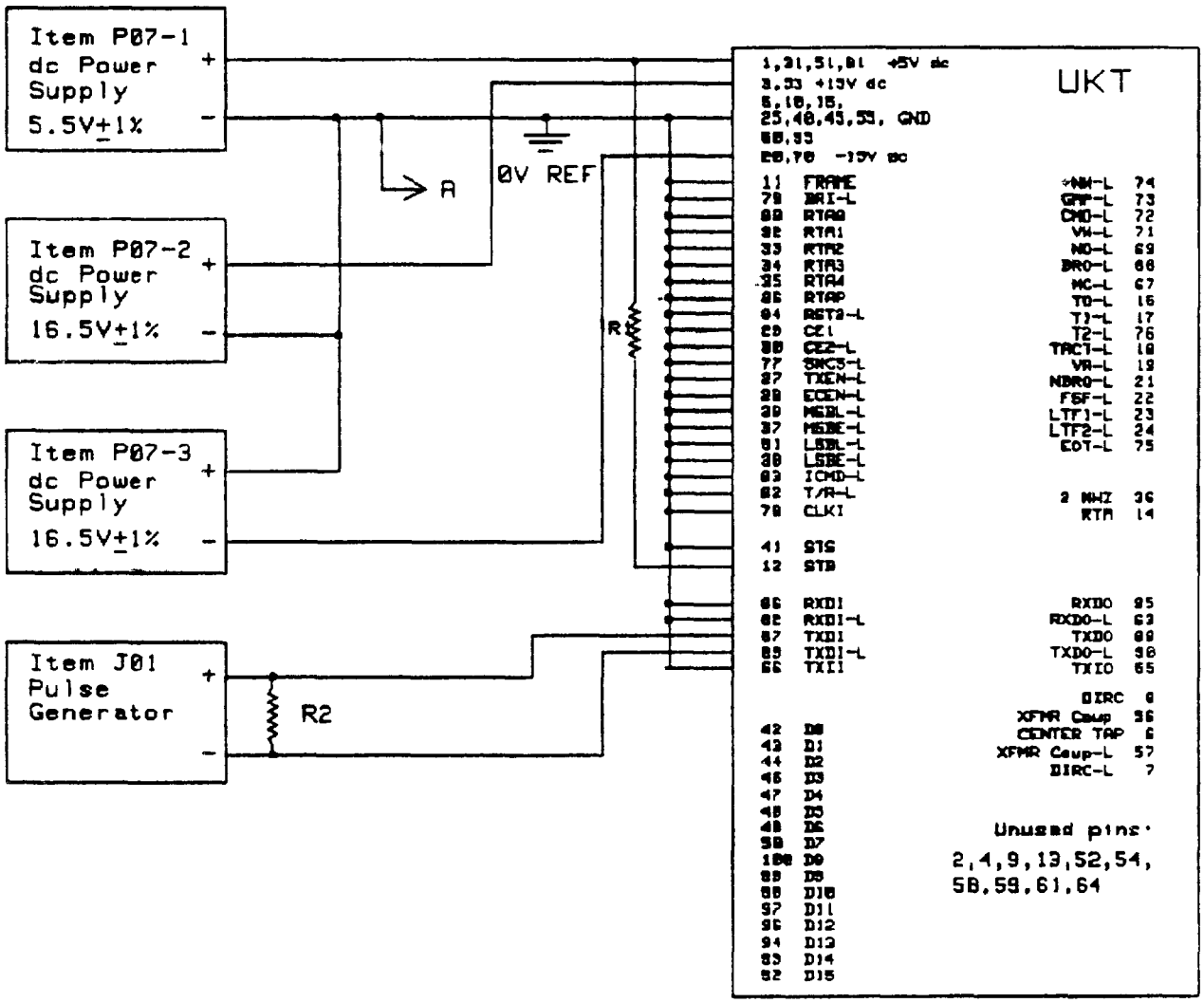
2. There shall be no power supply voltage overshoot on this test.

3. The test circuit shall be as indicated on figure 12.

4. The pulse generator waveforms shall be as indicated on figure 13.

5. Abbreviations used are defined in 6.1.

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NOTES:

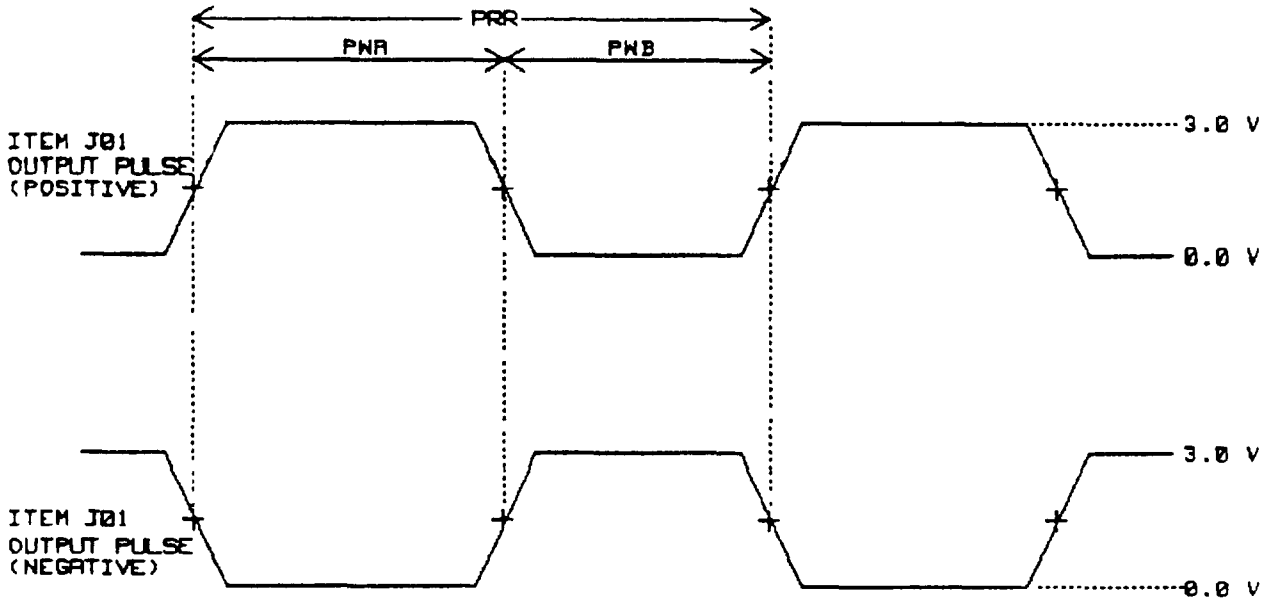
1. R1 = 1.0 KOhm ± 5 percent, 1/4 W.

2. R2 = 51.1 Ohm ± 1 percent, 1/2 W.

3. The duration of momentarily connecting point A for test conditions D1 and D2 shall not exceed one second.

FIGURE 12. Transformer short circuit stress test circuit for table V.

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NOTES:

1. PWA = PWB = 500 nS.

2. PRR = 1000 nS.

3. Transition times (10 percent - 90 percent) shall be less than 5 nS.

FIGURE 13. Test circuit waveforms for figure 12.

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TABLE VI. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	D0	42	5.5	0	100	uA	1-3
E2	D1	43					
E3	D2	44					
E4	D3	46					
E5	D4	47					
E6	D5	48					
E7	D6	49					
E8	D7	50					
E9	D8	100					
E10	D9	99					
E11	D10	98					
E12	D11	97					
E13	D12	96					
E14	D13	94					
E15	D14	93					
E16	D15	92					
E17	MSBE-L	37					
E18	LSBE-L	38					
E19	MSBL-L	39					
E20	LSBL-L	91					
E21	SNCS-L	77	5.5	0	100	uA	1-3

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TABLE VI. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	RTAP	26	5.5	0	150	uA	1-3
E23	TXII	66	5.5	0	200	uA	1-3
E24	STB	12	5.5	0	200	uA	1-3
E25	TXDI	87	5.5	0	400	uA	1-3
E26	TXDI-L	89	5.5	0	400	uA	1-3
E27	RTA0	80	5.5	0	150	uA	1-3
E28	RTA1	32					
E29	RTA2	33					
E30	RTA3	34					
E31	RTA4	35	5.5	0	150	uA	1-3
E32	BRI-L	79	5.5	0	40	uA	1-3
E33	ICMD-L	83					
E34	T/R-L	82	5.5	0	40	uA	1-3
E35	RXDI	86	5.5	0	20	uA	1-3
E36	RXDI-L	62					
E37	STS	41					
E38	CLKI	78					
E39	ECEN-L	28					
E40	TXEN-L	27	5.5	0	20	uA	1-3

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TABLE VI. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	CE2-L	30	5.5	0	20	uA	1-3
E42	CE1	29	5.5	0	20	uA	1-3
E43	RST2-L	84	5.5	0	40	uA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E21
(Vcc)	5.5 V	+ 0 V,-55mV	E22
(Vcc)	5.5 V	± 55 mV	E23-E26
(Vcc)	5.5 V	+ 0 V,-55mV	E27-E34
(Vcc)	5.5 V	± 55 mV	E35-E43
(V+)	+16.5 V	± 165 mV	E1-E43
(V-)	-16.5 V	± 165 mV	E1-E43
Power supply current limit			
(Vcc)	450 mA	NA	E1-E43
(V+)	375 mA	NA	E1-E43
(V-)	375 mA	NA	E1-E43
Forcing voltage, (Vin)	5.5 V	± 55 mV	E1-E21
(Vin)	5.5 V	+ 55mV,-0mV	E22
(Vin)	5.5 V	± 55 mV	E23-E26
(Vin)	5.5 V	+ 55mV,-0mV	E27-E34
(Vin)	5.5 V	± 55 mV	E35-E43
Input current measurement	NA	± 5.0 uA	E1-E21
	NA	± 7.5 uA	E22
	NA	± 10.0 uA	E23,E24
	NA	± 20.0 uA	E25,E26
	NA	± 7.5 uA	E27-E31
	NA	± 2.0 uA	E32-E34
	NA	± 1.0 uA	E35-E42
	NA	± 2.0 uA	E43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) For test conditions E1-E16, precondition the module by applying 3.0 V to pins 37 and 38 (MSBE-L and LSBE-L).

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	D0	42	0.0	-225	-40	mA	1-5
F2	D1	43					
F3	D2	44					
F4	D3	46					
F5	D4	47					
F6	D5	48					
F7	D6	49					
F8	D7	50					
F9	D8	100					
F10	D9	99					
F11	D10	98					
F12	D11	97					
F13	D12	96					
F14	D13	94					
F15	D14	93					
F16	D15	92	0.0	-225	-40	mA	1-5
F17	2MHZ	36	0.0	-100	-20	mA	1-5
F18	RTA	14					
F19	RXDO	85					
F20	RXDO-L	63					
F21	TXIO	65	0.0	-100	-20	mA	1-5

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F21
(V+)	16.5 V	± 165 mV	F1-F21
(V-)	-16.5 V	± 165 mV	F1-F21
Power supply current limit			
(Vcc)	450 mA	NA	F1-F21
(V+)	375 mA	NA	F1-F21
(V-)	375 mA	NA	F1-F21
Forcing voltage, (Vout)	0.0 V	± 50 mV	F1-F21
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F21
low, (VL)	0.0 V	± 100 mV	F1-F21
high, (VH) (See note 4)	5.5 V	± 55 mV	F1-F21
Output current measurement	NA	± 9.25 mA	F1-F16
	NA	± 4.0 mA	F17-F21

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. The high level pattern voltage for CLKI (pin 78) shall be 5.5 V.

5. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	D0	42	2.4	0	20	uA	1-3
G2	D1	43					
G3	D2	44					
G4	D3	46					
G5	D4	47					
G6	D5	48					
G7	D6	49					
G8	D7	50					
G9	D8	100					
G10	D9	99					
G11	D10	98					
G12	D11	97					
G13	D12	96					
G14	D13	94					
G15	D14	93					
G16	D15	92					
G17	MSBE-L	37					
G18	LSBE-L	38					
G19	MSBL-L	39					
G20	LSBL-L	91					
G21	SNCS-L	77	2.4	0	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G22	RTAP	26	2.4	-250	-40	uA	1-3
G23	TXII	66	2.4	0	40	uA	1-3
G24	STB	12	2.4	0	40	uA	1-3
G25	TXDI	87	2.4	0	80	uA	1-3
G26	TXDI-L	89	2.4	0	80	uA	1-3
G27	RTA0	80	2.4	-400	-10	uA	1-3
G28	RTA1	32					
G29	RTA2	33					
G30	RTA3	34					
G31	RTA4	35					
G32	BRI-L	79	2.4	-400	-10	uA	1-3
G33	ICMD-L	83	2.4	-250	-40	uA	1-3
G34	T/R-L	82	2.4	-250	-40	uA	1-3
G35	RXDI	86	2.4	-20	20	uA	1-3
G36	RXDI-L	62					
G37	STS	41					
G38	CLKI	78					
G39	ECEN-L	28					
G40	TXEN-L	27	2.4	-20	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G41	CE2-L	30	2.4	-20	20	uA	1-3
G42	CE1	29	2.4	-20	20	uA	1-3
G43	RST2-L	84	2.4	-40	40	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G43
(V+)	16.5 V	± 165 mV	G1-G43
(V-)	-16.5 V	± 165 mV	G1-G43
Power supply current limit (Vcc)	450 mA	NA	G1-G43
(V+)	375 mA	NA	G1-G43
(V-)	375 mA	NA	G1-G43
Forcing voltage, (Vih)	2.4 V	± 50 mV	G1-G37
(Vih)	5.0 V	± 50 mV	G38
(Vih)	2.4 V	± 50 mV	G39-G43
Input current measurement	NA	± 1.0 uA	G1-G21
	NA	± 10.5 uA	G22
	NA	± 2.0 uA	G23,G24
	NA	± 4.0 uA	G25,G26
	NA	± 19.5 uA	G27-G32
	NA	± 10.5 uA	G33,G34
	NA	± 2.0 uA	G35-G42
	NA	± 4.0 uA	G43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) For test conditions G1-G16, precondition the module by applying 3.0 V to pins 37 and 38 (MSBE-L and LSBE-L).

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	D0	42	0.4	-0.4	0.0	mA	1-3
H2	D1	43					
H3	D2	44					
H4	D3	46					
H5	D4	47					
H6	D5	48					
H7	D6	49					
H8	D7	50					
H9	D8	100					
H10	D9	99					
H11	D10	98					
H12	D11	97					
H13	D12	96					
H14	D13	94					
H15	D14	93					
H16	D15	92					
H17	MSBE-L	37					
H18	LSBE-L	38					
H19	MSBL-L	39					
H20	LSBL-L	91					
H21	SNCS-L	77	0.4	-0.4	0.0	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	RTAP	26	0.4	-0.8	-0.04	mA	1-3
H23	TXII	66	0.4	-0.8	0.0	mA	1-3
H24	STB	12	0.4	-0.8	0.0	mA	1-3
H25	TXDI	87	0.4	-1.6	0.0	mA	1-3
H26	TXDI-L	89	0.4	-1.6	0.0	mA	1-3
H27	RTA0	80	0.4	-1.2	-0.4	mA	1-3
H28	RTA1	32					
H29	RTA2	33					
H30	RTA3	34					
H31	RTA4	35	0.4	-1.2	-0.4	mA	1-3
H32	BRI-L	79	0.4	-0.8	-0.04	mA	1-3
H33	ICMD-L	83	0.4	-0.4	-0.04	mA	1-3
H34	T/R-L	82	0.4	-0.4	-0.04	mA	1-3
H35	RXDI	86	0.4	-0.02	0.02	mA	1-3
H36	RXDI-L	62					
H37	STS	41					
H38	CLKI	78					
H39	ECEN-L	28					
H40	TXEN-L	27	0.4	-0.02	0.02	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	CE2-L	30	0.4	-0.02	0.02	mA	1-3
H42	CE1	29	0.4	-0.02	0.02	mA	1-3
H43	RST2-L	84	0.4	-0.04	0.04	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H43
(V+)	16.5 V	± 165 mV	H1-H43
(V-)	-16.5 V	± 165 mV	H1-H43
Power supply current limit (Vcc)	450 mA	NA	H1-H43
(V+)	375 mA	NA	H1-H43
(V-)	375 mA	NA	H1-H43
Forcing voltage, (Vil)	0.4 V	± 50 mV	H1-H43
Input current measurement	NA	± 20 uA	H1-H21
	NA	± 38 uA	H22
	NA	± 40 uA	H23,H24
	NA	± 80 uA	H25,H26
	NA	± 40 uA	H27-H31
	NA	± 38 uA	H32
	NA	± 18 uA	H33,H34
	NA	± 2 uA	H35-H42
	NA	± 4 uA	H43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) For test conditions H1-H16, precondition the module by applying 3.0 V to pins 37 and 38 (MSBE-L and LSBE-L). For test condition H24, precondition the module by applying 0.0 V to the V+ and V- power supply inputs. For test condition H25, precondition the module by applying 3.0 V to pins 89 and 66 (TXDI-L and TXII). For test condition H26, precondition the module by applying 3.0 V to pins 87 and 66 (TXDI and TXII).

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	D0	42	-18.0	-1.5	-0.3	V	1-3
J2	D1	43					
J3	D2	44					
J4	D3	46					
J5	D4	47					
J6	D5	48					
J7	D6	49					
J8	D7	50					
J9	D8	100					
J10	D9	99					
J11	D10	98					
J12	D11	97					
J13	D12	96					
J14	D13	94					
J15	D14	93					
J16	D15	92					
J17	MSBE-L	37					
J18	LSBE-L	38					
J19	MSBL-L	39					
J20	LSBL-L	91					
J21	SNCS-L	77	-18.0	-1.5	-0.3	V	1-3

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TABLE X. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	RTAP	26	-18.0	-1.5	-0.3	V	1-3
J23	TXII	66					
J24	STB	12					
J25	TXDI	87					
J26	TXDI-L	89	-18.0	-1.5	-0.3	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J26
(V+)	13.5 V	± 135 mV	J1-J26
(V-)	-13.5 V	± 135 mV	J1-J26
Power supply current limit (Vcc)	450 mA	NA	J1-J26
(V+)	375 mA	NA	J1-J26
(V-)	375 mA	NA	J1-J26
Forcing current, (Iin)	-18 mA	± 1.8 mA	J1-J26
Input voltage measurement	NA	± 60 mV	J1-J26

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) For test conditions J1-J16, precondition the module by applying 3.0 V to pins 37 and 38 (MSBE-L and LSBE-L).

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XI. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	D0	42	-3.0	2.4	4.55	V	1-5
K2	D1	43					
K3	D2	44					
K4	D3	46					
K5	D4	47					
K6	D5	48					
K7	D6	49					
K8	D7	50					
K9	D8	100					
K10	D9	99					
K11	D10	98					
K12	D11	97					
K13	D12	96					
K14	D13	94					
K15	D14	93					
K16	D15	92	-3.0	2.4	4.55	V	1-5
K17	NW-L	74	-0.40	2.4	4.55	V	1-5
K18	GAP-L	73					
K19	CMD-L	72					
K20	VW-L	71					
K21	ND-L	69	-0.40	2.4	4.55	V	1-5

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TABLE XI. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K22	BRO-L	68	-0.40	2.4	4.55	V	1-5
K23	MC-L	67					
K24	TD-L	16					
K25	T1-L	17					
K26	T2-L	76					
K27	TACT-L	18					
K28	VA-L	19					
K29	NBRQ-L	21					
K30	FSF-L	22					
K31	LTF1-L	23					
K32	LTF2-L	24					
K33	EOT-L	75					
K34	2MHZ	36	-0.40	2.4	4.55	V	1-5
K35	RTA	14	-0.38	2.4	4.55	V	1-5
K36	RXDO	85	-0.40	2.4	4.55	V	1-5
K37	RXDO-L	63					
K38	TXDO	88					
K39	TXDO-L	90					
K40	TXIO	65	-0.40	2.4	4.55	V	1-5

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K40
(V+)	13.5 V	± 135 mV	K1-K40
(V-)	-13.5 V	± 135 mV	K1-K40
Power supply current limit (Vcc)	450 mA	NA	K1-K40
(V+)	375 mA	NA	K1-K40
(V-)	375 mA	NA	K1-K40
Forcing current, (Ioh)	-3.00 mA	± 30.0 uA	K1-K16
(Ioh)	-0.40 mA	± 4.0 uA	K17-K34
(Ioh)	-0.38 mA	± 3.8 uA	K35
(Ioh)	-0.40 mA	± 4.0 uA	K36-K40
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K40
low, (VL)	0.0 V	± 100 mV	K1-K40
high, (VH) (See note 4)	4.5 V	± 45 mV	K1-K40
Output voltage measurement	NA	± 107.5 mV	K1-K40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. For test conditions K36 and K37, precondition the module by applying 3.0 V to pin 12 (STB) and 0.0 V to V+ and V- supply pins.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

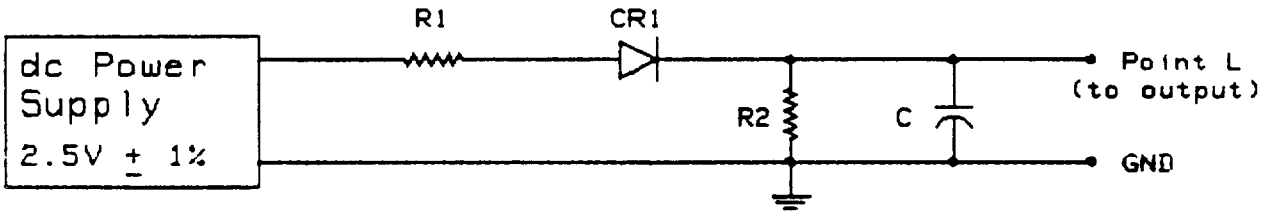
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 14.

4. The high level pattern voltage for CLKI (pin 78) shall be 4.5 V.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 pF ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

FIGURE 14. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
21,74,73,72,71, 69,68,67,16,17, 76,18,19,22,23, 24,75	6040	1.6	0.4
36,65,85,63	6040	4.0	0.4
14	6340	3.6	0.4
42,43,44,46,47, 48,49,50,100,99, 98,97,96,94,93,92	806	12.0	0.4
88,90	6040	2.4	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XII. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	D0	42	12.0	0.0	0.4	V	1-5
L2	D1	43					
L3	D2	44					
L4	D3	46					
L5	D4	47					
L6	D5	48					
L7	D6	49					
L8	D7	50					
L9	D8	100					
L10	D9	99					
L11	D10	98					
L12	D11	97					
L13	D12	96					
L14	D13	94					
L15	D14	93					
L16	D15	92	12.0	0.0	0.4	V	1-5
L17	NW-L	74	1.6	0.0	0.4	V	1-5
L18	GAP-L	73					
L19	CMD-L	72					
L20	VW-L	71					
L21	ND-L	69	1.6	0.0	0.4	V	1-5

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TABLE XII. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L22	BRO-L	68	1.6	0.0	0.4	V	1-5
L23	MC-L	67					
L24	TD-L	16					
L25	T1-L	17					
L26	T2-L	76					
L27	TACT-L	18					
L28	VA-L	19					
L29	NBRQ-L	21					
L30	FSF-L	22					
L31	LTF1-L	23					
L32	LTF2-L	24					
L33	EOT-L	75	1.6	0.0	0.4	V	1-5
L34	2MHZ	36	4.0	0.0	0.4	V	1-5
L35	RTA	14	3.6	0.0	0.4	V	1-5
L36	RXDO	85	4.0	0.0	0.4	V	1-5
L37	RXDO-L	63	4.0	0.0	0.4	V	1-5
L38	TXDO	88	2.4	0.0	0.4	V	1-5
L39	TXDO-L	90	2.4	0.0	0.4	V	1-5
L40	TXIO	65	4.0	0.0	0.4	V	1-5

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	L1-L40
(V+)	13.5 V	± 135 mV	L1-L40
(V-)	-13.5 V	± 135 mV	L1-L40
Power supply current limit (Vcc)	450 mA	NA	L1-L40
(V+)	375 mA	NA	L1-L40
(V-)	375 mA	NA	L1-L40
Forcing current, (Iol)	12.0 mA	± 120 uA	L1-L16
(Iol)	1.6 mA	± 16 uA	L17-L33
(Iol)	4.0 mA	± 40 uA	L34
(Iol)	3.6 mA	± 36 uA	L35
(Iol)	4.0 mA	± 40 uA	L36,L37
(Iol)	2.4 mA	± 24 uA	L38,L39
(Iol)	4.0 mA	± 40 uA	L40
Pattern voltage high, (VH)	3.0 V	± 100 mV	L1-L40
low, (VL)	0.0 V	± 100 mV	L1-L40
high, (VH) (See note 4)	4.5 V	± 45 mV	L1-L40
Output voltage measurement	NA	± 20 mV	L1-L40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. The load shall be as indicated on figure 14.

4. The high level pattern voltage for CLKI (pin 78) shall be 4.5 V.

5. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	NW-L	74	2.4	-20	20	uA	1-4
M2	GAP-L	73					
M3	CMD-L	72					
M4	VW-L	71					
M5	ND-L	69					
M6	BRO-L	68					
M7	MC-L	67					
M8	TD-L	16					
M9	T1-L	17					
M10	T2-L	76					
M11	TACT-L	18					
M12	VA-L	19					
M13	FSF-L	22					
M14	LTF1-L	23					
M15	LTF2-L	24					
M16	EOT-L	75	2.4	-20	20	uA	1-4

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M16
(V+)	16.5 V	± 165 mV	M1-M16
(V-)	-16.5 V	± 165 mV	M1-M16
Power supply current limit (Vcc)	450 mA	NA	M1-M16
(V+)	375 mA	NA	M1-M16
(V-)	375 mA	NA	M1-M16
Forcing voltage, (Voh)	2.4 V	± 50 mV	M1-M16
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M16
low, (VL)	0.0 V	± 100 mV	M1-M16
high, (VH) (See note 3)	5.5 V	± 55 mV	M1-M16
Output current measurement	NA	± 2 uA	M1-M16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve a high level at the appropriate output pin before applying 0.0 V to pin 29 (CE1) and 3.0 V to pin 30 (CE2-L).

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. The high level pattern voltage for CLKI (pin 78) shall be 5.5 V.

4. Abbreviations used are defined in 6.1.

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TABLE XIV. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	NW-L	74	0.4	-20	20	uA	1-4
N2	GAP-L	73					
N3	CMD-L	72					
N4	VW-L	71					
N5	ND-L	69					
N6	BRO-L	68					
N7	MC-L	67					
N8	TD-L	16					
N9	T1-L	17					
N10	T2-L	76					
N11	TACT-L	18					
N12	VA-L	19					
N13	FSF-L	22					
N14	LTF1-L	23					
N15	LTF2-L	24					
N16	EOT-L	75	0.4	-20	20	uA	1-4

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1-N16
(V+)	16.5 V	± 165 mV	N1-N16
(V-)	-16.5 V	± 165 mV	N1-N16
Power supply current limit (Vcc)	450 mA	NA	N1-N16
(V+)	375 mA	NA	N1-N16
(V-)	375 mA	NA	N1-N16
Forcing voltage, (Vol)	0.4 V	± 50 mV	N1-N16
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1-N16
low, (VL)	0.0 V	± 100 mV	N1-N16
high, (VH) (See note 3)	5.5 V	± 55 mV	N1-N16
Output current measurement	NA	± 2 uA	N1-N16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve a high level at the appropriate output pin before applying 0.0 V to pin 29 (CE1) and 3.0 V to pin 30 (CE2-L).

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIV and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. The high level pattern voltage for CLKI (pin 78) shall be 5.5 V.

4. Abbreviations used are defined in 6.1.

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TABLE XV. Power supply current test.

Test cond	Power supply	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
P1	Vcc	1,31, 51,81	30	300	mA	1-4
P2	V+	3,53	25	250	mA	1-4
P3	V-	20,70	25	250	mA	1-4

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	P1-P3
(V+)	16.5 V	± 165 mV	P1-P3
(V-)	-16.5 V	± 165 mV	P1-P3
Power supply current limit (Vcc)	450 mA	NA	P1-P3
(V+)	375 mA	NA	P1-P3
(V-)	375 mA	NA	P1-P3
Pattern voltage high, (VH)	3.0 V	± 100 mV	P1-P3
low, (VL)	0.0 V	± 100 mV	P1-P3
high, (VH) (See note 3)	5.5 V	± 55 mV	P1-P3
Power supply current measurement	NA	± 13.5 mA	P1
	NA	± 11.3 mA	P2,P3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XV.

3. The high level pattern voltage for CLKI (pin 78) shall be 5.5 V.

4. Abbreviations used are defined in 6.1.

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TABLE XVI. Functional test.

Test cond	Test description	Notes	Figs
Q1	Functional test minimum power supply voltage maximum clock frequency (12 MHz)	1-4	14,15
Q2	Functional test maximum power supply voltage maximum clock frequency (12 MHz)	1-4	14,15

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1
(V+)	13.5 V	± 135 mV	Q1
(V-)	-13.5 V	± 135 mV	Q1
(Vcc)	5.5 V	± 55 mV	Q2
(V+)	16.5 V	± 165 mV	Q2
(V-)	-16.5 V	± 165 mV	Q2
Power supply current limit (Vcc)	450 mA	NA	Q1,Q2
(V+)	375 mA	NA	Q1,Q2
(V-)	375 mA	NA	Q1,Q2
Pattern voltage high, (VH)	See figure 15	± 1 percent	Q1,Q2
low, (VL)		± 1 percent	Q1,Q2

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1
high, (Voh)	NA	± 155 mV	Q2
low, (Vol)	NA	± 20 mV	Q1,Q2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

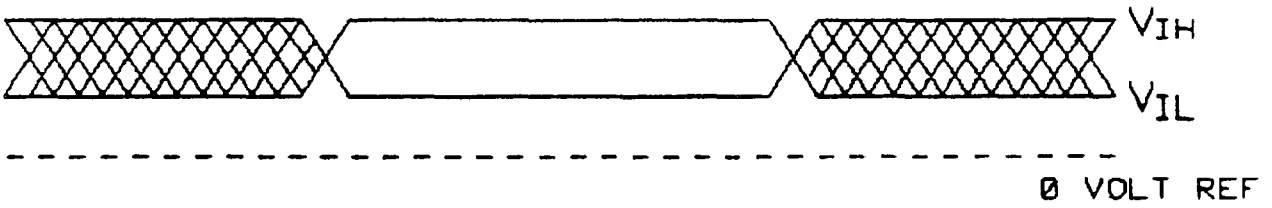
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XVI using only a capacitive load of 150 pF (see figure 14). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

3. Input signal voltages are as specified on figure 15.

4. Abbreviations used are defined in 6.1.

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NOTE:

1. The input pins shall have the following pattern voltages.

Signal name	Vil (V)	Vih (V)
RXDI, RXDI-L, STS, CE1, CE2-L, TXEN-L, ECEN-L	0.8	2.4
BRI-L, T/R-L, ICMD-L	0.8	2.4
MSBE-L, LSBE-L, MSBL-L, LSBL-L, SNCS-L, STB, TXII, TXDI, TXDI-L	0.7	2.0
RTA0, RTA1, RTA2, RTA3, RTA4, RTAP	0.8	2.0
RST2-L	0.7	2.4
D0, D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15	0.8	2.0
CLKI	0.5	Vcc-0.5

FIGURE 15. Input signal voltages for table XVI.

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TABLE XVII. Dynamic test.

Test cond	Test description	Notes	Figs
R1	Dynamic test maximum clock frequency (12 MHz) minimum power supply	1-6	14,16, 17
R2	Dynamic test maximum clock frequency (12 MHz) maximum power supply	1-6	14,16, 17

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	R1
(V+)	13.5 V	± 135 mV	R1
(V-)	-13.5 V	± 135 mV	R1
(Vcc)	5.5 V	± 55 mV	R2
(V+)	16.5 V	± 165 mV	R2
(V-)	-16.5 V	± 165 mV	R2
Power supply current limit (Vcc)	450 mA	NA	R1,R2
(V+)	375 mA	NA	R1,R2
(V-)	375 mA	NA	R1,R2
Pattern voltage high, (VH)	3.0 V	± 100 mV	R1,R2
low, (VL)	0.0 V	± 100 mV	R1,R2
high, (VH)	4.5 V	± 45 mV	R1
(See note 4)	5.5 V	± 55 mV	R2

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Transformer pins 7,8 or 56,57			
high, (VH)	3.0V	± 100 mV	R1,R2
low, (VL)	-3.0V	± 100 mV	R1,R2
Output voltage verification			
high, (Voh)	NA	± 105 mV	R1
high, (Voh)	NA	± 155 mV	R2
low, (Vol)	NA	± 20 mV	R1,R2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

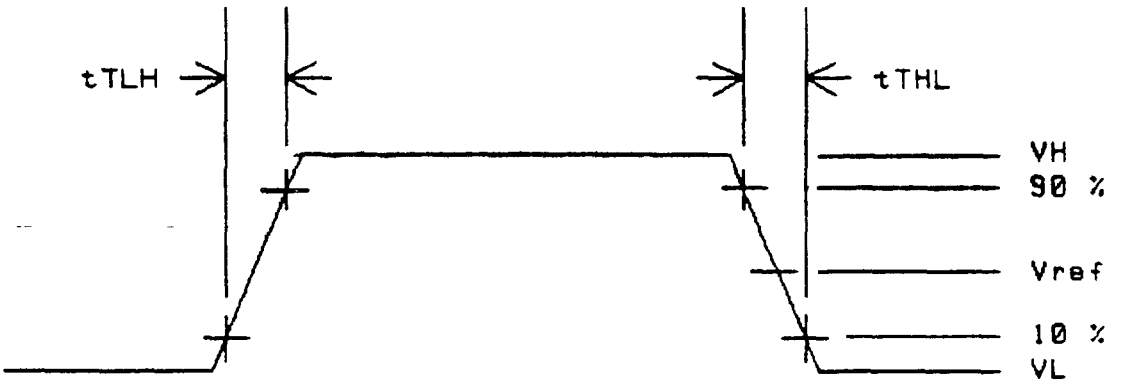
except transformer Input/Output (I/O) voltages are as specified on figure 16.

(c) The dynamic parameters are found in 6.3.3.

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NOTES: - Continued.

3. I/O waveforms are as specified on figures 6, 7 and 8. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

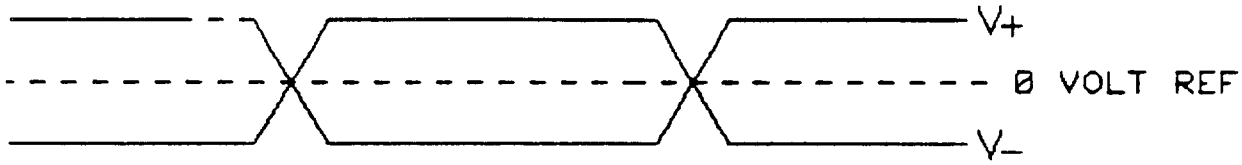
(c) Vref = 1.3 V.

4. The high level pattern voltage for CLKI (pin 78) shall be 4.5 V for test condition R1 and 5.5 V for test condition R2.

5. The transformer load circuit is as indicated on figure 17.

6. Abbreviations used are defined in 6.1.

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NOTES:

1. Receiver input

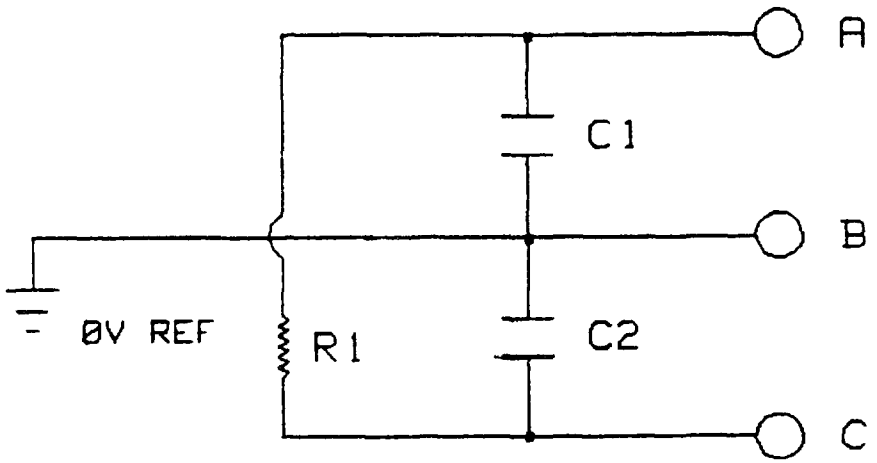
(a)	Direct coupled	1.2 V p-p min 20.0 V p-p max.
(b)	Transformer coupled	0.86 V p-p min 14.0 V p-p max.

2. Transmitter output

(a)	Direct coupled	6.0 V p-p min @ 35 ohm ± 2 percent, 9.0 V p-p max.
(b)	Transformer coupled	18.0 V p-p min @ 70 ohm ± 2 percent, 27.0 V p-p max.

FIGURE 16. Transformer I/O voltages for table XVII.

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NOTES:

1. C1 = C2 = 150 pF ± 10 percent (including fixture capacitance).
2. R1 = 35 Ohm ± 2 percent, 2 W (direct coupled).
3. R1 = 70 Ohm ± 2 percent, 2 W (transformer coupled).

FIGURE 17. Transformer load circuit.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S1	PSFC, Vcc	10	1,31,51 81	0.33	0.61	uF	1-3
S2	PSFC, V-	10	20,70	4.8	8.8	uF	1-3
S3	PSFC, V+	10	3,53	4.8	8.8	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	S1-S3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	S1-S3
Capacitance measurement	NA NA	± 14 nF ± 200 nF	S1 S2,S3

2. The capacitance bridge used in this test shall be item C01. The negative terminal 'A' shall be connected to the module ground and the positive terminal 'B' to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	D0	42	2	50	pF	1-3
T2	D1	43				
T3	D2	44				
T4	D3	46				
T5	D4	47				
T6	D5	48				
T7	D6	49				
T8	D7	50				
T9	D8	100				
T10	D9	99				
T11	D10	98				
T12	D11	97				
T13	D12	96				
T14	D13	94				
T15	D14	93				
T16	D15	92				
T17	MSBE-L	37				
T18	LSBE-L	38				
T19	MSBL-L	39				
T20	LSBL-L	91				
T21	SNCS-L	77	2	50	pF	1-3

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TABLE XIX. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T22	RTAP	26	2	50	pF	1-3
T23	TXII	66				
T24	STB	12				
T25	TXDI	87				
T26	TXDI-L	89				
T27	RTA0	80				
T28	RTA1	32				
T29	RTA2	33				
T30	RTA3	34				
T31	RTA4	35				
T32	BRI-L	79				
T33	ICMD-L	83				
T34	T/R-L	82				
T35	RXDI	86				
T36	RXDI-L	62				
T37	STS	41				
T38	CLKI	78				
T39	ECEN-L	28				
T40	TXEN-L	27	2	50	pF	1-3

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TABLE XIX. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T41	CE2-L	30	2	50	pF	1-3
T42	CE1	29	"	"	"	"
T43	RST2-L	84	2	50	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	T1-T43
Capacitance meter voltage	15 mV	± 1.5 mV	T1-T43
Capacitance measurement	NA	± 2.40 pF	T1-T43

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIX.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
U1	NW-L	74	2	50	pF	1-3
U2	GAP-L	73				
U3	CMD-L	72				
U4	VW-L	71				
U5	ND-L	69				
U6	BRO-L	68				
U7	MC-L	67				
U8	TD-L	16				
U9	T1-L	17				
U10	T2-L	76				
U11	TACT-L	18				
U12	VA-L	19				
U13	FSF-L	22				
U14	LTF1-L	23				
U15	LTF2-L	24				
U16	EOT-L	75	2	50	pF	1-3

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NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	U1-U16
Capacitance meter voltage	15 mV	± 1.5 mV	U1-U16
Capacitance measurement	NA	± 2.40 pF	U1-U16

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XX.

3. Abbreviations used are defined in 6.1.

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TABLE XXI. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
V1	PSTA, Vcc	0	400	mV	1-4	18
V2	PSTA, V+	0	400	mV	1-4	18
V3	PSTA, V-	0	400	mV	1-4	18

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 V	V1-V3
(V+)	15.0 V	± 150 V	V1-V3
(V-)	-15.0 V	± 150 V	V1-V3
Power supply current limit (Vcc)	450 mA	NA	V1-V3
(V+)	375 mA	NA	V1-V3
(V-)	375 mA	NA	V1-V3
Pattern voltage high, (VH)	3.0 V	± 100 mV	V1-V3
low, (VL)	0.0 V	± 100 mV	V1-V3
high, (VH) (See note 3)	5.0 V	± 50 mV	V1-V3
Power supply transient amplitude measurement	NA	± 20 mV	V1-V3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XXI.

(c) All Vcc pins shall be shorted together and the scope point, Y01, connected to Vcc at the point of shorting.

(d) Using the oscilloscope, D07, set at a maximum strobe rate of one-tenth of the cycle time per division, visually inspect each pattern row for the most negative spike as indicated on figure 18.

(e) The maximum probe capacitance is 10 pF. There shall be no other capacitance present on Vcc to ground.

3. The high level pattern voltage for CLKI (pin 78) shall be 5.0 V.

4. Abbreviations used are defined in 6.1.

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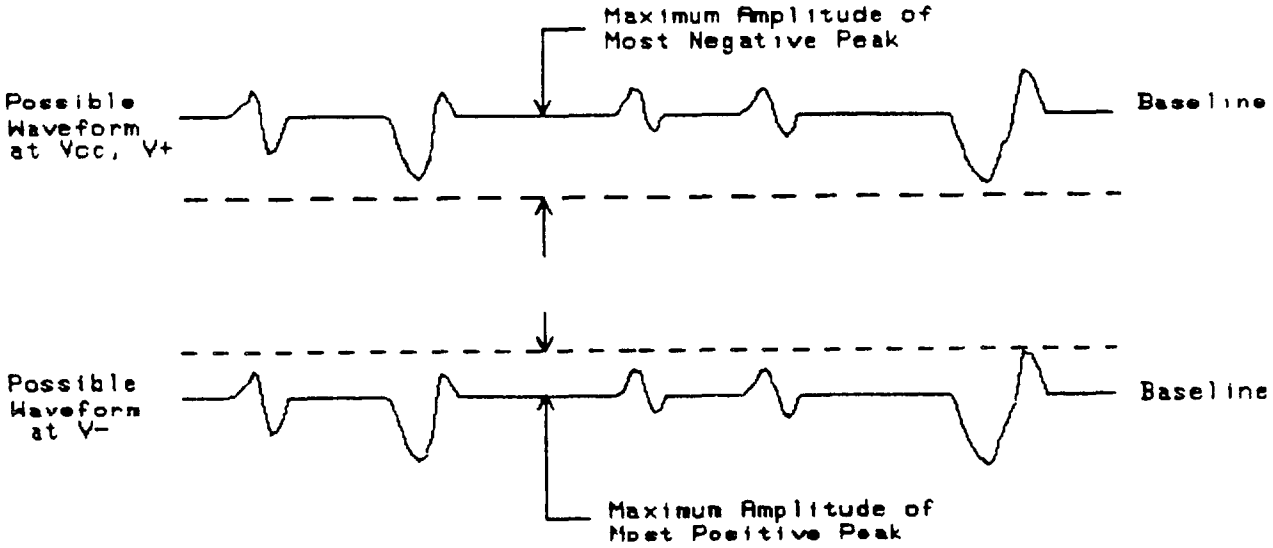


FIGURE 18. Power supply transient amplitude waveform for table XXI.

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TABLE XXII. Transformer impedance test.

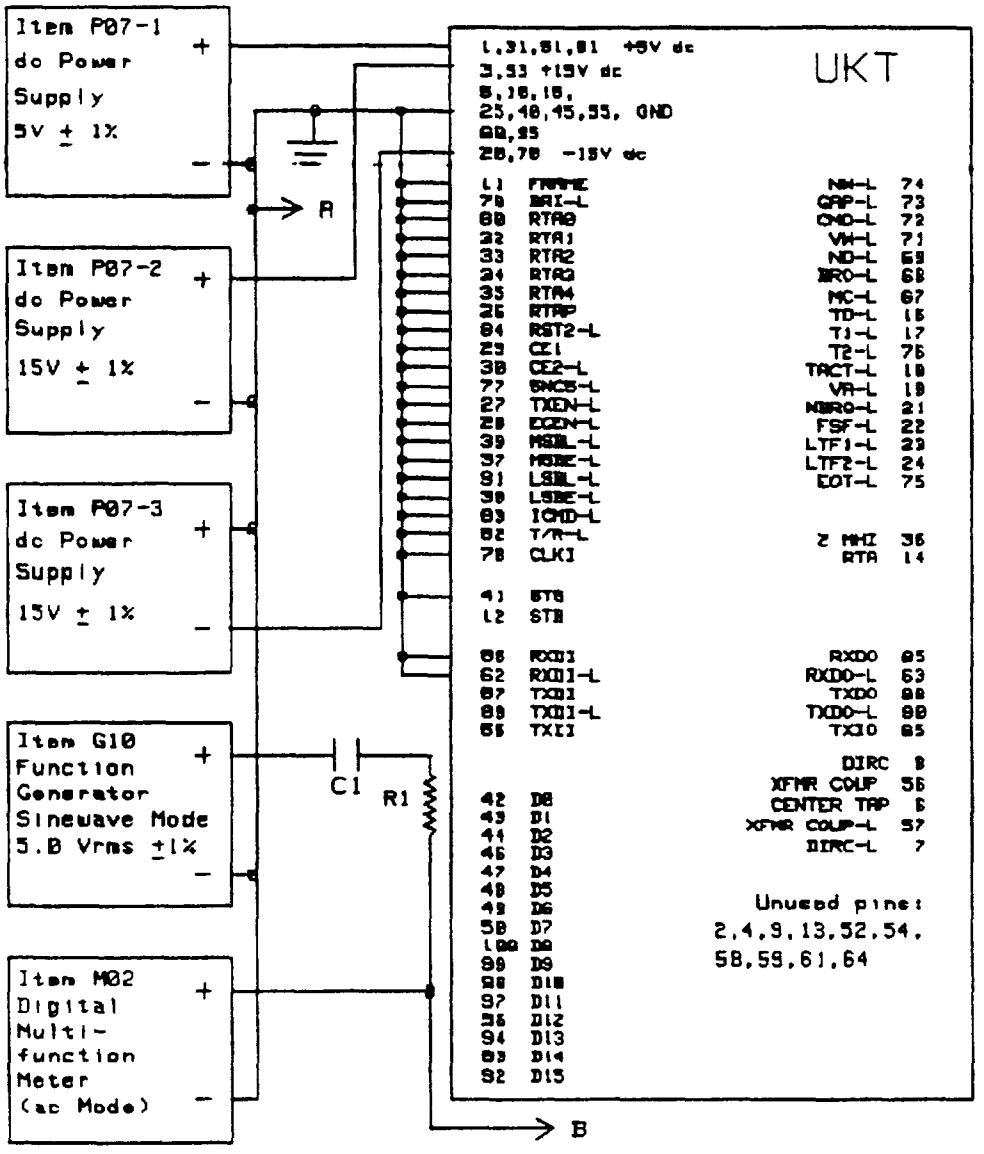
Test cond	Parameter	Item G10 frequency ± 1 percent	Connect		25 ± 5°C Initial limits		Units	Fig
			A to pin	B to pin	Min	Max		
W1	Direct coupled transformer impedance	75 KHz	7	8	2.5	5.0	Vrms	19
W2	Direct coupled transformer impedance	1 MHz	7	8	2.5	5.0	Vrms	19
W3	Transformer coupled transformer impedance	75 KHz	57	56	2.5	5.0	Vrms	19
W4	Transformer coupled transformer impedance	1 MHz	57	56	2.5	5.0	Vrms	19

NOTES:

1. The test circuit shall be as indicated on figure 19.

2. Connect the module as indicated in table XXII and make the indicated measurements.

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NOTES:

1. R1 = 2 KOhm ± 1 percent for test conditions W1 and W2. R1 = 1 KOhm ± 1 percent for test conditions W3 and W4.

2. C1 = 1 uF ± 10 percent.

FIGURE 19. Transformer impedance test circuit for table XXII.

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TABLE XXIII. Receiver common mode test.

Test cond	Parameter	G10-2 frequency ±1 percent	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
X1	Voh1,RXDO	2.0 MHz	85	7	8	2.4	5.0	V	20,21
X2	Vol1,RXDO	2.0 MHz	85	7	8	0.0	0.4	V	20,21
X3	Tpd1,prop delay	2.0 MHz	85	7	8	250.0	400.0	nS	20,21
X4	Voh2,RXDO-L	2.0 MHz	63	7	8	2.4	5.0	V	20,21
X5	Vol2,RXDO-L	2.0 MHz	63	7	8	0.0	0.4	V	20,21
X6	Tpd2,prop delay	2.0 MHz	63	7	8	250.0	400.0	nS	20,21
X7	Voh1,RXDO	2.0 MHz	85	57	56	2.4	5.0	V	20,21
X8	Vol1,RXDO	2.0 MHz	85	57	56	0.0	0.4	V	20,21
X9	Tpd1,prop delay	2.0 MHz	85	57	56	250.0	400.0	nS	20,21
X10	Voh2,RXDO-L	2.0 MHz	63	57	56	2.4	5.0	V	20,21
X11	Vol2,RXDO-L	2.0 MHz	63	57	56	0.0	0.4	V	20,21
X12	Tpd2,prop delay	2.0 MHz	63	57	56	250.0	400.0	nS	20,21

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NOTES:

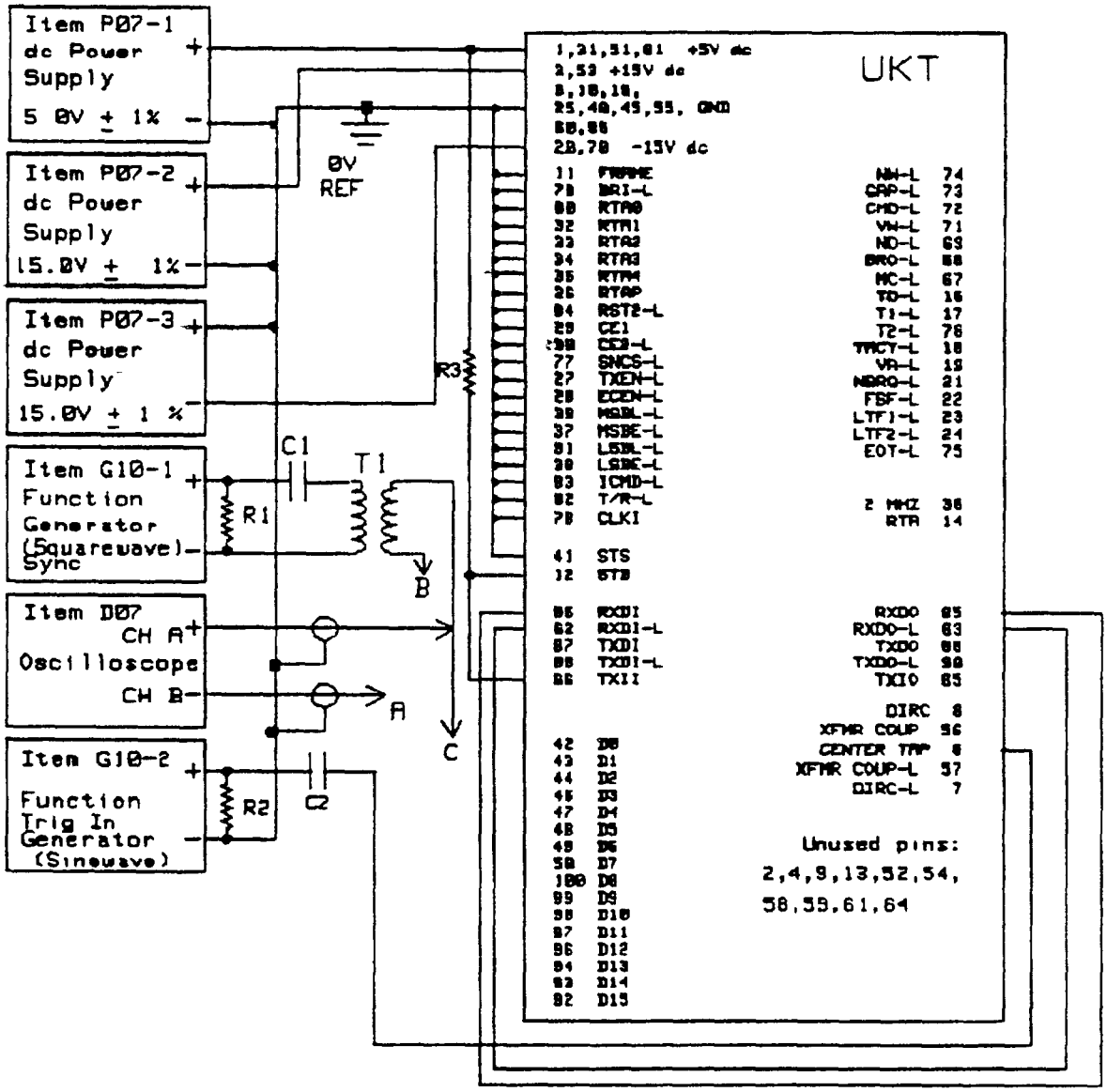
1. The test circuit shall be as indicated on figure 20.

2. Pins 63 and 85 shall be loaded as specified on figure 14.

3. Connect the module as indicated in table XXIII and make the indicated measurement.

4. The input and output waveforms shall be as indicated on figure 21.

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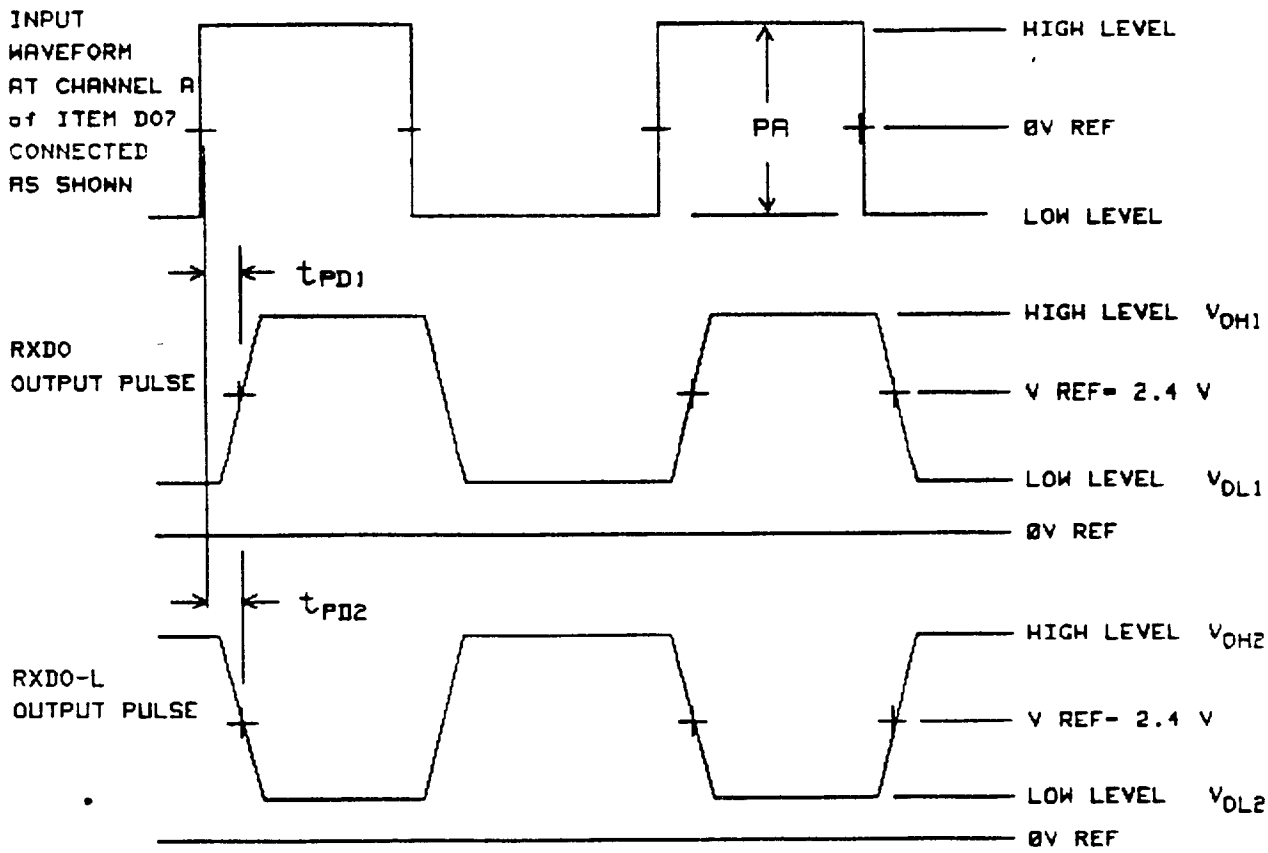


NOTES:

1. R1 = R2 = 51.1 ± 1 percent.
2. R3 = 1 KOhm ± 1 percent.
3. C1 = C2 = 1 uF ± 10 percent.
4. T1 = 1:1 isolation transformer (BUS 37070 or equivalent).

FIGURE 20. Receiver common mode test circuit for table XXIII.

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NOTES:

1. ITEM G10-1 output is a 1 MHz ± 10 percent squarewave. Item G10-2 output is a 6.5 Vrms sinewave with frequency specified in table XXIII. PA = 7 V p-p ± 1 percent.

2. Use offset capability of G10 to obtain bipolar waveform.

FIGURE 21. Receiver common mode test circuit waveforms for table XXIII.

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TABLE XXIV. Transmitter functional test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 + °C Initial limits Min	25 + °C Initial limits Max	Units	Figs
Y1	Zero crossing time (tZ)	8	7,66	87	89	Open	475.0	525.0	nS	22,23
Y2	Transition time (tTHL)	8	7,66	87	89	Open	100.0	300.0	nS	22,23
Y3	Transition time (tTLH)	8	7,66	87	89	Open	100.0	300.0	nS	22,23
Y4	TXDI to DIRC-L propagation delay time (tpd)	8	7,66	87	89	Open	50.0	250.0	nS	22,23
Y5	TXDI-L to DIRC-L propagation delay time (tpd)	8	7,66	87	89	Open	50.0	250.0	nS	22,23
Y6	Overshoot (VOS)	8	7,66	87	89	Open	0.0	300.0	mV	22,23
Y7	Undershoot (VUS)	8	7,66	87	89	Open	0.0	300.0	mV	22,23
Y8	Output voltage (high level)	8	7,66	87	89	Open	3.0	4.5	Vpk	22,23
Y9	Output voltage (low level)	8	7,66	87	89	Open	-4.5	-3.0	Vpk	22,23
Y10	Zero crossing time (tz)	56	57,66	87	89	Open	475.0	525.0	nS	22,23
Y11	Transition time (tTHL)	56	57,66	87	89	Open	100.0	300.0	nS	22,23
Y12	Transition time (tTLH)	56	57,66	87	89	Open	100.0	300.0	nS	22,23

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TABLE XXIV. Transmitter functional test - Continued.

Test cond	Parameter	Connect					25 ± 5°C Initial limits		Units	Figs
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y13	TXDI to XFMR COUP propagation delay (tpd)	56	57,66	87	89	Open	50.0	250.0	nS	22,23
Y14	TXDI-L to XFMR COUP propagation delay (tpd)	56	57,66	87	89	Open	50.0	250.0	nS	22,23
Y15	Overshoot (VOS)	56	57,66	87	89	Open	0.0	900.0	mV	22,23
Y16	Undershoot (VUS)	56	57,66	87	89	Open	0.0	900.0	mV	22,23
Y17	Output voltage (high level)	56	57,66	87	89	Open	9.0	13.5	Vpk	22,23
Y18	Output voltage (low level)	56	57,66	87	89	Open	-13.5	-9.0	Vpk	22,23
Y19	No response, TXII low	56	57,66	87, 89	Open	Open	No response			22,23
Y20	No response, TXII high	56	57	87	89	66	No response			22,23

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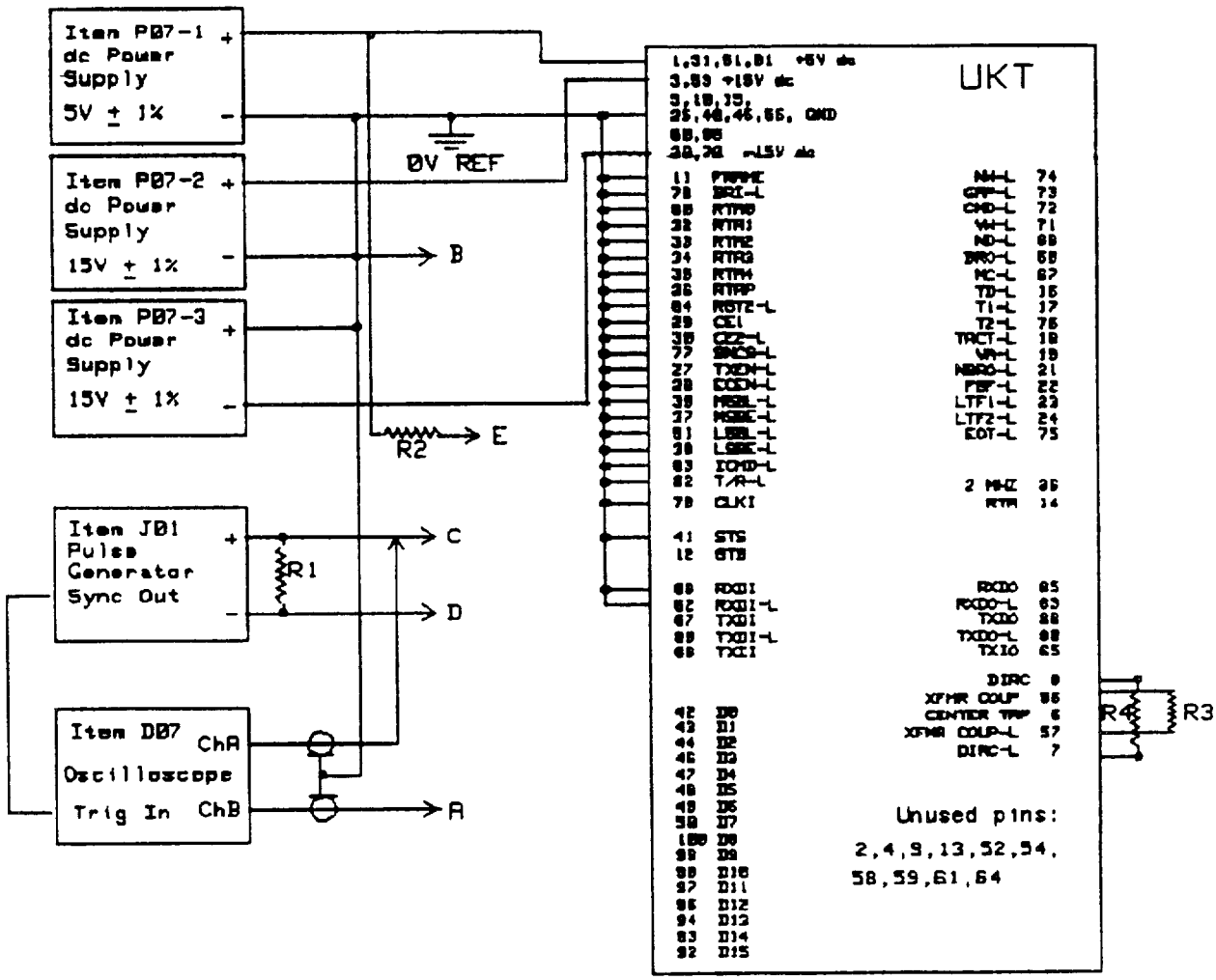
NOTES:

1. The test circuit shall be as indicated on figure 22.

2. Connect the module as indicated in table XXIV and make the indicated measurement.

3. The input and output waveforms shall be as indicated on figure 23.

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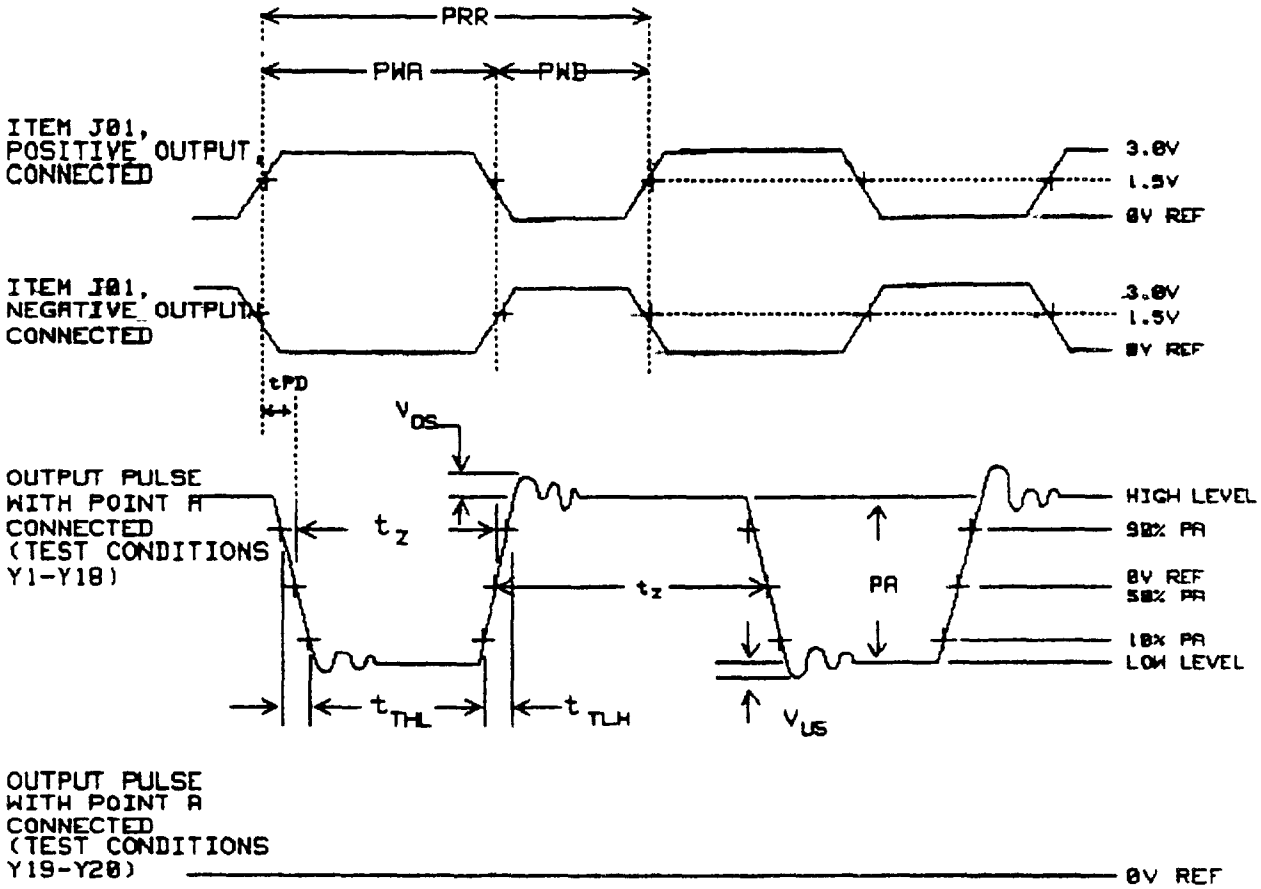
NOTES:

1. Disconnect R3 for test conditions Y1-Y9. Disconnect R4 for test conditions Y10-Y20.

2. R1 = 51.1 Ohm ± 5 percent, 1 W. R2 = 1 KOhm ± 5 percent, 1/4 W. R3 = 70 Ohm ± 2 percent, 1/8 W. R4 = 35 Ohm ± 2 percent, 1/8 W.

FIGURE 22. Transmitter functional test circuit for table XXIV.

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NOTES:

1. Input pulse parameters:

   PWA = PWB = 500 ± 25 nS.
   PRR = 1000 ± 50 nS.
   Transition times (10 percent - 90 percent of PA) shall be less than 5 nS.

2. Delay between positive and negative outputs of J01 shall be less than 5 nS.

FIGURE 23. Transmitter functional test circuit waveforms for table XXIV.

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TABLE XXV. Receiver no response test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z1	Vol1, RXDO	85	7	8	0.0	0.4	V	24,25
Z2	Vol2, RXDO-L	63	7	8	0.0	0.4	V	24,25
Z3	Vol1, RXDO	85	57	56	0.0	0.4	V	24,25
Z4	Vol2, RXDO-L	63	57	56	0.0	0.4	V	24,25

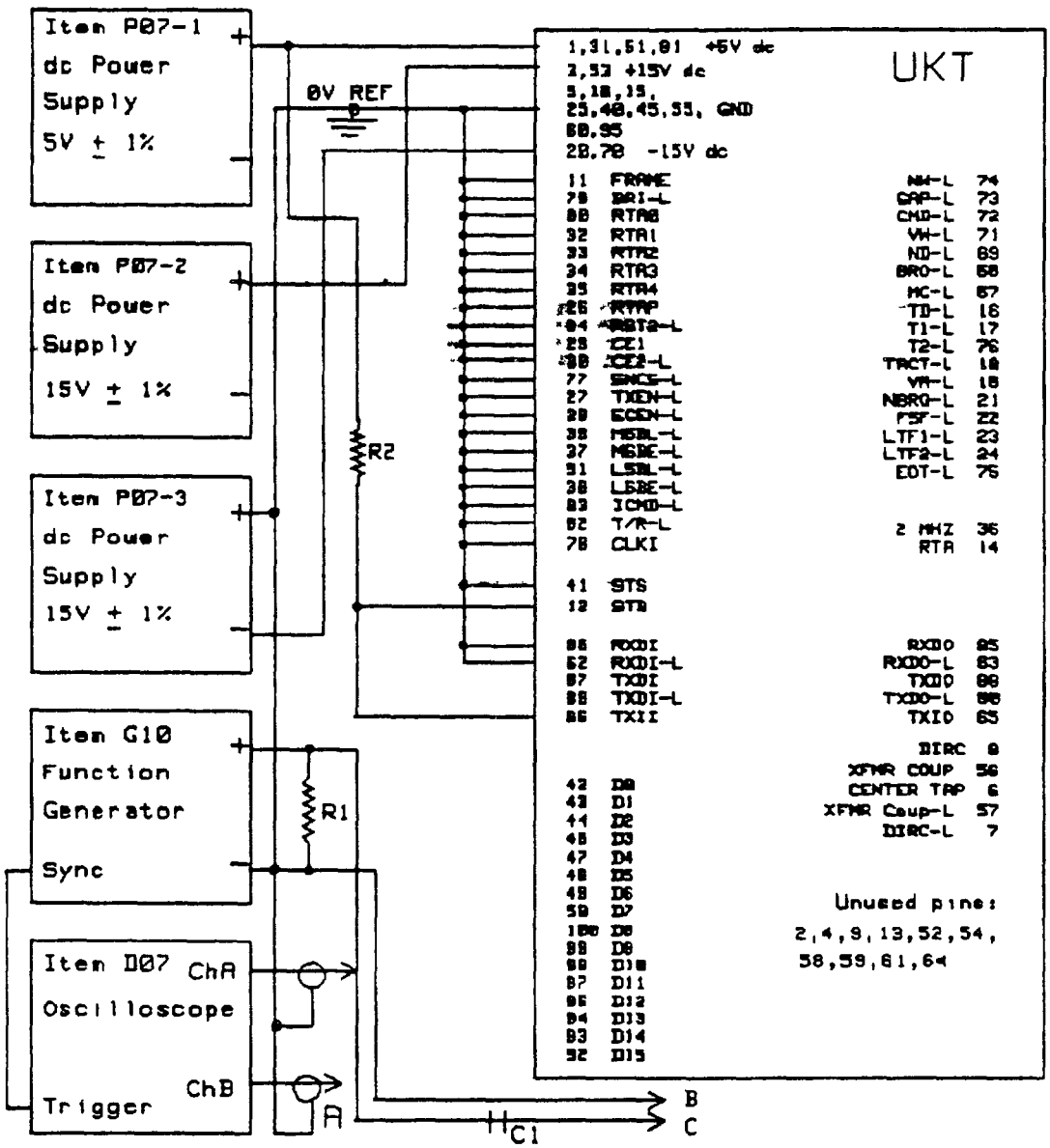
NOTES:

1. The test circuit shall be as indicated on figure 24.

2. Connect the module as specified in table XXV and make the indicated measurement.

3. The input and output waveforms shall be as indicated on figure 25.

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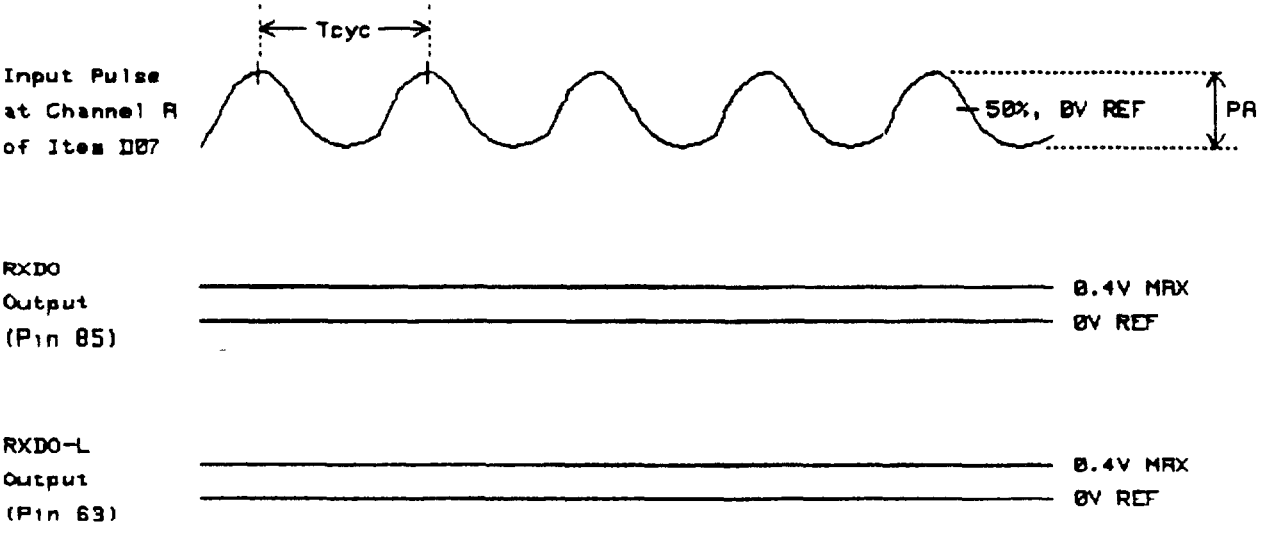


NOTES:

1. R1 = 51.1 Ohm ± 5 percent, 1.0 W.
2. R2 = 1 KOhm ± 5 percent, 1/4 W.
3. C1 = 1 uF ± 10 percent.

FIGURE 24. Receiver no response test circuit for table XXV.

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NOTES:

1. Input pulse parameters (item G-10 setting).
   Tcyc = 1 ± 0.3 uS.
   PA = 0.28 V p-p for test conditions Z1 and Z2.
   PA = 0.2 V p-p for test conditions Z3 and Z4.

2. Insure there is no 1 MHz square wave at the RXDO (pin 85) and RXDO-L (pin 63) outputs.

FIGURE 25. Receiver no response test circuit waveforms for table XXV.

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TABLE XXVI. Output symmetry test.

Test cond	Parameter	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
AA1	DIRC	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	- 90	90	mVpk	26,27
AA2		L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
AA3		H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
AA4		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
AA5		L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	L				
AA6	DIRC	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	- 90	90	mVpk	26,27
AA7	Transformer coupled symmetry	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	-250	250	mVpk	28,27
AA8		L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
AA9		H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
AA10		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
AA11		L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	L				
AA12	Transformer coupled symmetry	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	-250	250	mVpk	26,27

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NOTES:

1. The test circuit shall be as indicated on figure 26.

(a) For test conditions AA1-AA6 disconnect R1 and connect point 'A' to pin 8.

(b) For test conditions AA7-AA12 disconnect R2 and connect point 'A' to pin 56.

2. The pattern voltages shall be as follows:

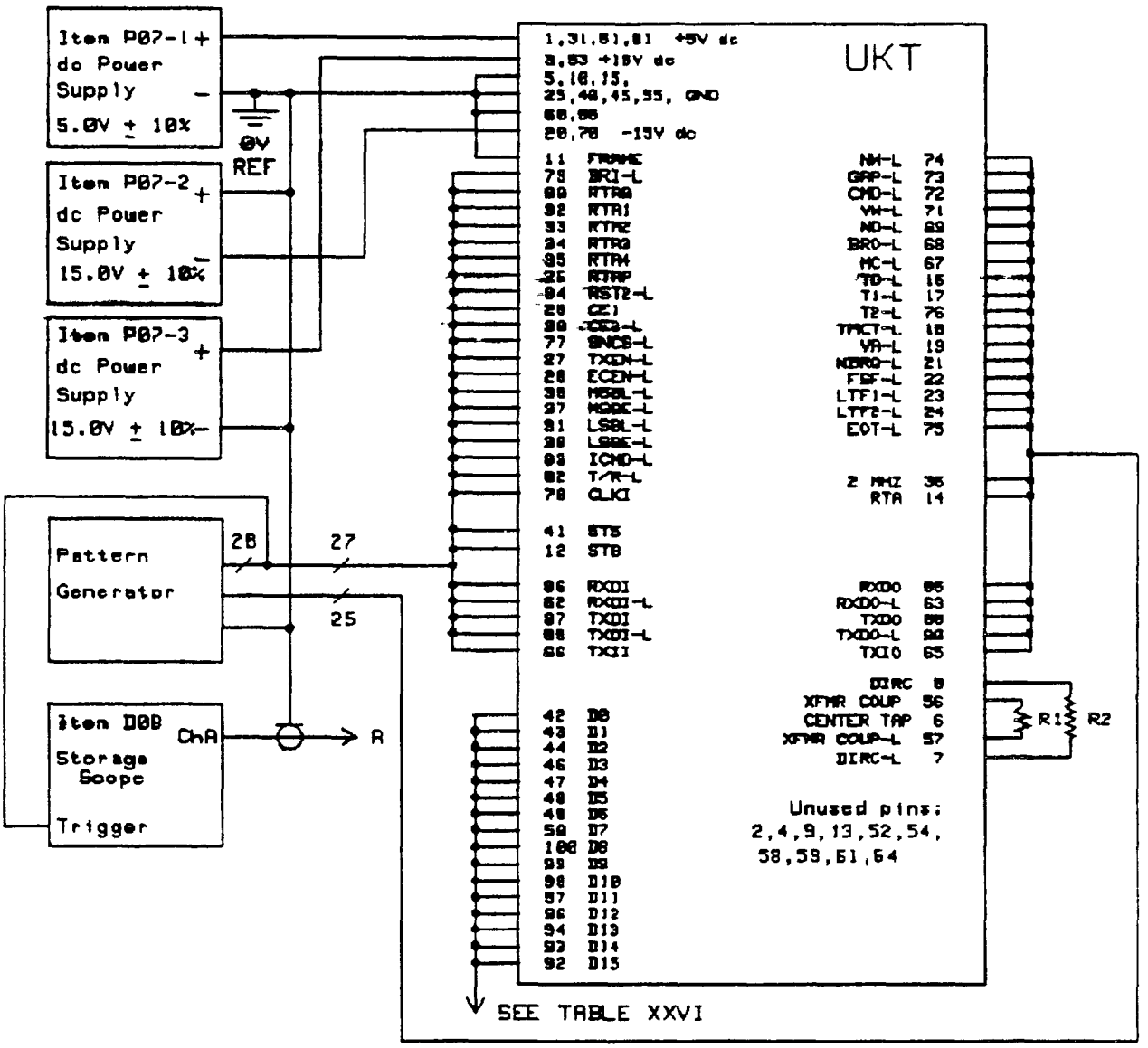
High level (VH) = 3.0 V for all pins except CLKI (pin 78).
High level (VH) = 5.0 V for CLKI (pin 78).
Low level (VL) = 0.4 V for all pins.

3. The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.4 V

4. From the time beginning 2.5 uS after the mid-bit crossing of the parity bit of the last word transmitted by the UKT, the maximum and minimum voltage at point 'A' of item D08 shall be within the limits specified in table XXVI. The output waveform shall be as indicated on figure 27. This condition shall be tested with the UKT transmitting the bit pattern indicated in table XXVI in a contiguous block of 33 words, all with that same bit pattern indicated for the test condition.

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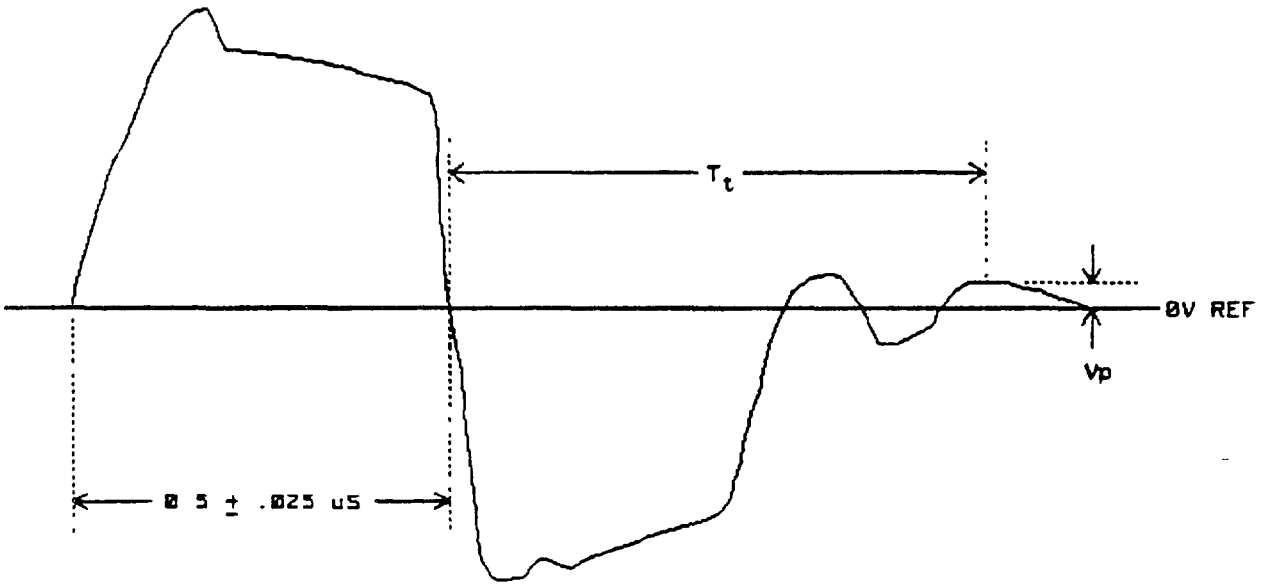


NOTE:

1. R1 = 70 Ohm ± 2 percent. R2 = 35 Ohm ± 2 percent.

FIGURE 26. Output symmetry test circuit for table XXVI.

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NOTES:

1. Tt = waveform tailoff time = 2.5 uS.
2. Vp = peak waveform voltage as specified in table XXVI.

FIGURE 27. Output symmetry test circuit waveform for table XXVI.

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TABLE XXVII. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
BB1	D0	42	0	400	mV	1-6	28,29
BB2	D1	43					
BB3	D2	44					
BB4	D3	46					
BB5	D4	47					
BB6	D5	48					
BB7	D6	49					
BB8	D7	50					
BB9	D8	100					
BB10	D9	99					
BB11	D10	98					
BB12	D11	97					
BB13	D12	96					
BB14	D13	94					
BB15	D14	93					
BB16	D15	92					
BB17	NW-L	74					
BB18	GAP-L	73					
BB19	CMD-L	72					
BB20	VW-L	71					
BB21	ND-L	69	0	400	mV	1-6	28,29

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TABLE XXVII. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
BB22	BRO-L	68	0	400	mV	1-6	28,29
BB23	MC-L	67					
BB24	TD-L	16					
BB25	T1-L	17					
BB26	T2-L	76					
BB27	TACT-L	18					
BB28	VA-L	19					
BB29	NBRQ-L	21					
BB30	FSF-L	22					
BB31	LTF1-L	23					
BB32	LTF2-L	24					
BB33	EOT-L	75					
BB34	2MHZ	36					
BB35	RTA	14					
BB36	RXDO	85					
BB37	RXDO-L	63					
BB38	TXDO	88					
BB39	TXDO-L	90					
BB40	TXIO	65	0	400	mV	1-6	28,29

See notes following table XXVIII.

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TABLE XXVIII. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
CC1	D0	42	0	400	mV	1-6	28,29
CC2	D1	43					
CC3	D2	44					
CC4	D3	46					
CC5	D4	47					
CC6	D5	48					
CC7	D6	49					
CC8	D7	50					
CC9	D8	100					
CC10	D9	99					
CC11	D10	98					
CC12	D11	97					
CC13	D12	96					
CC14	D13	94					
CC15	D14	93					
CC16	D15	92	0	400	mV	1-6	28,29
CC17	NW-L	74	0	700	mV	1-6	28,29
CC18	GAP-L	73					
CC19	CMD-L	72					
CC20	VW-L	71					
CC21	ND-L	69	0	700	mV	1-6	28,29

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TABLE XXVIII. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
CC22	BRO-L	68	0	400	mV	1-6	28,29
CC23	MC-L	67	0	700	mV	1-6	28,29
CC24	TD-L	16					
CC25	T1-L	17					
CC26	T2-L	76					
CC27	TACT-L	18	0	700	mV	1-6	28,29
CC28	VA-L	19	0	400	mV	1-6	28,29
CC29	NBRQ-L	21	0	700	mV	1-6	28,29
CC30	FSF-L	22					
CC31	LTF1-L	23					
CC32	LTF2-L	24					
CC33	EOT-L	75	0	700	mV	1-6	28,29
CC34	2MHZ	36	0	400	mV	1-6	28,29
CC35	RTA	14					
CC36	RXDO	85					
CC37	RXDO-L	63					
CC38	TXDO	88					
CC39	TXDO-L	90					
CC40	TXIO	65	0	400	mV	1-6	28,29

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NOTES:

1. Output crosstalk, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	BB1-BB40,CC1-CC40
(V+)	15.0 V	± 150 mV	BB1-BB40,CC1-CC40
(V-)	-15.0 V	± 150 mV	BB1-BB40,CC1-CC40
Power supply current limit (Vcc)	450 mA	NA	BB1-BB40,CC1-CC40
(V+)	375 mA	NA	BB1-BB40,CC1-CC40
(V-)	375 mA	NA	BB1-BB40,CC1-CC40
Pattern voltage high, (VH)	3.0 V	± 100 mV	BB1-BB40,CC1-CC40
low, (VL)	0.0 V	± 100 mV	BB1-BB40,CC1-CC40
high, (VH) (See note 4)	5.0 V	± 50 mV	BB1-BB40,CC1-CC40
Output crosstalk measurement	NA	± 20 mV	BB1-BB40
	NA	± 20 mV	CC1-CC16
	NA	± 35 mV	CC17-CC21
	NA	± 20 mV	CC22
	NA	± 35 mV	CC23-CC27
	NA	± 20 mV	CC28
	NA	± 35 mV	CC29-CC33
	NA	± 20 mV	CC34-CC40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in tables XXVII and XXVIII.

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NOTES: - Continued.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in tables XXVII and XXVIII, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 28.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 14 with C = 50 pF.

4. The high level pattern voltage for CLKI (pin 78) shall be 5.0 V.

5. The test circuit shall be as specified on figure 29.

6. Abbreviations used are defined in 6.1.

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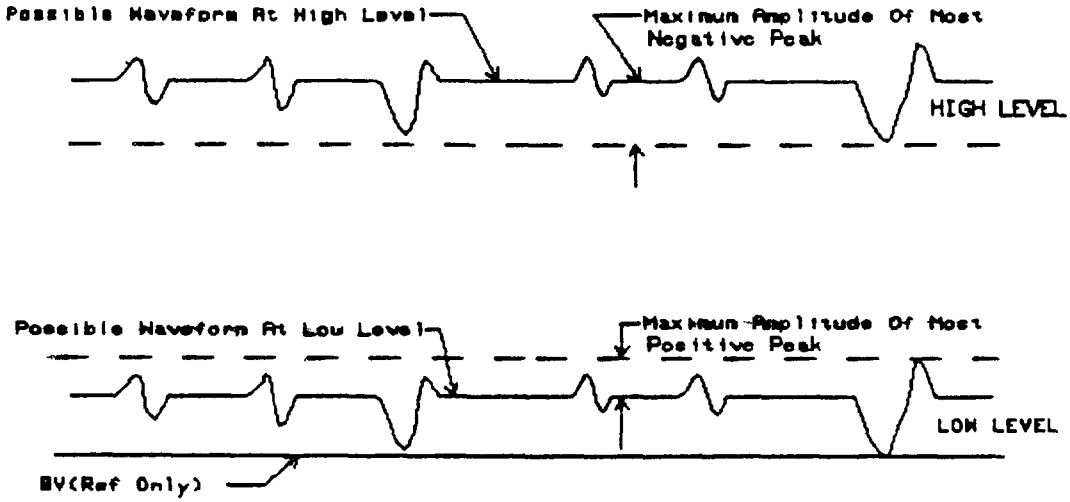


FIGURE 28. Output crosstalk test circuit waveforms, high and low level, for tables XXVII and XXVIII.

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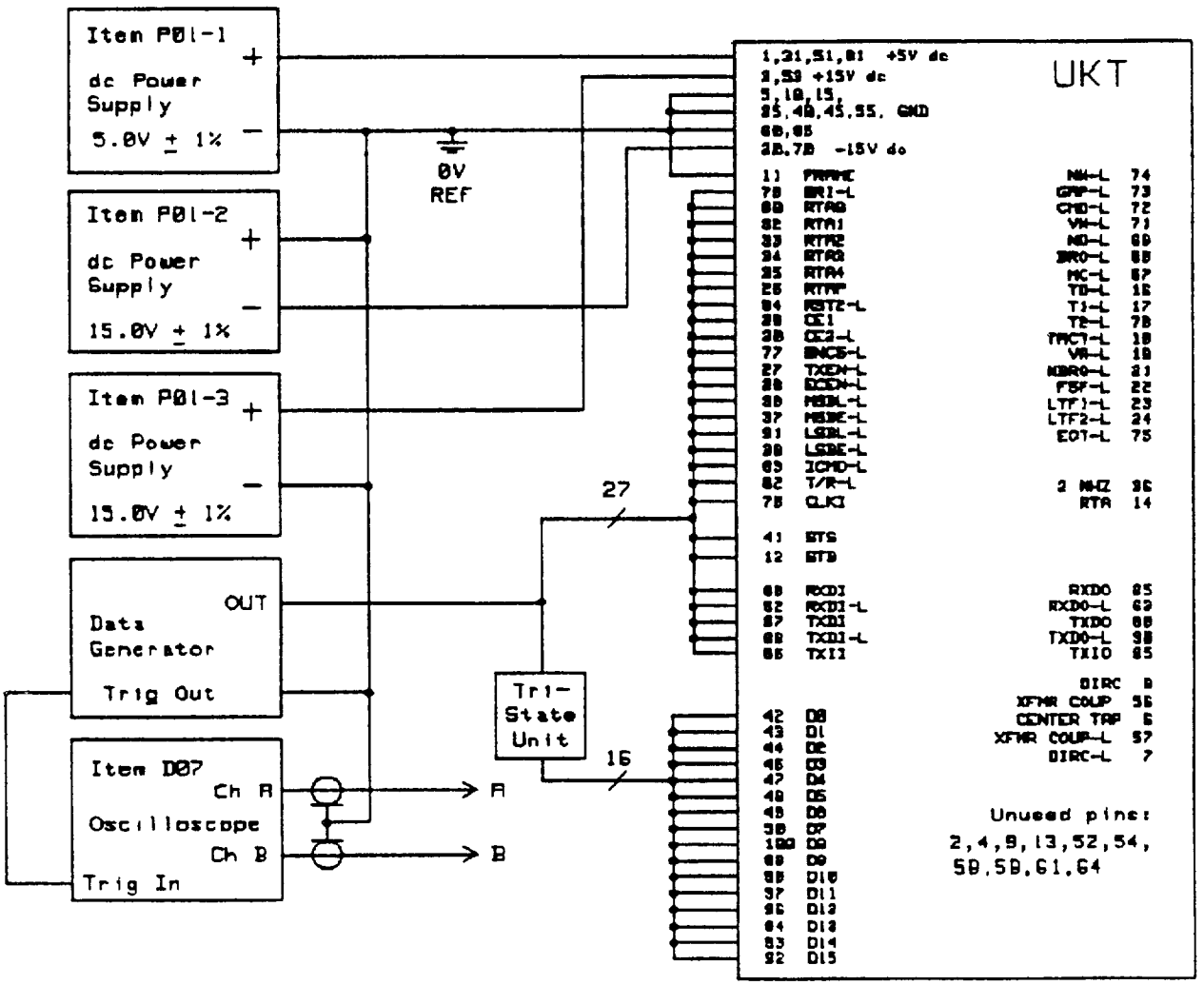


FIGURE 29. Output crosstalk, high and low level, test circuit for tables XXVII and XXVIII.

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TABLE XXIX. Receiver functional test.

Test cond	Parameter	G10-1 frequency ± 1 percent	Connect A to	Connect B to	Connect C to	Connect D to	Connect E to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test figs
DD1	Voh1, RXDO	1.0 MHz	85	7	8	12	Open	2.4	5.0	V	30,31
DD2	Vol1, RXDO	1.0 MHz	85	7	8	12	Open	0.0	0.4	V	30,31
DD3	tpd1, prop delay	1.0 MHz	85	7	8	12	Open	250.0	350.0	nS	30,31
DD4	Voh2, RXDO-L	1.0 MHz	63	7	8	12	Open	2.4	5.0	V	30,31
DD5	Vol2, RXDO-L	1.0 MHz	63	7	8	12	Open	0.0	0.4	V	30,31
DD6	tpd2, prop delay	1.0 MHz	63	7	8	12	Open	250.0	350.0	nS	30,31
DD7	Voh1, RXDO	1.0 MHz	85	57	56	12	Open	2.4	5.0	V	30,31
DD8	Vol1, RXDO	1.0 MHz	85	57	56	12	Open	0.0	0.4	V	30,31
DD9	tpd1, prop delay	1.0 MHz	85	57	56	12	Open	250.0	350.0	nS	30,31
DD10	Voh2, RXDO-L	1.0 MHz	63	57	56	12	Open	2.4	5.0	V	30,31
DD11	Vol2, RXDO-L	1.0 MHz	63	57	56	12	Open	0.0	0.4	V	30,31
DD12	tpd2, prop delay	1.0 MHz	63	57	56	12	Open	250.0	350.0	nS	30,31
DD13	No response	1.0 MHz	85	7	8	Open	12	No response			30,31
DD14	No response	1.0 MHz	63	7	8	Open	12	No response			30,31
DD15	No response	1.0 MHz	85	57	56	Open	12	No response			30,31
DD16	No response	1.0 MHz	63	57	56	Open	12	No response			30,31

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NOTES:

1. The test circuit shall be as indicated on figure 30.

2. Connect the module as indicated in table XXIX and make the indicated measurement.

3. Pins 63 and 58 shall be loaded as specified on figure 17.

4. Input and output waveforms shall be as indicated on figure 31.

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TABLE XXIX. Receiver functional test.

Test cond	Parameter	G10-1 frequency ± 1 percent	Connect A to	B to	C to	D to	E to	25 ± 5°C Initial limits Min	Max	Units	Test figs
DD1	Voh1, RXDO	1.0 MHz	85	7	8	12	Open	2.4	5.0	V	30,31
DD2	Vol1, RXDO	1.0 MHz	85	7	8	12	Open	0.0	0.4	V	30,31
DD3	tpd1, prop delay	1.0 MHz	85	7	8	12	Open	250.0	350.0	nS	30,31
DD4	Voh2, RXDO-L	1.0 MHz	63	7	8	12	Open	2.4	5.0	V	30,31
DD5	Vol2, RXDO-L	1.0 MHz	63	7	8	12	Open	0.0	0.4	V	30,31
DD6	tpd2, prop delay	1.0 MHz	63	7	8	12	Open	250.0	350.0	nS	30,31
DD7	Voh1, RXDO	1.0 MHz	85	57	56	12	Open	2.4	5.0	V	30,31
DD8	Vol1, RXDO	1.0 MHz	85	57	56	12	Open	0.0	0.4	V	30,31
DD9	tpd1, prop delay	1.0 MHz	85	57	56	12	Open	250.0	350.0	nS	30,31
DD10	Voh2, RXDO-L	1.0 MHz	63	57	56	12	Open	2.4	5.0	V	30,31
DD11	Vol2, RXDO-L	1.0 MHz	63	57	56	12	Open	0.0	0.4	V	30,31
DD12	tpd2, prop delay	1.0 MHz	63	57	56	12	Open	250.0	350.0	nS	30,31
DD13	No response	1.0 MHz	85	7	8	Open	12	No response			30,31
DD14	No response	1.0 MHz	63	7	8	Open	12	No response			30,31
DD15	No response	1.0 MHz	85	57	56	Open	12	No response			30,31
DD16	No response	1.0 MHz	63	57	56	Open	12	No response			30,31

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NOTES:

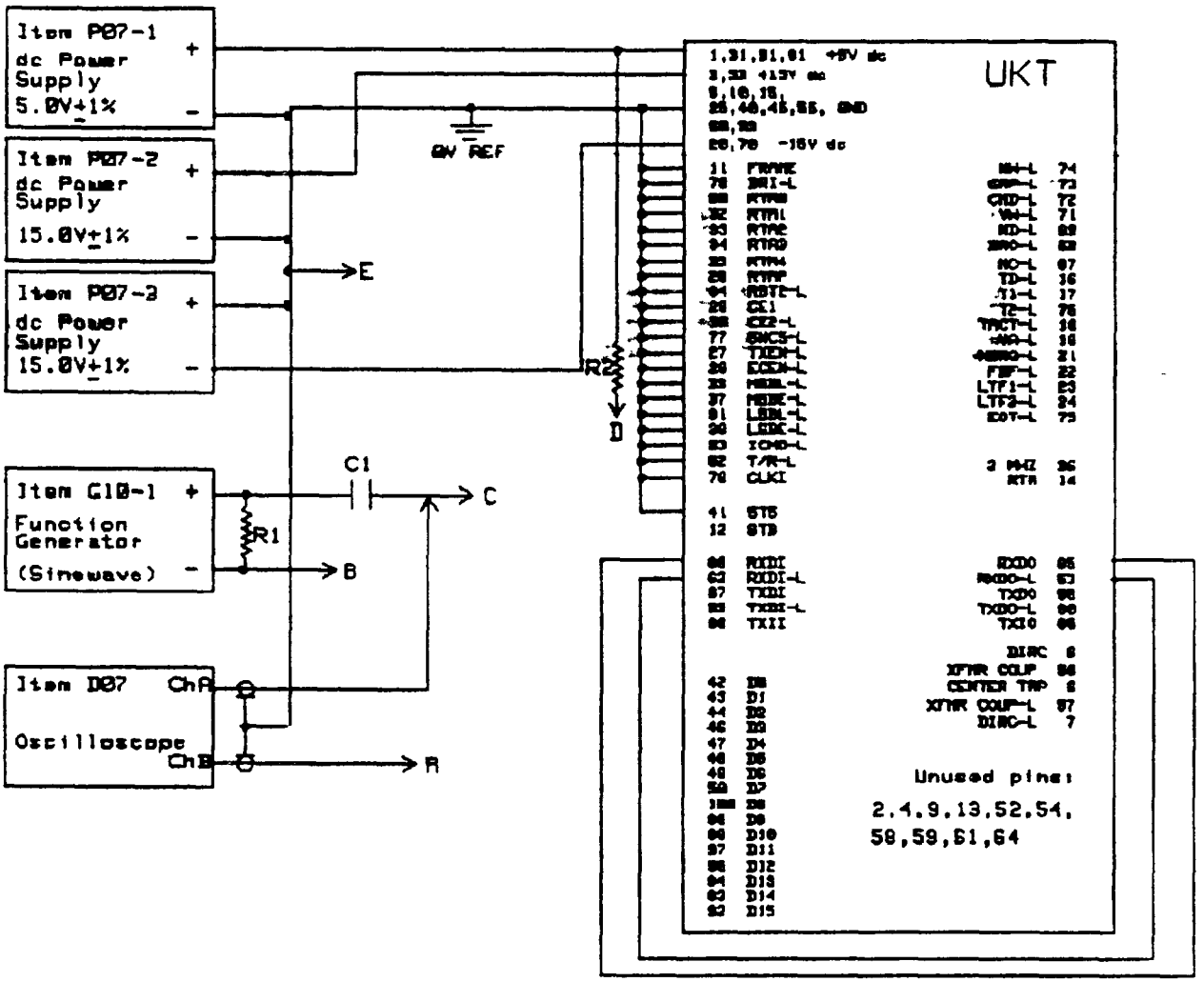
1. The test circuit shall be as indicated on figure 30.

2. Connect the module as indicated in table XXIX and make the indicated measurement.

3. Pins 63 and 58 shall be loaded as specified on figure 17.

4. Input and output waveforms shall be as indicated on figure 31.

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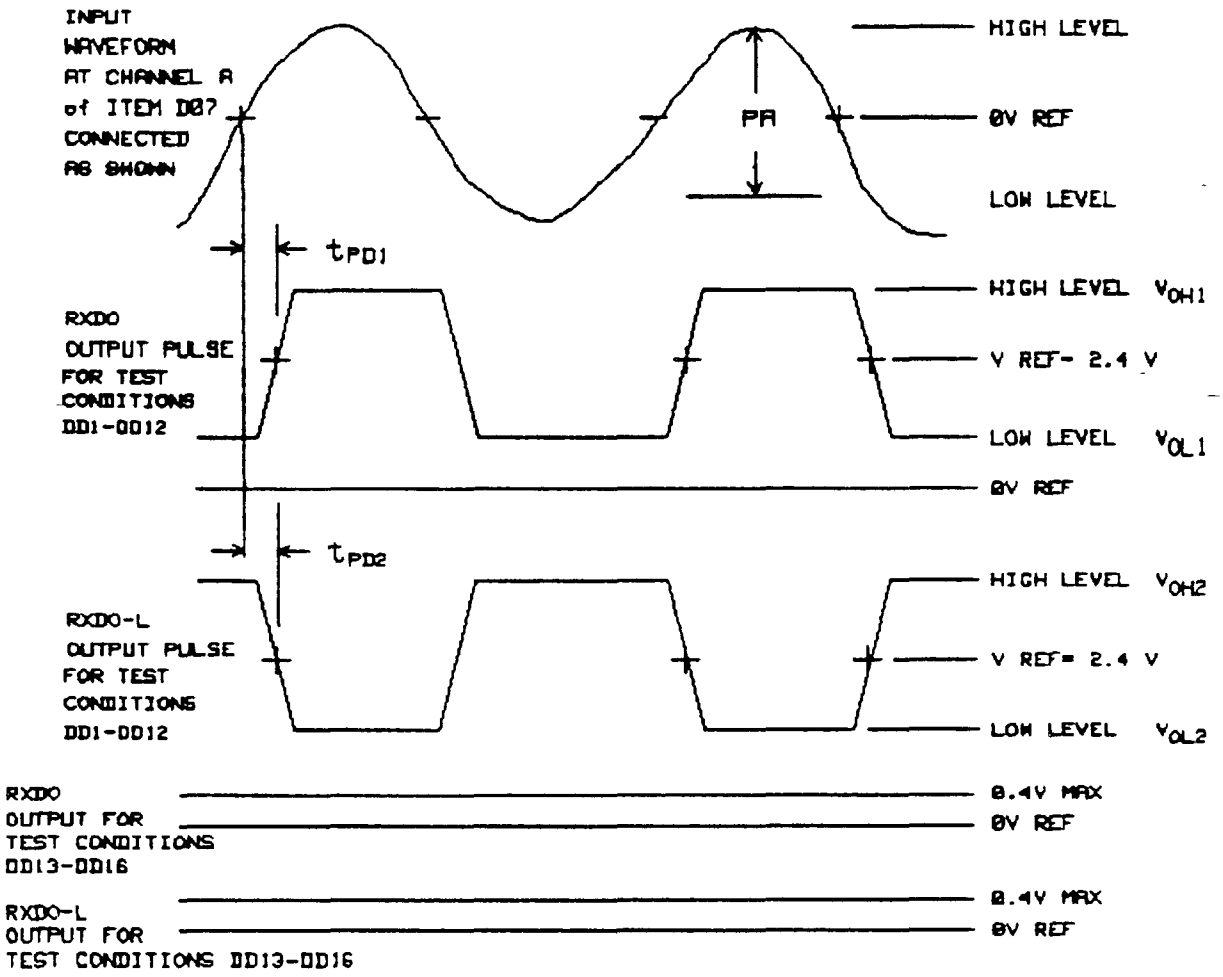


NOTES:

1. R1 = 51.1 Ohm ± 1 percent, 1 W.
2. R2 = 1 KOhm ± 5 percent, 1/4 W.
3. C1 = 1 uF ± 10 percent.

FIGURE 30. Receiver functional test circuit for table XXIX.

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NOTES:

1. Item G10-1 is a 1 MHz ± 10 percent sinewave. PA = 7 V p-p ± 1 percent.

2. Use offset capability of G10 to obtain bipolar waveform.

FIGURE 31. Receiver functional test circuit waveforms for table XXIX.

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